

Lecture Notes for Algebraic Structures (Use at your own peril!)

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1 Introduction

I found Algebraic Structures to be a hard but rewarding and fun course. To keep my memory fresh and my $\text{\LaTeX}$ skills sharp I found that writing this would be a fun challenge. Since there are no (what I know of) lecture notes (in PDF form) on this course I also see this as an opportunity to help new students taking this course. I will take heavily inspiration from my lecture notes I took during the course and also some from the course literature (which uses different notation from what the course uses).

Feel free to contact me and report fault or recommendation of things that could be added or updated. There is now a private Github for the project making it easier for contributions. As of 15th September 2026, I would consider this version a draft-stage.

I also want to give a big thanks to those who have helped in any way with the project.

2 Groups

2.1 Definition

We start by defining the notion of a group.

Definition 2.1. A *group* is a tuple $(G, *, e)$ consisting of

- a set G ,
- a binary operator $*$,
- and an identity element $e \in G$,

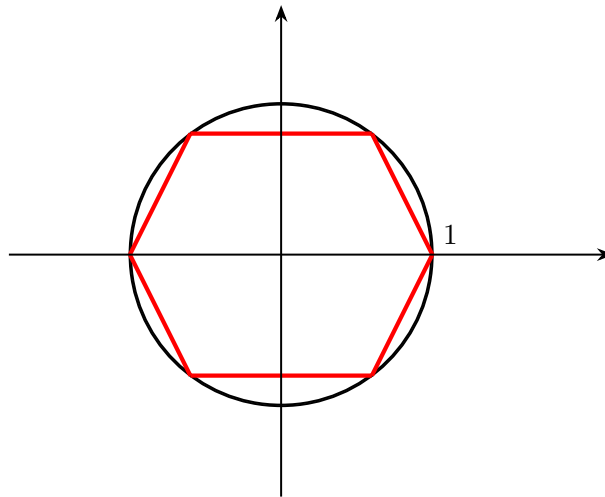
which satisfy:

- $\forall a, b, c \in G : (a * b) * c = a * (b * c)$ (associativity)
- $\forall a \in G : e * a = a * e = a$ (identity)
- $\forall a \in G \exists b \in G : a * b = b * a = e$ (inverse)

Remark 2.2. We often say “let G be a group” instead of “let $(G, *, e)$ be a group”, when the group operation and identity element can be inferred from context.

Example 2.3 (Groups and non-groups).

1. $G = \{e\}$ since $e * e = e$. Known as the “trivial group” or “singleton”. Note that the inverse of an element can be the element itself, they do not have to be different.
2. $(\mathbb{Z}, +, 0)$ is a group, where the inverse element of $x \in \mathbb{Z}$ is $-x$.
3. $(\mathbb{Z}, \cdot, 1)$ is not a group, since only 1 and -1 have inverses.
4. $(\mathbb{Q}, \cdot, 1)$ is not a group, since 0 lacks an inverse.
5. $(\mathbb{Q}^*, \cdot, 1)$, where $\mathbb{Q}^* := \mathbb{Q} \setminus \{0\}$, is a group.
6. a) Let $X \subset \mathbb{R}^2$ and let $\text{Sym}(X) := \{f : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \mid f(X) = X\}$ be the set of transformations on $\mathbb{R}^2$ which leave X invariant. A transformation can be e.g. an isometry, a linear bijection or a rotational symmetry. Then $(\text{Sym}(X), \circ, \text{id}_{\mathbb{R}^2})$ is a group, where $\circ$ is composition of functions.
 b) (Dihedral groups) Consider a regular n -gon $X \subset \mathbb{R}^2$ inscribed in the unit circle, and all bijective linear maps $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ which preserve X . For example, consider the hexagon:



The hexagon can be rotated 0, 60, 120, 180, 240, and 300 degrees around the origin. We can also take reflections in the axes offset from the x -axis by 0, 30, 60, 90, 120, and 150 degrees, i.e. the axes passing through vertices or the middle of edges. These are the elements of the dihedral group $D_{2,6}$, with “do nothing” being the identity element and composition of rotations and reflections being the group operation.

A general n -gon will have corresponding Dihedral group $D_{2,n}$. We write $2 \cdot n$ since there are n rotations and n reflections, so $|D_{2,n}| = 2 \cdot n$, but you may come across the notation D_n also.

2.2 Basic properties

We prove a few basic properties of groups in general.

Proposition 2.4 (Uniqueness of the identity). If G is a group and $e' \in G$ such that $e' * a = a * e' \forall a \in G$ then $e = e'$

Proof. $e = e * e' = e'$ □

Proposition 2.5 (Uniqueness of inverses). If G is a group and $a, a', b, b' \in G$ such that $a * b = b * a = e$ and $a * b' = b' * a = e$ then $b = b'$

Proof. $b' = e * b' = (b * a) * b' = b * (a * b') = b * e = b$ □

Notation 2.6. We usually write a^{-1} for the inverse of a , if the group operation is identified as addition we may also write $-a$.

Proposition 2.7. $(a * b)^{-1} = b^{-1} * a^{-1}$

Proof. $(a * b) * (a * b)^{-1} = e = a * b * b^{-1} * a^{-1}$, Proposition 2.5 gives the desired result. □

Example 2.8.

1. (Symmetric groups) Let X be a set and set $S(X) := \{f : X \rightarrow X \mid f \text{ is bijective}\}$. Then $(S(X), \circ, \text{id}_X)$ is the symmetric group on X . If $X = \{1, 2, \dots, n\}$, then $S(X) = S_n$. We will talk more about these later.
2. $(\text{Mat}_{n \times n}(\mathbb{R}), +, 0_{n \times n})$ is a group, where $\text{Mat}_{n \times n}(\mathbb{R})$ denotes $n \times n$ -matrices with real elements.

3. $(\text{Mat}_{n \times n}(\mathbb{R}), \cdot, I_n)$ is not a group, since some matrices are not invertible!
4. $(\text{GL}_n(\mathbb{R}), \cdot, I_n)$ is a group where $\text{GL}_n(\mathbb{R}) = \{M \in \text{Mat}_{n \times n} \mid \det(M) \neq 0\}$. Here, “GL” is short for “General Linear”. This is closed because $\det(AB) = \det(A)\det(B)$, so $\det(AB) \neq 0$ when $\det(A) \neq 0$ and $\det(B) \neq 0$.

Remark 2.9. That $a * b = b * a$ does not hold in general.

Definition 2.10 (Abelian group). If G is a group such that $a * b = b * a$ for all $a, b \in G$, then G is called abelian (after the Norwegian mathematician Niels Henrik Abel) or commutative.

Example 2.11.

- The integers under addition, $(\mathbb{Z}, +, 0)$, forms an abelian group.
- $(\text{GL}_n(\mathbb{R}), \cdot, I_n)$ is not abelian for $n > 1$ since it is generally not the case that $AB = BA$.

Remark 2.12.

- Instead of writing the tuple every time, the usual notation is the set, e.g. G , $\text{GL}_n(\mathbb{R})$, or $\mathbb{Z}$ instead of $(G, *, e)$, $(\text{GL}_n(\mathbb{R}), \cdot, I_n)$, or $(\mathbb{Z}, +, 0)$.
- When the $*$ -operator is multiplication $(\cdot)$, $a \cdot b$ is often written as ab .

2.3 Subgroups

Let $(G, *, e)$ be a group.

Definition 2.13. A subset $H \subset G$ is called a *subgroup* provided that:

- $e \in H$
- For all $a, b \in H$, we have $a * b \in H$
- For all $a \in H$, we have $a^{-1} \in H$

Remark 2.14. Any subgroup is itself a group with respect to $*|_{H \times H}$ ($*$ restricted to elements of H) and e , creating the group $(H, *|_{H \times H}, e)$.

Notation 2.15. $H < G$ means that H is a subgroup of G .

Example 2.16.

1. $G < G$ and $\{e\} < G$ are the two trivial subgroups. The first one is clear and the latter is easy to check for since there is only one element to check: $e * e = e$.
2. $n\mathbb{Z} < \mathbb{Z}, n \in \mathbb{Z}$
3. $\text{Rot}_n < D_{2 \cdot n}$ is subgroup containing only the rotational symmetries.
4. L  gg massa figurer
5. $\mathbb{R}^n \curvearrowright (\text{Iso}(\mathbb{R}^n), \circ, \text{id}_{\mathbb{R}^n})$ — the group of isometries of $\mathbb{R}^n$. Reminder that an isometry is a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $d(f(x), f(y)) = d(x, y)$ for all $x, y \in \mathbb{R}^n$.

Now let $X \subset \mathbb{R}^n$, then $\text{Sym}(X) := \{f \in \text{Iso}(\mathbb{R}^n) \mid f(X) = X\}$ induces a subgroup of $\text{Iso}(\mathbb{R}^n)$.

Proposition 2.17. G -group, $H < G$, $K < G$ then $H \cap K < G$.

More generally: $H_I < G, i \in I$ then $\bigcap_{i \in I} H_I < G$

Proof. Here one only need to check if $H \cap K$ is a subgroup of G from the definition of a subgroup: if $e \in H, K$ then $e \in H \cap K$. Then if $a, b \in H \cap K$ then $a * b \in H, K$ and so $a * b \in H \cap K$. Lastly, if $a \in H \cap K \implies a^{-1} \in H, K \implies a^{-1} \in H \cap K$. The general case is left as an exercise for the reader. $\square$

2.4 Generating subgroups and order

Now we will start talking about generating subgroups, and so the set-up will be a group G and a subset $X \subset G$. Please note that ' $\subset$ ' may be used interchangeably with ' $\subseteq$ ' throughout this course. It is similar to how Walter Rudin does it in his book *Principles of Mathematical Analysis*.

Definition 2.18 (Generating subgroup).

$$\langle X \rangle := \bigcap_{X \subset H < G} H$$

is the subgroup of G generated by X . $\langle X \rangle$ is the minimum subgroup of G containing X .

Example 2.19.

1. $X = G \implies \langle G \rangle = G$
2. $X = \emptyset \implies \langle \emptyset \rangle = \{e\} = \langle e \rangle$
3. Consider $G = \mathbb{Z}$, then $\langle \{1\} \rangle = \mathbb{Z}$ and $\langle \{2\} \rangle = 2\mathbb{Z}$

Definition 2.20. The *order* of an element $a \in G$ denoted $o(a)$, is the cardinality of $\langle \{a\} \rangle$.

One can think of generating subgroups as a 'word-generator' that generates all 'words' from elements of the set X , e and inverses of elements in X . So for example, if $a, b \in G$ then $\langle \{a, b\} \rangle = \{e, a, b, aa, ab, bb, a^{-1}, a^{-1}b, \dots\}$. So in example 2.16, $\langle \{1\} \rangle = \{0, 1, 1+1, 0-1, 1+1+1, 0-1-1, \dots\} = \mathbb{Z}$.

Example 2.21.

1. $o(e) = 1$.
2. In $\mathbb{Z}$, $o(1) = \infty$.

Proposition 2.22. Let G be a group, $a \in G$, $a^n = \underbrace{a * \dots * a}_{n\text{-times}}$. Consider $X_a := \{a, a^2, a^3, \dots\}$ then we get two cases:

Case 1: $|X_a| = \infty \implies o(a) = \infty$

Case 2: $|X_a| < \infty \implies o(a) = |X_a| = \min\{k > 0 \mid a^k = e\}$

Proof. **Case 1:** No $k \in \mathbb{N}$ exists such that $a^k = a^{k+1} = a^{k+2} = \dots$ and hence $o(a) = \infty$.

Case 2: We assume that $|X_a|$ is finite and so there exists a k and m , $k < m$ such that $a^k = a^m$, let $\kappa = m - k$. Then $a^\kappa = e$ and so $X_a = \{a, a^2, \dots, a^{\kappa-1}, e, a, a^2, \dots\} = \langle a \rangle$, and now it is clear that $|X_a| = o(a) = \min\{k > 0 \mid a^k = e\}$. $\square$

2.5 Cosets

Set up: Let G be a group and $H < G$ subgroup of G . We'll start off by a definition:

Definition 2.23 (Left and right cosets).

- Left coset: $g \in G$ $gH := \{gh | h \in H\}$
- Right coset: $g \in G$ $Hg := \{hg | h \in H\}$

In some literature they will define left and right cosets in the opposite order, but all theorems and propositions will still be the same.

Example 2.24.

1. $H = G \implies$ all $gH = G = Hg \forall g \in G$
2. $H = \{e\} \implies$ all $gH = g = Hg \forall g \in G$
3. If $H = 2\mathbb{Z} < \mathbb{Z} = G$, then the cosets are: $2\mathbb{Z}$ and $1 + 2\mathbb{Z}$.

Proposition 2.25.

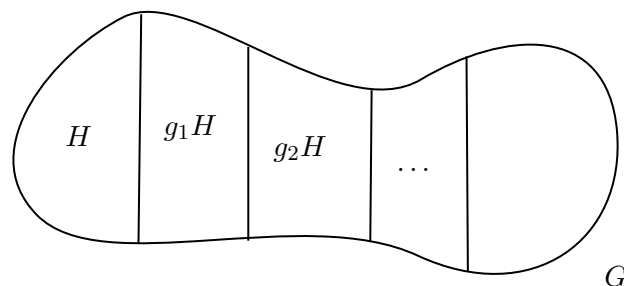
1. $g_1, g_2 \in G \implies g_1H = g_2H$ or $g_1H \cap g_2H = \emptyset$
2. $g_1, g_2 \in G \implies Hg_1 = Hg_2$ or $Hg_1 \cap Hg_2 = \emptyset$

Proof. We will only prove 1) because 2) is (almost) exactly the same. A good exercise would be to go back to the proposition later and try to prove 2).

Assume $x \in g_1H \cap g_2H \implies \exists h_1, h_2 \in H$ such that $x = g_1h_1 = g_2h_2$. Now multiply by h_1^{-1} on both sides: $g_1e = g_1h_1h_1^{-1} = g_2 \underbrace{h_2h_1^{-1}}_{\in H} \implies g_1 \in g_2H$. Hence we have $g_1H \subset g_2H$. By symmetry we get $g_2H \subset g_1H$ and

hence $g_1H = g_2H$. □

One can look at the cosets of a subgroup as partitions of the group G . This will become useful later.



Notation 2.26.

- $G / H := \{gH | g \in G\}$
- $H \setminus G := \{Hg | g \in G\}$

Definition 2.27. $H < G$ is called *normal* (notation: $H \triangleleft G$) provided that $\forall g \in G \ gH = Hg^{-1}$, or equivalently if $H = gHg^{-1}$

Example 2.28.

$$S_3 = \left\{ \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}}_{=e}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}}_{=s}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}}_{=t}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}}_{=st}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}}_{=ts}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}}_{=\omega_0=sts=tst} \right\}$$

The way to read this notation is to look at what elements go to where. So each element on the top row goes to the element under it on the bottom row. Multiplying two elements, say

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

where one reads from right to left. Here 1 goes to 1 (reading from the first matrix) and then we carry the result to the next matrix, 1 goes to 2, so we write in the resulting matrix 1 at the top and 2 under. The next number we do 2 goes to 3, and 3 goes to 3; 2 goes to 3, so 2 on top and 3 under. Lastly 3 goes to 2 and 2 goes to 1, so 3 goes to 1 and hence we write 3 on top and 1 under.

Now consider

$$H := \left\{ \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}}_{=e}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}}_{=s} \right\} < S_3$$

Why H is a subgroup of S_3 is left as an exercise. Then:

$$S_3 / H = \left\{ \underbrace{\left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \end{pmatrix} \right\}}_{=e}, \underbrace{\left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \right\}}_{=t}, \underbrace{\left\{ \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\}}_{=\omega_0} \right\}$$

and writing a shorter version for the right quotient:

$$H \setminus S_3 = \{H, \{t, ts\}, \{st, \omega_0\}\}$$

and now since these sets are not equal it doesn't follow that for *all* $g \in S_3$ we have $gH = Hg^{-1}$, hence $H \not\triangleleft S_3$.

Proposition 2.29. $H < G, g \in G$

The map:

$$\begin{array}{ccc} \varphi: H & \longrightarrow & gH \\ \downarrow & & \downarrow \\ h & \longmapsto & gh \end{array} \quad \text{is a bijection}$$

Proof. φ is surjective by definition, but we need to check injectivity. Recall that a function f is injective if $f(x) = f(y) \implies x = y$. We assume $\varphi(h_1) = \varphi(h_2)$ for some fixed $g \in G$ and show $h_1 = h_2$:

$$\begin{aligned} gh_1 = gh_2 &\implies \underbrace{g^{-1}gh_1 = g^{-1}gh_2}_{\text{left-multiply by } g^{-1}} \implies eh_1 = eh_2 \implies h_1 = h_2 \end{aligned}$$

Since φ is injective and surjective, it is bijective. □

I want to give a short anecdote that is not relevant to this course but something I came to think about when writing this proof. The statement “if there exists an injection $X \hookrightarrow Y$ and an injection $Y \hookrightarrow X$ then there exists a bijection between X and Y ”, is something that feel very trivial but is actually a somewhat big theorem called *Cantor-Bernstein-Schröder theorem* and is brought up in the Mathematical Logic course, which I highly recommend.

Corollary 2.30. If $H < G$ and $g_1, g_2 \in G$ then $|g_1 H| = |g_2 H| = |H|$

Proof. Left as an exercise □

Notation 2.31. The *index* $[G : H]$ is the number of (left) H cosets in G

Theorem 2.32 (Lagrange’s theorem). $H < G, |G| < \infty \implies |H| \mid |G|$ and in fact $|G| = |H| \cdot [G : H]$

Corollary 2.33. If $|G| < \infty$ then $o(a) \mid |G|$ for all $a \in G$

2.6 Multiplicity of groups

Proposition 2.34. $K < H < G \implies [G : K] = [G : H] \cdot [H : K]$

Proof. We assume $|G| < \infty$. By Lagrange’s theorem:

$$\begin{aligned} [G : H] &= \frac{|G|}{|H|} & [H : K] &= \frac{|H|}{|K|} \\ [G : K] &= \frac{|G|}{|K|} & \text{and} & \quad \frac{|G|}{|K|} = \frac{|G|}{|H|} \cdot \frac{|H|}{|K|} \end{aligned}$$

□

Remark 2.35. G -abelian, then if $\forall H < G, \forall g \in G \ gH = Hg$ holds, then $H \triangleleft G$

3 Homomorphisms

Setup: $(G, *, e)$ and $(H, \diamond, f)$ are groups.

Definition 3.1.

A map $\varphi : G \rightarrow H$ is called a (group) *homomorphism* provided that $\varphi(a * b) = \varphi(a) \diamond \varphi(b) \quad \forall a, b \in G$

Example 3.2.

1. $\forall G \quad \text{id}_G : G \rightarrow G$ is a homomorphism.
2. $\forall G, H \quad \varphi : G \rightarrow H, \varphi(g) = f, \forall g \in G$ is a homomorphism

Proposition 3.3. If $\varphi : G \rightarrow H$ is a homomorphism, then:

1. $\varphi(e) = f$
2. $\forall a \in G, \varphi(a^{-1}) = \varphi(a)^{-1}$

Proof.

1. $\varphi(e) = \varphi(e * e) = \varphi(e) \diamond \varphi(e)$. By multiplying with $\varphi(e)^{-1}$ on both sides we get $f = \varphi(e)$.
2. $f = \varphi(a) \diamond \varphi(a^{-1}) = \varphi(a * a^{-1}) = \varphi(e) = f$. Then $\varphi(a^{-1}) = \varphi(a)^{-1}$ by the uniqueness of inverses.

□

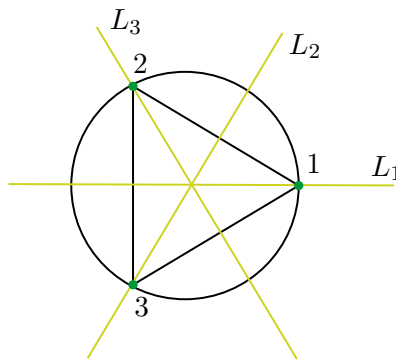
Example 3.4.

1. If $\mathbb{K} \in \{\mathbb{Q}, \mathbb{R}, \mathbb{C}\}$, then

$$\begin{array}{ccc} \det : \mathrm{GL}_n(\mathbb{K}) & \longrightarrow & \mathbb{K}^+ \\ \psi & & \psi \\ M & \longmapsto & \det(M) \end{array}$$

is a homomorphism. Check: $\det(MN) = \det(M) \cdot \det(N)$.

2. Consider $D_{2,3} = \{\mathrm{id}, \mathrm{rot}_{\frac{2\pi}{3}}, \mathrm{rot}_{\frac{4\pi}{3}}, s_1, s_2, s_3\}$, where s_1, s_2 , and s_3 are reflections with respect to the lines L_1, L_2 and L_3 .



Then define $\varphi : D_{2,3} \rightarrow S_3$ by

$$\begin{array}{ll} \mathrm{id} \mapsto \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} & \mathrm{rot}_{\frac{2\pi}{3}} \mapsto \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \\ \mathrm{rot}_{\frac{4\pi}{3}} \mapsto \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} & s_1 \mapsto \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \\ s_2 \mapsto \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} & s_3 \mapsto \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \end{array}$$

showing that there are other ways to realise $D_{2,3}$ than for example using rotational and reflective matrices in $\mathbb{R}^2$.

Definition 3.5. Two groups G and H are *isomorphic* if $\exists \varphi : G \rightarrow H$, φ is a homomorphism and φ is bijective.

Proposition 3.6. If $\varphi : G \rightarrow H$ & $\psi : H \rightarrow K$ are group homomorphisms, then $\psi \circ \varphi : G \rightarrow K$ is also a group homomorphism.

Proof. $\psi \circ \varphi(ab) = \psi(\varphi(ab)) = \psi(\varphi(a)\varphi(b)) = \psi(\varphi(a))\psi(\varphi(b)) = \psi \circ \varphi(a) \cdot \psi \circ \varphi(b)$

□

3.1 Kernel & images

Set-up: $\varphi : G \rightarrow H$ is a group homomorphism.

Definition 3.7.

1. The *kernel* of φ is defined as $\ker(\varphi) := \{a \in G \mid \varphi(a) = e_H\}$
2. The *image* of φ is defined as $\operatorname{im}(\varphi) := \{\varphi(a) \mid a \in G\}$

Example 3.8. 1. The identity map $\operatorname{id}_G : G \rightarrow G$ has $\ker(\operatorname{id}_G) = \{e\}$ and $\operatorname{im}(\operatorname{id}_G) = G$.

2. For any groups G and H , the function $\begin{array}{ccc} \varphi : G & \longrightarrow & H \\ \wr & & \wr \\ g & \longmapsto & e_H \end{array}$ has $\ker(\varphi) = G$ and $\operatorname{im}(\varphi) = \{e_H\}$.

3. For the map $\det : \operatorname{GL}_n(\mathbb{K}) \rightarrow \mathbb{K}^*$, then $\operatorname{im}(\det) = \mathbb{K}^*$ and $\ker(\det) = \operatorname{SL}_n(\mathbb{K}) = \{M \mid \det(M) = 1\}$.

Proposition 3.9 (Noether's first isomorphism theorem). Let $\varphi : G \rightarrow H$ be a group homomorphism. Then φ induces a bijection:

$$\begin{array}{ccc} \bar{\varphi} : G / \ker(\varphi) & \longrightarrow & \operatorname{im}(\varphi) \\ \wr & & \wr \\ a \ker(\varphi) & \longmapsto & \varphi(a) \end{array}$$

Proposition 3.10.

1. $\ker(\varphi) < G$

2. $\operatorname{im}(\varphi) < H$

Proof.

1. The strategy is to check the three criteria for being a subgroup: $e \in \ker(\varphi)$ so $\varphi(e_G) = e_H$. And for $a, b \in \ker(\varphi)$, $\varphi(\underbrace{ab}) = \varphi(a)\varphi(b) = e_H e_H, \implies ab \in \ker(\varphi)$. And lastly, for $a \in \ker(\varphi)$ $\varphi(a) = e_H$

and $\varphi(a^{-1}) = \varphi(a)^{-1} \stackrel{\in G}{=} e_H^{-1} = e_H$. All the criterium hold, so $\ker(\varphi) < G$.

2. Exercise.

□

Proposition 3.11. Let $\varphi : G \rightarrow H$ be a homomorphism. Then:

1. φ is injective $\iff \operatorname{Ker}(\varphi) = \{e_G\}$
2. φ is surjective $\iff \operatorname{Im}(\varphi) = H$

Proof. Exercise.

□

Remark 3.12. G -abelian $\iff Z(G) = G$

3.2 Inner automorphism

Definition 3.13. G is a group, and $a \in G$ an element of G , we define φ_a as

$$\begin{aligned}\varphi_a : G &\rightarrow G \\ x &\mapsto axa^{-1}\end{aligned}$$

We claim that φ_a is an *automorphism* of G .

Proof. Recall that an automorphism is a bijective homomorphism.

- **Injectivity:** $\varphi_a(x) = \varphi_a(y) \iff axa^{-1} = aya^{-1} \implies x = y$.
- **Surjectivity:** $\forall y \exists x : \varphi_a(x) = y$ that is $axa^{-1} = y \iff x = a^{-1}ya$. Then we observe that

$$\varphi_a(x) = a(a^{-1}ya)a^{-1},$$

hence φ_a is bijective.

- **Homomorphism:**

$$\varphi_a(xy) = a(xy)a^{-1} = (axa^{-1})(aya^{-1}) = \varphi_a(x)\varphi_a(y).$$

□

Proposition 3.14. Let $H < G$, then H normal $\iff \varphi_a(H) \subset H, \forall a \in G$.

Proof.

$$H \triangleleft G \iff aH = Ha \quad \forall a \in G \iff aHa^{-1} = H \quad \forall a \in G.$$

□

Remark 3.15. $\subset$ can be swapped to $=$.

3.3 Automorphism

Definition 3.16. An *automorphism* of G is an isomorphism $\varphi : G \rightarrow G$.

Additionally, denote $\text{Aut}(G) = \{\text{all automorphisms of } G\}$. We then claim that $\text{Aut}(G) < S(G)$.

Proof. We start simply by checking the criteria for being a subgroup:

- **Identity** $\text{id}_G \in \text{Aut}(G)$. Ok!
- **Closedness:** Assume that $\varphi, \psi \in \text{Aut}(G)$, then

$$\varphi \circ \psi(xy) = \varphi(\psi(xy)) = \varphi(\psi(x)\psi(y)) = \varphi(\psi(x))\varphi(\psi(y)) = \varphi \circ \psi(x) \cdot \varphi \circ \psi(y).$$

- **Inverse:** $\varphi \in \text{Aut}(G) \implies \varphi(xy) = \varphi(x)\varphi(y)$. If $\varphi(x) = a$ and $\varphi(y) = b$ we have $x = \varphi^{-1}(a)$ and $y = \varphi^{-1}(b)$.

$$xy = \varphi^{-1} \circ \varphi(xy) = \varphi^{-1}(\underbrace{\varphi(x)}_{=a} \underbrace{\varphi(y)}_{=b})$$

which is the same as saying

$$\varphi^{-1}(a)\varphi^{-1}(b) = \varphi^{-1}(ab)$$

□

Proposition 3.17. $\{\varphi_a \mid a \in G\} =: \text{Inn}(G) \triangleleft \text{Aut}(G)$.

Proof. Proof is fairly straight forward from the definitions. It's a good idea to be familiar with this way of going about to prove.

➤ $\text{id}_G = \varphi_e$.

➤ Closedness:

$$\begin{aligned}\varphi_a \circ \varphi_b(x) &= a(bxb^{-1})a^{-1} = (ab)x(ab)^{-1} \\ &= \varphi_{ab}(x).\end{aligned}$$

➤ Normality:

$$\begin{aligned}\forall \psi \in \text{Aut}(G), \text{ we get that } \psi \circ \varphi_a \circ \psi^{-1}(x) &= \psi(a\psi^{-1}(x)a^{-1}) \\ &= \psi(a)x\psi(a^{-1}) \\ &= \varphi_{\psi(a)}(x).\end{aligned}$$

□

3.4 Quotients (factor groups)

Setup: G -group, $H \triangleleft G$, and $G/H = H \setminus G$.

Proposition 3.18. The set G/H has the natural structure of a group with respect to:

- $(aH)(bH) := (ab)H$
- $(aH)^{-1} := a^{-1}H$

and identity $eH = H$.

We state this as a proposition as we now need to do the important step of checking if these definitions are well defined.

Proof. Let $a' \in aH$ and $b' \in bH$, we want to prove $a'b' \in abH$. We note that for some $h_1, h_2 \in H$ $a' = ah_1$ and $b' = bh_2$. Now from the normality of H , there is some $\tilde{h} \in H$ such that $h_1b = b\tilde{h} \in bH$. Then

$$a'b' = ah_1bh_2 = ab \underbrace{\tilde{h}h_2}_{\in H}$$

and more precisely

$$a'b' = ah_1bh_2 = ab \underbrace{b^{-1}h_1b}_{=\varphi_{b^{-1}}(h_1) \in H} h_2,$$

from proposition (3.14).

What is left to check is that $(G/H, \cdot, eH)$ is a group.

- Identity:

$$aHeH = aeH = aH.$$

- Associativity:

$$aH(bHcH) = aH(bcH) = abcH = (ab)cH = (aH)cH = (aHbH)cH.$$

- Inverses:

$$(aH)(a^{-1}H) = aa^{-1}H = eH = H.$$

□

Terminology: G/H -quotient group.

Remark 3.19. $H \triangleleft G$, the usually so called "canonical projection"

$$\begin{array}{ccc} \pi: G & \longrightarrow & G/H \\ \wr & & \wr \\ a & \longmapsto & aH \end{array}$$

is a homomorphism with $\pi(ab) = abH = aHbH = \pi(a)\pi(b)$.

4 Noether's first isomorphism theorem

Theorem 4.1. Let $\varphi: G \rightarrow H$ be a homomorphism. Then φ induces an *isomorphism*

$$\begin{array}{ccc} \bar{\varphi}: G/\ker(\varphi) & \longrightarrow & \text{im}(\varphi) \\ \wr & & \wr \\ aH & \longmapsto & \varphi(a) \end{array}$$

Proof. To check: $\bar{\varphi}$ is a homomorphism, since we know from proposition 3.9 earlier that $\bar{\varphi}$ is bijective.

$$\bar{\varphi}(aHbH) = \bar{\varphi}(abH) = \varphi(ab) = \varphi(a)\varphi(b) = \bar{\varphi}(aH)\bar{\varphi}(bH).$$

□

Example 4.2.

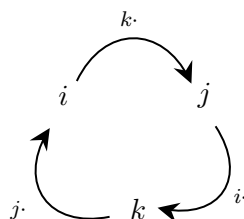
1. Consider the cyclic group on $\mathbb{Z}_4$, $G = (\mathbb{Z}_4, +, \bar{0}) = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$, then $H = \{\bar{0}, \bar{2}\}$ is a normal subgroup $H \triangleleft G$ that is commutative with respect to $+$. Note

$$G/H = \{\{\bar{0}, \bar{2}\}, \{\bar{1}, \bar{3}\}\}$$

as $\{\bar{1}, \bar{3}\} = \bar{1} + \{\bar{0}, \bar{2}\}$. We can also see how the subgroup is commutative by drawing a table

+	$\{\bar{0}, \bar{2}\}$	$\{\bar{1}, \bar{3}\}$
$\{\bar{0}, \bar{2}\}$	$\{\bar{0}, \bar{2}\}$	$\{\bar{1}, \bar{3}\}$
$\{\bar{1}, \bar{3}\}$	$\{\bar{1}, \bar{3}\}$	$\{\bar{0}, \bar{2}\}$

2. $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ the octonion group. Here $ij = k, jk = i, ki = j$



and $i^2 = j^2 = k^2 = -1$, but we see that $ij = k \implies i^2j = ik \implies -j = ik \neq ki = j$ Q_8 is NOT abelian!

Note that $Z(Q_8) = \{\pm 1\} \triangleleft Q_8$, and moreover $Q_8 / Z(Q_8) = \{\{\pm 1\}, \{\pm i\}, \{\pm j\}, \{\pm k\}\}$, which has $\frac{|Q_8|}{|Z(Q_8)|}$ elements. The table of $Q_8 / Z(Q_8)$:

$\cdot$	$\{\pm 1\}$	$\{\pm i\}$	$\{\pm j\}$	$\{\pm k\}$
$\{\pm 1\}$	$\{\pm 1\}$	$\{\pm i\}$	$\{\pm j\}$	$\{\pm k\}$
$\{\pm i\}$	$\{\pm i\}$	$\{\pm 1\}$	$\{\pm k\}$	$\{\pm j\}$
$\{\pm j\}$	$\{\pm j\}$	$\{\pm k\}$	$\{\pm 1\}$	$\{\pm i\}$
$\{\pm k\}$	$\{\pm k\}$	$\{\pm j\}$	$\{\pm i\}$	$\{\pm 1\}$

Remark 4.3. $Q_8 / Z(Q_8)$ is abelian.

A group is called *simple* if the only subgroups are the trivial ones. There are only finitely many of these simple groups and the theorem is popularly called the "enormous theorem" as the proof consists of several hundred journal articles and about 100 authors.

5 Group actions

5.1 Actions

Setup: G -group and X a non-empty set.

Definition 5.1. An *action* of G on X is a homomorphism $\varphi : G \rightarrow S(X)$.

This means that each element of G is realised/represented via a bijective transformation $\varphi(g)$ of X such that the group operation on G is realised via group operations of groups. In other words, G is abstract and we concretize it with $S(X)$ which is concrete.

We rewrite the definition a bit more concretely to

Definition 5.2 (Left group action). An *action* of G on X is given by an action map

$$\begin{array}{ccc} _ \cdot _ : G \times X & \longrightarrow & X \\ \downarrow & & \downarrow \\ (g, x) & \longmapsto & g \cdot x \end{array}$$

satisfying

1. $(hg) \cdot x = h(g \cdot x)$ (funny associativity).
2. $e \cdot x = x$ for all $x \in X$.

Remark 5.3. Right group action is defined in the exact same fashion.

Example 5.4. 1. For all $x \in X$ there exists a trivial action of G on X , $g \mapsto id_X$, that is $g \cdot x = x$.

Remark 5.5. Notation: we write $G \curvearrowright X$ (" G acts on X "), $X - G$ -act, $g \cdot x$ to describe an action.

2. Any group act on itself by left multiplication (left regular act).

$$g \cdot a := ga \text{ for } g, a \in G$$

or $g \in G$ gives $l_g \in S(G)$, $l_g(x) := gx$ for all $x \in G$. $[l_g \circ l_h](x) = g(hx) = (gh)x = l_{gh}(x)$ and $l_e(x) = ex = x$. OK!

3. Each group acts on itself by *conjugation*, that is for all $a, x \in G$ $a \cdot x := axa^{-1}$. Check if this definition is ok:

$$(ab) \cdot x = abx(ab)^{-1} = a(bxb^{-1})a^{-1} = a(b \cdot x)a^{-1} = a \cdot (b \cdot x) \quad \text{OK!}$$

$$e \cdot x = exe^{-1} = x \quad \text{OK!}$$

4. $S(x) \curvearrowright X$ by definition. For $\alpha \in S(X)$ and $x \in X$, $\alpha \cdot x := \alpha(x)$.

5. $D_{2n} \curvearrowright \mathbb{R}$ by definition. Recall how we defined the dihedral groups in example ??.

Definition 5.6. A subset $Y \subset X$ is *G-invariant* if $\forall a \in Y, \forall g \in G \quad g \cdot a \in Y$.

Definition 5.7. If $x \in X$, then the *orbit* of x is

$$\mathcal{O}_x := \{a \cdot x \mid a \in G\}.$$

Definition 5.8. If $x \in X$, the *stabiliser* of x in G is

$$\text{St}_G(x) := \{a \in G \mid a \cdot x = x\}.$$

Definition 5.9. An action of G on X is called *transitive* provided that $\forall x, y \in X \quad \exists g \in G \quad g \cdot x = y$.

Example 5.10.

1. $G \curvearrowright X$ trivially. Each action of G is the identity. The invariant subsets are then all subsets. That is, $x \in X \quad \mathcal{O}_x = \{x\}$ and $\text{St}_G(x) = G$. Also note that if $|X| > 1$ then it's not transitive.
2. $S_3 \curvearrowright \{1, 2, 3\}$. Invariant subsets of $\{1, 2, 3\}$ (the empty set is always invariant, but we can ignore it):

$$\underbrace{\{1\} \quad \{2\} \quad \{3\} \quad \{1, 2\} \quad \{1, 3\} \quad \{2, 3\}}_{2^k - 1 \quad k=3}$$

There are no invariant subsets of $\{1, 2, 3\}$ as

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} A \neq A \text{ for all } A \subsetneq \{1, 2, 3\}$$

$$\mathcal{O}_1 = \{1, 2, 3\}$$

$$\text{St}_{S_3}(\{1\}) = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \right\}$$

Transitive? Yes!

Proposition 5.11. $G \curvearrowright X, x \in X \implies \mathcal{O}_x$ is G -invariant

Proof. Recall $\mathcal{O}_x = \{b \cdot x \mid b \in G\}$. If $a \in G$ then $a \cdot (b \cdot x) = (ab) \cdot x \in \mathcal{O}_x$. $\square$

Proposition 5.12. $G \curvearrowright X, x, y \in X \implies \mathcal{O}_x = \mathcal{O}_y$ or $\mathcal{O}_x \cap \mathcal{O}_y = \emptyset$

Proof. Assume the intersection is non-empty, then $z \in \mathcal{O}_x \cap \mathcal{O}_y \implies z = ax, z = by$ for $a, b \in G \implies a \cdot x = b \cdot y \implies a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (b \cdot y) \implies x = (a^{-1}b) \cdot y \implies x \in \mathcal{O}_y \implies \mathcal{O}_x \subset \mathcal{O}_y$. By symmetry we also get $\mathcal{O}_y \subset \mathcal{O}_x$, and hence $\mathcal{O}_x = \mathcal{O}_y$. $\square$

Example 5.13. $H < G$ $H \curvearrowright G$ by left multiplication. Orbits: $x \in G \mathcal{O}_x = \{hx \mid h \in H\} = Hx \in H \backslash G$.

On the other side, what happens if $H \curvearrowright G$ by right multiplication?

$$h \cdot x := xh, \quad h \in H, x \in G$$

$$e \cdot x = xe = x$$

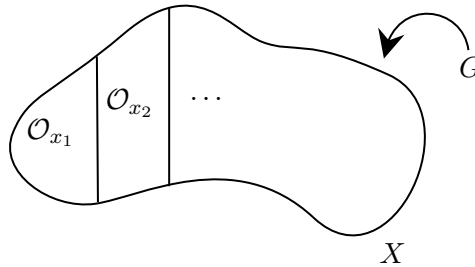
We need $(h_1 h_2) \cdot x = h_1 \cdot (h_2 \cdot x)$. But $(h_1 h_2) \cdot x = x h_1 h_2$ and $h_1 \cdot (h_2 \cdot x) = h_1 \cdot (h_2 x) = h_1 (h_2 x)$. So the equality only holds if G is commutative!

So we change the definition!

$$h \cdot x := x h^{-1}$$

Now $(h_1 h_2) \cdot x = x (h_1 h_2)^{-1} = x h_2^{-1} h_1^{-1} = h_2 \cdot (x h_1^{-1}) = h_1 \cdot (h_2 \cdot x)$.

Remark 5.14. Orbits = minimal invariant subsets. If $\emptyset \neq Y \subset \mathcal{O}$, Y is G -invariant $\implies Y = \mathcal{O}$.



Corollary 5.15. $G \curvearrowright X$ is transitive if $X = \mathcal{O}_x$ ("for some x " $\iff$ "for any x ").

Proposition 5.16. $G \curvearrowright X, x \in X$ then $\text{St}_G(x) < G$.

Proof. $e \cdot x = x \implies e \in \text{St}_G(x)$. $a, b \in \text{St}_G(x) \implies (ab) \cdot x = a \cdot (b \cdot x) = a \cdot x = x \implies ab \in \text{St}_G(x)$. $a \cdot x = x \implies x = (a^{-1}a) \cdot x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot x \implies a^{-1} \in \text{St}_G(x)$. $\square$

Theorem 5.17. Let $G \curvearrowright X, x \in X, a \in G$, then there exists a bijection

$$G / \text{St}_G(x) \longrightarrow \mathcal{O}_x$$

$$a \text{St}_G(x) \longmapsto a \cdot x.$$

Proof. Before trying to prove anything we must first check if the mapping is well defined. $h \in \text{St}_G(x)$ then $ah \mapsto (ah) \cdot x = a \cdot (h \cdot x) = a \cdot x$ since $h \in \text{St}_G(x)$. OK!

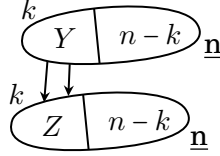
Injectivity: $a \cdot x = b \cdot x$ for some $a, b \in G$. Then applying a^{-1} on both sides $x = (a^{-1}b) \cdot x \implies (a^{-1}b) \in \text{St}_G(x)$. Then $b = a(a^{-1}b) \implies b \in a\text{St}_G(x) \implies b\text{St}_G(x) = a\text{St}_G(x)$. OK!

Surjectivity is left as an exercise. Hint: recall the definition of orbits. □

Corollary 5.18. Let $G \curvearrowright X$ with $|G|, |X| < \infty$, then $\forall x \in X \quad |\mathcal{O}_x| = \frac{|G|}{|\text{St}_G(x)|} = [G : \text{St}_G(x)]$
In particular $|\mathcal{O}_x| \mid |G|$.

Example 5.19. 1. $S_n \curvearrowright \underline{n} := \{1, 2, \dots, n\}, n \geq 1$. Orital i : $|\mathcal{O}_i| = |\underline{n}| = n \mid |S_n| = n!$.

2. Consider $2^{\underline{n}} = \mathcal{P}(\underline{n})$ with $S_n \curvearrowright 2^{\underline{n}}$, then for $\sigma \in S_n$ and $Y \in 2^{\underline{n}}$ (that is $Y \subset \underline{n}$), and the $\cdot$ operator is defined to be $\sigma \cdot Y = \{\sigma(g) \mid g \in Y\}$. What is the size of $\mathcal{O}_Y = \{Z \subset \underline{n} \mid |Y| = |Z|\}$?



Observe that the binomial coefficient $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the amount of ways one can select k distinct elements from a set of n elements. Hence an equivalent definition to the binomial coefficient is $\binom{n}{k} := |\{Y \subset \underline{n} \mid |Y| = k\}|$. It then becomes clear that $|\mathcal{O}_Y| = |\{Z \subset \underline{n} \mid |Y| = |Z|\}| = \binom{n}{|Y|}$.

Proposition 5.20. If $G \curvearrowright X$, $x \in X$, $y \in \mathcal{O}_x \implies \exists a \in G$ such that $a \cdot x = y$ ^(*), then $\text{St}_G(x) = a^{-1}\text{St}_G(y)a$.

Proof. $\supset \text{St}_G(x) \supset a^{-1}\text{St}_G(y)a$

$b \in \text{St}_G(x)$, take $a^{-1}ba = a^{-1} \cdot (b \cdot (a \cdot x)) = a^{-1} \cdot (b \cdot y) = a^{-1} \cdot y \stackrel{(*)}{=} a^{-1} \cdot (a \cdot x) = x \in \text{St}_G(x)$.

$\supset$ For $\text{St}_G(x) \subset a^{-1}\text{St}_G(y)a$, we note that we can rewrite $b \in \text{St}_G(x)$ as $a^{-1}(aba^{-1})a$ ^(**), then we only need to prove that $aba^{-1} \in \text{St}_G(y)$.

$(aba^{-1}) \cdot y = ab \cdot (a^{-1} \cdot y) \stackrel{(*)}{=} a \cdot (b \cdot x) \stackrel{(**)}{=} a \cdot x \stackrel{(*)}{=} y \in \text{St}_G(y)$.

□

Example 5.21. $G \curvearrowright G$ by left multiplication: $a \cdot x = ax$. $\mathcal{O}_x = \{a \cdot x \mid a \in G\} = G$.

For all $y \in G \exists a \in G$ such that $a \cdot x = y$ as $a = yx^{-1}$ and hence $a \cdot x = \underbrace{(yx^{-1})}_{\in G} x = y$. So this action is transitive!

$\text{St}_G(x) = \{e\}$ as $a \cdot x = x \implies a = x$.

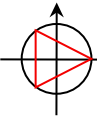
5.2 Fixed points

Definition 5.22. $x \in X$ is a *fixed point* for $g \in G$ if $g \cdot x = x$.

$$X^g := \{x \in X \mid g \cdot x = x\}$$

Example 5.23.

1. $G \curvearrowright X$ trivially, that is $g \cdot x = x \implies X^g = X$ for all $g \in G$.
2. $G \curvearrowright$ by left multiplication. For $g \in G$ what is G^g ? Solve for $gx = x$, then $g = e$, and so $G^g = G$ if $g = e$, if $g \neq e$ $G^g = \emptyset$.

3.  $D_{2,3}$ with the action $D_{2,3} \curvearrowright \mathbb{R}^2$. Then for $g \in D_{2,3}$

$$(\mathbb{R}^2)^g = \begin{cases} \mathbb{R}^2, & \text{if } g = e \\ \{0\}, & \text{if } g\text{-rotation} \\ \text{reflex line,} & \text{if } g\text{-reflection} \end{cases}.$$

Theorem 5.24 (Burnside's lemma). Let a finite group G act on a finite set X , then

$$\text{"The number of orbits"} = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

Proof. Consider $G \times X \supset \Omega := \{(g, x) \mid g \cdot x = x\}$. Then to give a picture,

$G \times X$	x_1	x_2	$\dots$	x_n
g_1	(g_1, x_1)	(g_1, x_2)	$\dots$	(g_1, x_n)
g_2	(g_2, x_1)	(g_2, x_2)	$\dots$	(g_2, x_n)
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
g_k	(g_k, x_1)	(g_k, x_2)	$\dots$	(g_k, x_n)

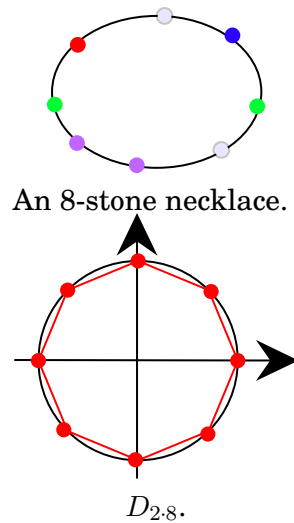
where $(,) \in \Omega$.

By looking at the rows $|\Omega| = \sum_{g \in G} |X^g|$, and by looking at the columns $|\Omega| = \sum_{g \in G} |\text{St}_G(x)|$, thus $\sum_{g \in G} |X^g| = \sum_{g \in G} |\text{St}_G(x)|$.

$$\begin{aligned} \frac{1}{|G|} \sum_{g \in G} |X^g| &= \frac{1}{|G|} \sum_{g \in G} |\text{St}_G(x)| = \\ &= \frac{1}{|G|} \sum_{\mathcal{O}\text{-orbits}} \sum_{x \in \mathcal{O}} |\text{St}_G(x)| = \\ &= \sum_{\mathcal{O}\text{-orbits}} \sum_{x \in \mathcal{O}} \frac{|\text{St}_G(x)|}{|G|} \stackrel{??}{=} \sum_{\mathcal{O}\text{-orbits}} \sum_{x \in \mathcal{O}} \frac{1}{|\mathcal{O}|} = \\ &= \sum_{\mathcal{O}\text{-orbits}} \underbrace{\frac{1}{|\mathcal{O}|} \sum_{x \in \mathcal{O}} 1}_{=1} = \sum_{\mathcal{O}\text{-orbits}} 1. \end{aligned}$$

□

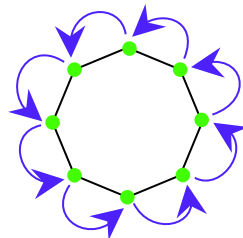
Example 5.25. How many different eight stones necklaces can one create using: diamonds, sapphires, rubies, emeralds, and amethysts?



Observe that two necklaces are equivalent $\iff$ they differ by an action of $D_{2,8}$. Let $X = \{\text{All possible colorings of the vertices of a regular 8-gon using 5 colors.}\}$. We let $D_{2,8} \curvearrowright X$. To start we note that $|X| = 5^8$ as one can choose 5 colors for each 8 vertex. An action on a necklace will not change the necklace, hence each unique necklace will be represented as an orbit. The number of different necklaces will then be the number of orbits.

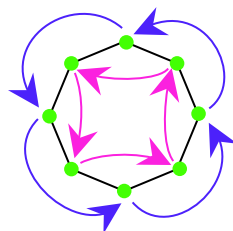
To do: calculate $|X^g| \forall g \in D_{2,8}$.

1. $g = e$, $X^e = X$ so $|X^g| = 5^8$.
2. $g = \text{rot}(\pm \frac{\pi}{4})$



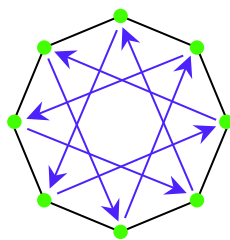
in order to be a fixed point, all vertices need to be the same color, hence $|X^g| = 5$.

3. $g = \text{rot}(\pm \frac{\pi}{2})$



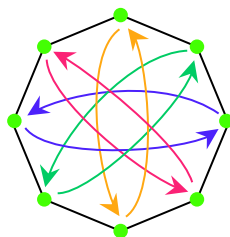
$$|X^g| = 5^2.$$

4. $g = \text{rot}(\pm \frac{3\pi}{4})$



$$|X^g| = 5.$$

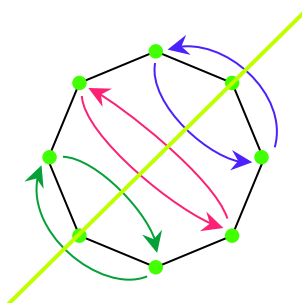
5. $g = \text{rot}(\pi)$



$$|X^g| = 5^4.$$

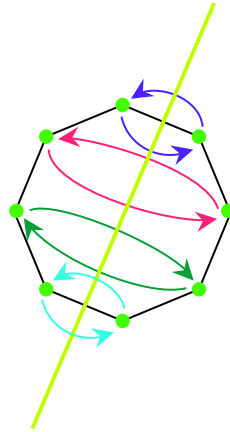
For reflection we note that there are two types. One where the reflexive line passes through two vertices, and one where it passes through two edges.

- $g = \text{refl}_{\text{vertex}}$ (there are 4 such elements)



$$|X^g| = 5^5.$$

- $g = \text{refl}_{\text{edge}}$ (there are 4 such elements)



$$|X^g| = 5^4.$$

Now lastly, using Burnside's lemma 5.24 we find that #different necklaces = $\frac{1}{|D_{2 \cdot 8}|} (5^8 + 2 \cdot 5 + 2 \cdot 5^2 + 2 \cdot 5 + 5^4 + 4 \cdot 5^5 + 4 \cdot 5^4) = \frac{406\,320}{16} = 25\,395$.

6 Generators

G -group, $X \subset G$ and let

$$\langle X \rangle := \bigcap_{X \subset H < G} H$$

Definition 6.1. X is called a *generating set* for G if $\langle X \rangle = G$.

Example 6.2.

1. $\langle G \rangle = G$
2. $\langle \{1\} \rangle = \mathbb{Z}$ because $\langle \{1\} \rangle = \{0, 1, 1+1, \dots, -1, -1+(-1), \dots\}$.
3. $\langle \{2, 3\} \rangle = \mathbb{Z}$.

Remark 6.3. 1. $\langle X \rangle = \{g_1 g_2 \dots g_k \mid g_i \in X \cup X^{-1}\}$, where $X^{-1} := \{g^{-1} \mid g \in X\}$. The claim is left as an exercise.

2. $\langle X \rangle = G \iff$ every element in G is a product of elements in X or their inverses.

Question: Given some generating set X of a group G , how does one describe G in terms of X in an "efficient" way?

Answer: We will define "relations".

7 Free groups

Let X be a set.

Definition 7.1 (The universal property of free groups). A *free group* over X is a pair $(\mathcal{F}(X), i)$ where $\mathcal{F}(X)$ is a group and $i: X \rightarrow \mathcal{F}(X)$ a map such that

$$\forall G \text{ and } \forall f: X \rightarrow G, \exists! \text{ a unique homomorphism.}$$

$\varphi : \mathcal{F}(X) \rightarrow G$ such that $\varphi \circ i = f$

that is

$$\begin{array}{ccc} X & \xrightarrow{i} & \mathcal{F}(X) \\ f \downarrow & \swarrow \exists! \varphi & \\ G & & \end{array}$$

Free groups are like generating sets consisting of all words that can be made from the elements of X , but unlike generating sets, two words are equal if equality follows from the group axioms. The construction is made later which hopefully make this more concrete.

Example 7.2.

1. $X = \{1\}$, $i : \{1\} \rightarrow \mathbb{Z}$, $\mathcal{F}(X) = \mathbb{Z}$.
2. **Claim:** $(\mathbb{Z}, i)$ is a free group over $\{1\}$.

Proof. Take any group G , $f : \{1\} \rightarrow G$, and let $f(1) = a \in G$. Define $\varphi : \mathbb{Z} \rightarrow G$ via $\varphi(n) = a^n \in G$. Then φ is a homomorphism as $\varphi(nm) = a^{nm} = a^n a^m = \varphi(n)\varphi(m)$, and unique as $\varphi(n) = \varphi(1)^n = a^n$. $\square$

Remark 7.3. $\mathbb{Z}$ is not free over $\{2, 3\}$ because $2 + 3 = 3 + 2$, if a group G is not commutative and $a, b \in G$, $ab \neq ba \implies \nexists$ a homomorphism.

7.1 Uniqueness

If $(\mathcal{F}(X), i)$ and $(\mathcal{F}'(X), j)$ are two free groups on $X \implies \exists!$ group homomorphism $\eta : \mathcal{F}(X) \rightarrow \mathcal{F}'(X)$ such that $\eta \circ i = j$.

Proof.

$$\begin{array}{ccc} X & \xrightarrow{i} & \mathcal{F}(X) \\ j \downarrow & \swarrow \exists! \varphi & \nearrow \exists! \varphi' \\ \mathcal{F}'(X) & & \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{i} & \mathcal{F}(X) \\ i \downarrow & \swarrow \exists! & \nearrow \text{id}_{\mathcal{F}(X)} \\ \mathcal{F}(X) & & \end{array}$$

so

$$\begin{array}{ccc} X & \xrightarrow{i} & \mathcal{F}(X) \\ i \downarrow & \swarrow \varphi' \circ \varphi = \text{id}_{\mathcal{F}(X)} & \\ \mathcal{F}(X) & & \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{j} & \mathcal{F}'(X) \\ j \downarrow & \swarrow \varphi \circ \varphi' = \text{id}_{\mathcal{F}'(X)} & \\ \mathcal{F}'(X) & & \end{array}$$

implying that φ is unique up to isomorphism. $\square$

7.2 Construction

X -set, let $A := X \sqcup X^{-1}$ be the alphabet.

Remark 7.4. The disjoint union in is defined as

$$\bigsqcup_{i \in I} A_i = \bigcup_{i \in I} \{(x, i) \mid x \in A_i\}$$

By convention this will be most often be treated as the normal union.

Denote $A^* = \{\text{all finite words in } A\} = \{a_1 a_2 \dots a_k \mid a_i \in A\}$ (includes $\emptyset$). We denote the words $w_1, w_2 \in A^*$ to be equivalent ' $w_1 \sim w_2$ ' if w_1 can be obtained from w_2 by a finite sequence of insertions or removals of xx^{-1} or $x^{-1}x, x \in X$.

Example 7.5. $X = \{a, b, c\}$, $A = \{a, b, c, a^{-1}, b^{-1}, c^{-1}\}$. $w_1 = ab\cancel{c}c^{-1}ab^{-1}c \sim abb^{-1}bab^{-1}a \sim abb^{-1}bab^{-1}a$.

Definition 7.6.

- Let $\mathcal{F}(X) := A^* / \sim$ and denote $\bar{f}(X) = \bar{f}(X) := \{f(x) \mid x \in X\}$.
- For $w \in A^*$ we denote $\bar{w}$ to be the equivalent class of w .
-

$$\begin{array}{ccc} i: X & \longrightarrow & \bar{f}(X) \\ \downarrow \psi & & \downarrow \psi \\ a & \longmapsto & \bar{a} \end{array}$$

Claim: $(\bar{f}(X), i)$ is a free group on X .

Proof. The only natural way of defining a binary operation $_ \cdot _$ is $\bar{w}_1 \cdot \bar{w}_2 := \overline{w_1 w_2}$. Check: $w_1 \sim w'_1$ and $w_2 \sim w'_2 \implies w_1 w_2 \sim w'_1 w'_2$ so it is well defined.

It is associative as $(\bar{w}_1 \cdot \bar{w}_2) \cdot \bar{w}_3 = \overline{w_1 w_2} \cdot \bar{w}_3 = \overline{w_1 w_2 w_3} = \bar{w}_1 \cdot \overline{w_2 w_3} = \bar{w}_1 \cdot (\bar{w}_2 \cdot \bar{w}_3)$.

Inverses exists as $(\overline{a_1 \dots a_k})^{-1} = \overline{a_k^{-1} \dots a_1^{-1}}$. Note that $\bar{\emptyset}$ is the empty word, so $\bar{x} \cdot \bar{x}^{-1} = \overline{xx^{-1}} = \bar{\emptyset}$, hence $\bar{f}(X)$ is also closed.

$\implies \bar{f}(X)$ is a group!

For the universal property:

$$\begin{array}{ccc} X & \xrightarrow{i} & \bar{f}(X) \\ f \downarrow & \swarrow \exists! \varphi & \\ G & & \end{array}$$

We seek $\varphi(\overline{a_1 a_2 \dots a_k}) = \varphi(\bar{a}_1) \varphi(\bar{a}_2) \dots \varphi(\bar{a}_k)$, where $a_i \in X \sqcup X^{-1}$ and $\varphi(\bar{a}_i) = \begin{cases} f(x), & \text{if } a_i = x \in X \\ f(x)^{-1}, & \text{if } a_i = x^{-1} \in X^{-1} \end{cases}$

$\varphi(\overline{x_1^{\pm 1} x_2^{\pm 1} \dots x_k^{\pm 1}}) = f(x_1)^{\pm 1} f(x_2)^{\pm 1} \dots f(x_k)^{\pm 1} = \varphi(\bar{a}_1) \varphi(\bar{a}_2) \dots \varphi(\bar{a}_k)$ so φ is a homomorphism. $\square$

Remark 7.7. 1. $\mathcal{F}(\emptyset) = \{e\}$, the trivial group.

2. $|\bar{f}(X)| = \infty \implies X \neq \emptyset$.

3. $\bar{f}(x)$ so not abelian if $|X| > 1$. $\leftarrow$ (Difficult to show)

4. Every group is a quotient of a free group.

Proof. Let G be a group. Let $X \subset G$ be a generating set $\langle X \rangle = G$.

$$\begin{array}{ccc} X & \xrightarrow{i} & \mathcal{F}(X) \\ f \downarrow & \swarrow \exists! \varphi & \\ G & & \end{array}$$

$\exists \varphi: \bar{f} \rightarrow G$, and $X < \text{im}(\varphi) < G$, because $\varphi \circ i = f$, and hence $\text{im} = G$. Then by the first isomorphism theorem 4.1 $G = \text{im}(\varphi) \simeq \bar{f}(X) / \ker(\varphi)$. $\square$

7.3 Presentation

Presentation means defining groups by generators and relations. Let G be a group, $X \subset G$ a generating set $\langle X \rangle = G$. Then $G \simeq \bar{f}(X) / H$ where $H = \ker(\varphi)$ with $\varphi : \bar{f}(X) \twoheadrightarrow G$ (surjective). $H \triangleleft \bar{f}(X)$: Let $Y \subset H$ be any generating set of H as a normal subgroup of $\bar{f}(X)$. That is

$$H = \bigcap_{Y \subset K \triangleleft \bar{f}(X)} K.$$

In this situation, once says that G is given by generators X and relations Y . This is denoted as

$$G = \langle X \mid Y \rangle.$$

This is the presentation of G .

Example 7.8.

1. $\bar{f}(X) = \langle X \mid \emptyset \rangle$.
2. $\mathbb{Z} \simeq \langle \{1\} \mid \emptyset \rangle$.
3. $\mathbb{Z} \simeq \langle \{a, b\} \mid \{1\} \rangle$ where

$$\begin{aligned} \varphi : \bar{f}(\{a, b\}) &\longrightarrow \mathbb{Z} \\ a &\longmapsto 2 \\ b &\longmapsto 3 \end{aligned}$$

then $\mathbb{Z} \simeq \bar{f}(\{a, b\}) / \ker(\varphi)$. We note that $aba^{-1}b^{-1} \in \ker \varphi$ and $aaab^{-1}b^{-1} \in \ker \varphi$ since $2+3-2-3=0$ and $2+2+2-3-3=0$, that is, we expect $ab=ba$ as $2+3=3+2$ and $aaa=bb$ as $2+2+2=3+3$. So the presentation for $\mathbb{Z}$ becomes

$$\mathbb{Z} \simeq \langle \{a, b\} \mid \{ab=ba, aaa=bb\} \rangle.$$

More rigorously, given any word $w \in A^* = \{a, b, a^{-1}, b^{-1}\}^*$, w can be written as $a^i a^{-j} b^s b^{-t}$, $i, j, s, t \geq 0$. If $i, j \neq 0$ then w can be written as $a^{i-j} b^{s-t}$, now use the fact that $a^3 = b^2$ to get that any element $w \in \bar{f}(\{a, b\}) / \underbrace{\langle \{aba^{-1}b^{-1}, aaab^{-1}b^{-1}\} \rangle}_{=H \triangleleft \mathbb{Z}}$ can be written as either a^i or $a^i b$. $a^i \rightarrow$

even integers, and $a^i b \rightarrow$ odd integers.

$$\implies \bar{f}(\{a, b\}) / H \xrightarrow{\simeq} \mathbb{Z}.$$

7.4 How do we use presentations?

Given two groups G and H . How does one construct homomorphisms $G \rightarrow H$?

recipe: Write $G = \langle X \mid Y \rangle$, some presentation. Take any map $f : X \rightarrow H$. By the universal property, this extends to a group homomorphism $\bar{f}(X)$ to H :

$$G \simeq \bar{f}(X) / \text{normal subgroup } \langle Y \rangle.$$

When does φ define a map $\bar{\varphi} : G \rightarrow H$?

Answer: If and only if the image $f(x)$, $x \in X$ satisfy the relations from Y in H . Any relation must be preserved!

$$G \longrightarrow H$$

$$a, b \longmapsto f(a)f(b)$$

some relation $\mapsto$ satisfies relations

Example 7.9.

$$\mathbb{Z} \simeq \langle \{a, b\} \mid \{ab = ba, aaaa = bb\} \rangle$$

implies that for any H and any $f(a), f(b) \in H$, f extends to $\mathbb{Z} \rightarrow H$, that is

$$f(a)f(b) = f(b)f(a) \quad \text{and} \quad f(a)f(a)f(a) = f(b)f(b).$$

8 Classification of finite abelian groups

Building blocks: Finite cyclic groups

$$\mathbb{Z}_n = \mathbb{Z} / n\mathbb{Z} \quad |\mathbb{Z}| = n \quad n > 1.$$

$$\mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$$

Remark 8.1. Cyclic groups are often denoted as C_n where $n > 1$ is length of the cycle. A cyclic groups can be generated by one element $C_i = \langle a \mid a^n = a^0 \rangle$.

We build abelian groups using direct products of groups. If G and H are groups, then

$$G \times H = \{(a, b) \mid a \in G, b \in H\}$$

$$(a, b) \cdot (a', b') := (aa', bb')$$

$$(e, e) \in G \times H$$

making $G \times H$ a group, and if both G and H are abelian, then so is $G \times H$.

Theorem 8.2. Let G be a finite abelian group

- (a) There exists primes $p_1, p_2, \dots, p_k > 0$ and positive integers $m_1, m_2, \dots, m_k$ such that $G \simeq \mathbb{Z}_{p_1}^{m_1} \times \mathbb{Z}_{p_2}^{m_2} \times \dots \times \mathbb{Z}_{p_k}^{m_k}$.
- (b) This *decomposition* is unique up to permutation of factors, i.e. $\mathbb{Z}_{p_1}^{m_1} \times \mathbb{Z}_{p_2}^{m_2} \times \dots \times \mathbb{Z}_{p_k}^{m_k} \simeq \mathbb{Z}_{q_1}^{r_1} \times \mathbb{Z}_{q_2}^{r_2} \times \dots \times \mathbb{Z}_{q_l}^{r_l} \implies k = l$ and that there exists some $\sigma \in S_k$ such that $p_i = q_{\sigma(i)}$ and $m_i = r_{\sigma(i)} \forall i = 1, 2, \dots, k$.

Before a proof is given, we some results.

- Example 8.3.**
1. $\mathbb{Z}_6 \simeq \mathbb{Z}_2 \times \mathbb{Z}_3$: $1 \cdot (\bar{1}, \bar{1}) = (\bar{1}, \bar{1}), 2 \cdot (\bar{1}, \bar{1}) = (\bar{0}, \bar{2}), 3 \cdot (\bar{1}, \bar{1}) = (\bar{1}, \bar{0}), 4 \cdot (\bar{1}, \bar{1}) = (\bar{0}, \bar{1}), 5 \cdot (\bar{1}, \bar{1}) = (\bar{1}, \bar{2}), 6 \cdot (\bar{1}, \bar{1}) = (\bar{0}, \bar{0}) \implies \text{order}(\bar{1}, \bar{1}) = 6 \implies \mathbb{Z}_2 \times \mathbb{Z}_3 \text{ is cyclic!}$
 2. $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_4 = \mathbb{Z}_{2^2}$ are NOT isomorphic by b) of 8.2! One can check that $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic since no element generates the whole group.

8.1 Decomposition

Let G be an abelian group with $|G| = m \cdot n$ and the greatest common denominator $(m, n) = 1$.

Definition 8.4.

- $G(n) := \{x \in G \mid nx = 0\}$
- $G(m) := \{x \in G \mid mx = 0\}$

Proposition 8.5. $G \simeq G(m) \times G(n)$

Proof. Consider

$$\begin{array}{ccc} \varphi: G(m) \times G(n) & \longrightarrow & G \\ \downarrow & & \downarrow \\ (a, b) & \longmapsto & a + b \end{array}$$

then $\varphi((a, b) + (a', b')) = \varphi((a + a', b + b')) = a + a' + b + b' = a + b + a' + b' = \varphi((a, b)) + \varphi((a', b')) \implies \varphi$ is a homomorphism.

Is φ injective & surjective?

φ is injective $\iff \ker(\varphi) = \{(\bar{0}, \bar{0})\}$. $\varphi((a, b)) = 0 \iff a + b = 0 \iff \underbrace{a}_{\in G(m)} = - \underbrace{b}_{G(n)}$. That is φ is

injective $\iff G(m) \cap G(n) = \{0\}$, i.e.

$$\begin{cases} ma = 0 \\ na = 0 \end{cases}$$

with $(m, n) = 1$ implying (by Euclides algorithm) that $\exists \alpha, \beta \in \mathbb{Z}$ such that $\alpha m + \beta n = 1$. That is

$$a = 1 \cdot a = (\alpha m + \beta n)a = \alpha(ma) + \beta(na) = \alpha \cdot 0 + \beta \cdot 0 = 0$$

$\implies \varphi$ is injective.

Surjectivity: $\forall a \in G \ a = 1 \cdot a = (\alpha m + \beta n)a = \alpha(ma) + \beta(na)$. $ma \in G$, $n(ma) = 0$ by Lagrange's theorem 2.32 $\implies ma \in G(n) \implies \alpha(ma) \in G(n)$. Similarly $na \in G(m) \implies \beta(na) \in G(m) \implies a \in \text{im}(\varphi) \implies \varphi$ is surjective.

$\implies \varphi$ is bijective. □

Corollary 8.6 (Chinese remainder theorem). $n > 1$, $n = p_1^{k_1} p_2^{k_2} \dots p_m^{k_m}$, $p_i \neq p_j$ if $i \neq j$, then

$$\mathbb{Z}_n \simeq \mathbb{Z}_{p_1^{k_1}} \times \mathbb{Z}_{p_2^{k_2}} \times \dots \times \mathbb{Z}_{p_m^{k_m}}$$

Proof. G is an abelian group with $|G| = n = p_1^{k_1} p_2^{k_2} \dots p_m^{k_m}$, $p_i \neq p_j$ if $i \neq j$. By previous proposition 8.5 $G \simeq G_1 \times G_2 \times \dots \times G_m$, where $|G_i| = p_i^{k_i}$. □

Assume $|G| = p^k$, we then say that G is a p -group.

Proposition 8.7. Let m be a maximum integer such that G contains an element of order p^m , then $G \simeq \mathbb{Z}_{p^m} \times H$ for some H .

Remark 8.8. This proposition 8.7 implies part a) of theorem 8.2.

Proof. $K = \langle a \rangle < G$, $|K| = p^m$. We are going to construct a map $\varphi: G \rightarrow \mathbb{Q}/\mathbb{Z} \simeq \Psi = \{z \in \mathbb{C} \mid \exists a \in \mathbb{Z}^+, z^a = 1\}$, such that $\text{im}(\varphi) = \{0, \frac{1}{p^m}, \frac{2}{p^m}, \dots\}$ and $\varphi|_K: K \rightarrow \{0, \frac{1}{p^m}, \frac{2}{p^m}, \dots\}$ is an isomorphism. The first isomorphism theorem 3.9 tells us that $G / \ker \varphi \simeq \text{im}(\varphi) \simeq K \simeq \mathbb{Z}_{p^m}$. $\ker(\varphi) \cap K = \{0\} \implies G \simeq \ker \varphi \times K$.

Construction of φ . Step 1: Define $\varphi(a) = \frac{1}{p^m} \implies \varphi(ia) = \frac{i}{p^m} \ \forall i = 1, 2, \dots, p^m$. Step 2: If $K = G$ we are done! If not, there exists $b \in G \setminus K$, $K_2 := \langle a, b \rangle < G$. We want $\varphi(b)$.

Case 1: $\langle b \rangle \cap \langle a \rangle = \{0\}$, set $\varphi(b) = 0$.

Case 2: Assume there exists a minimal i such that $ib \in \langle a \rangle$, $ib = ra$. Then $\varphi(b) = \frac{1}{i} \cdot \underbrace{\frac{r}{p^m}}_{\varphi(ra)}$, where (not sure

what my notes said): $\text{order}(b) \leq \text{order}(a) = p^m$. This implies that $\frac{1}{i} \cdot \frac{r}{p^m} = \frac{\frac{r}{i}}{p^m} \in \{0, \frac{1}{p^m}, \frac{2}{p^m}, \dots\}$ as $\frac{r}{i} \geq 1$. This defines φ on K_2 .

Step 3: If $K_2 = G$ then we are done. Otherwise $\exists c \in G \setminus K_2$, $K_3 := \dots$ repeating step 2. As G is finite, this process will terminate in finitely many steps. $\square$

8.2 Uniqueness

Proposition 8.9. Let p be a prime, and $k_1, k_2, \dots, k_m \in \mathbb{Z}^+$, with $k_1 \leq k_2 \leq \dots \leq k_m$. Then the group $\mathbb{Z}_{p^{k_1}} \times \mathbb{Z}_{p^{k_2}} \times \dots \times \mathbb{Z}_{p^{k_m}} = G$ has $p^{k_1+k_2+\dots+k_m} - \frac{p^{k_1+k_2+\dots+k_m}}{p^{|\{i|k_i=k_m\}|}}$ elements of order p^{k_m} .

Proof. If $(a_1, a_2, \dots, a_m) \in G$ has order p^{k_m} , then

1. $p^{k_m}(a_1, a_2, \dots, a_m) = 0 = (p^{k_m}a_1, p^{k_m}a_2, \dots, p^{k_m}a_m) = 0$. Always ok!
2. $0 \neq p^{k_m-1}(a_1, a_2, \dots, a_m) = (0, \dots, 0, p^{k_m-1}a_i, \dots, p^{k_m-1}a_m)$ [looks like "med:"] a_i has order p^{k_m} in $\mathbb{Z}_{p^{k_i}}$.

How many elements in $\mathbb{Z}_{p^{k_m}}$ have order p^{k_m} $\square$

Example 8.10. Is $\mathbb{Z}_6 \times \mathbb{Z}_{35} \simeq \mathbb{Z}_{10} \times \mathbb{Z}_{11}$?

8.3 Free abelian groups

Definition 8.11. A free abelian group G is an abelian group that has a basis. A basis is defined as $\underline{e} = (e_1, e_2, \dots, e_n)$ where $e_i \in G$ and $e_i \neq e_j$ for $i \neq j \forall i = 1, \dots, n$ and such that $\forall a \in G$, $a = \sum_{i=1}^n a_i e_i$, $a_i \in \mathbb{Z}$ is unique.

The free abelian groups we are going to study will be of the form

$$\mathbb{Z}^n = \underbrace{\mathbb{Z} \times \mathbb{Z} \times \dots \times \mathbb{Z}}_{n\text{-times}} = \left\{ \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \mid a_i \in \mathbb{Z} \right\}$$

with the so called standard basis

$$\underline{e} = (e_1, e_2, \dots, e_n) \quad e_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow i\text{:th position.}$$

Theorem 8.12 (Free property of $\mathbb{Z}^n$). For all abelian groups G and all $g_1, g_2, \dots, g_n \in G$, there exists a unique group homomorphism $\varphi : \mathbb{Z}^n \rightarrow G$ such that $\varphi(e_i) = g_i$, $i = 1, \dots, n$.

Proof. For $a \in \mathbb{Z}^n$, $a = \sum_{i=1}^n a_i e_i$. Then $\varphi(a) := \sum_{i=1}^n a_i g_i$ is unique and defines a homomorphism. The latter is left as an exercise. $\square$

Corollary 8.13. Every finitely generated abelian group is a quotient of $\mathbb{Z}^n$.

Proof. Assume $G = \langle x \rangle$, $|x| = n < \infty$ and G is abelian. Then $x = \{g_1, g_2, \dots, g_n\}$ and from last theorem ?? there exists $\varphi : \mathbb{Z}^n \rightarrow G$ with $e_i \mapsto g_i$ and so $\text{im}(\varphi) = G$. Then by the first isomorphism theorem ?? $G = \text{im}\varphi \simeq \mathbb{Z}^n / \ker \varphi$. $\square$

8.4 Subgroups of $\mathbb{Z}^n$

Proposition 8.14. Every subgroup of $\mathbb{Z}^n$ is isomorphic to $\mathbb{Z}^m$ for some $m \leq n$.

Proof. We do induction on n .

Base case: $n = 1$, we have $H < \mathbb{Z}$, $a, b \in H \implies (a, b) = \alpha a + \beta b$ for some $\alpha, \beta \in \mathbb{Z} \implies \langle a, b \rangle = \langle (a, b) \rangle \implies H = \langle q \rangle$ where $q = \gcd(\langle a \mid a \in H \rangle)$.

Induction step: $n - 1 \implies n$. $H < \mathbb{Z}^n$. Consider $\pi : \mathbb{Z}^n \rightarrow \mathbb{Z}$, $\pi \left(\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \right) := a_n$, this is a homomorphism

(exercise). $\text{im}(\pi) < \mathbb{Z}$, $\exists a \in \mathbb{Z}$ such that $\text{im}(\pi) = \langle a \rangle$, $\ker(\pi) = \left\{ \begin{pmatrix} a_1 \\ \vdots \\ a_{n-1} \\ 0 \end{pmatrix} \in \mathbb{Z}^n \right\} < \mathbb{Z}^n \simeq \mathbb{Z}^{n-1}$, $\text{im}(\pi) = \mathbb{Z}$

so for $\pi|_H : H \rightarrow \mathbb{Z}$, $\ker(\pi|_H) < \mathbb{Z}^{n-1} \implies \ker(\pi|_H) \simeq \mathbb{Z}^r$ for some $r \leq n - 1$. $\exists k \in H$ such that

$\pi(k) = a$, $K := \langle a \rangle \subset H$. If $i \in \mathbb{Z}$, $\pi(ik) = i\pi(k) = ia$, $i \neq j \implies ia \neq ja$ ($a \neq 0$) $\implies \pi|_K : \mathbb{Z} \xrightarrow{\sim} \langle a \rangle$, $K \simeq \mathbb{Z}$, for $a \neq 0$ and $K \simeq e$ if $a = 0$. Let $h_1, h_2 \in H$ be such that $\pi(h_1) = \pi(h_2)$, then $(h_1 - h_2) \in$

$\ker(\pi) \implies \pi(h_1) = \pi(ik) \implies h_1 - ik \in \ker(\pi) \implies \text{every } h \in H \text{ can be written as } h = h - ik + ik$.
 $K \cap \ker(\pi) = 0 \implies H \simeq \underbrace{\ker(\pi|_H)}_{\simeq \mathbb{Z}^r, r \leq n-1} \times \underbrace{K}_{\simeq \text{Zor}\{e\}} \simeq \mathbb{Z}^m$ for some $m \leq n$. $\square$

Proposition 8.15. $\mathbb{Z}^n \simeq \mathbb{Z}^m \iff n = m$.

Proof. " $\Leftarrow$ " is clear.

" $\implies$ ": For a contradiction let $n > m$ and assume $\varphi : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$ is an isomorphism, that is

$$\begin{array}{ccc} \varphi: & \mathbb{Z}^n & \longrightarrow \mathbb{Z}^m \\ & \cup & \cup \\ & e_1, \dots, e_n & \longmapsto \varphi(e_1), \dots, \varphi(e_n) \in \mathbb{Q}^m \end{array}$$

where $\varphi(e_1), \dots, \varphi(e_n) \in \mathbb{Q}^m$ ($\mathbb{Q}^m$ is a vectorfield) are linear independent as $n > m$ that is; there exists $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{Q}$ not all zero, such that $\sum_{i=1}^n \alpha_i \varphi(e_i) = 0$, where α_i is a common denominator (not zero) of $\varphi(e_i)$ (why?) for $i = 1, \dots, n$, so actually there exists $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{Z}$ implying as φ is an isomorphism

that $\varphi \left(\underbrace{\sum_{i=1}^n \alpha_i e_i}_{\neq 0} \right) = 0 \implies \ker(\varphi) \neq 0$ which is a contradiction. $\square$

8.5 Finitely generated abelian groups

Definition 8.16.

- Let G be a finitely generated abelian group. If there exists a homomorphism $\varphi : \mathbb{Z}^n \rightarrow G$,

then $G \simeq \mathbb{Z}^n / \ker(\varphi)$. If $\ker(\varphi) \simeq \mathbb{Z}^m$, $m \leq n$, then there exists $\psi : \mathbb{Z}^m \rightarrow \mathbb{Z}^n \xrightarrow{\varphi} G$, then the *cokernel* of ψ is defined as

$$\text{coker}(\psi) := \mathbb{Z}^n / \text{im}(\psi).$$

with $\text{coker}(\psi) \simeq G$.

- If $e_1^{(m)}, e_2^{(m)}, \dots, e_m^{(m)}$ is the standard basis of $\mathbb{Z}^m$, then ψ is determined by $\psi(e_1^{(m)}), \psi(e_2^{(m)}), \dots, \psi(e_m^{(m)})$, where $\psi(e_i^{(m)}) = \sum_{j=1}^n a_{ij} e_j^{(n)}$, $a_{ij} \in \mathbb{Z}$ for $i = 1, \dots, m$. ψ can then be written as:

$$\psi \rightsquigarrow \begin{pmatrix} a_{11}^1 & a_{12}^1 & \dots & a_{1m}^1 \\ a_{11}^2 & a_{12}^2 & \dots & a_{1m}^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{11}^n & a_{12}^n & \dots & a_{1m}^n \end{pmatrix} =: [\psi].$$

- A basis in $\mathbb{Z}^n$ is a collection of elements $a_1, \dots, a_k$ such that every $b \in \mathbb{Z}^n$ can be *uniquely* written as $b = \sum_{i=1}^k \alpha_i a_i$, $\alpha_i \in \mathbb{Z}$. Remark that $k = n$.
- A change of basis over b_1, b_2 , and b_3 is given by a *transform* matrix $Q, T \in \text{Mat}_{n \times n}(\mathbb{Z})$ with $\det(Q) = \pm 1$ and $\det(T) = \pm 1$:

$$b_1 \xrightarrow{T} b_2 \xrightarrow{Q} b_3$$

$$\xleftarrow{T^{-1}} \quad \xleftarrow{Q^T}$$

- $[\psi]$ can be rewritten as $T[\psi]Q$.

Theorem 8.17 (Smith normal form). For all $M \in \text{Mat}_{n \times m}(\mathbb{Z})$, there exists $A \in \text{Mat}_{n \times n}(\mathbb{Z})$ and $B \in \text{Mat}_{m \times m}(\mathbb{Z})$ with $\det A, \det B = \pm 1$ such that

$$AMB = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & & 0 \\ \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \dots & d_m \end{pmatrix} \quad d_i \mid d_{i+1} \forall i$$

not necessarily a square matrix. The matrix AMB is on the Smith normal form (SNF).

Corollary 8.18. If $\mathbb{Z}^m \xrightarrow{\psi} \mathbb{Z}^n \rightarrow G$ and $[\psi]$ is of SNF, then $G(=?) \simeq \mathbb{Z} / d_1 \mathbb{Z} \times \mathbb{Z} / d_2 \mathbb{Z} \times \dots \times \mathbb{Z} / d_m \mathbb{Z}$.

Remark 8.19. $d_i \neq 0 \implies \mathbb{Z} / d_i \mathbb{Z} \simeq \mathbb{Z}_{d_i}$, and $d_i = 0 \implies \mathbb{Z} / d_i \mathbb{Z} \simeq \mathbb{Z}$.

Example 8.20.

$$\begin{pmatrix} 6 & 8 \\ 0 & 8 \\ 12 & -24 \end{pmatrix} \xrightarrow{-2R_2} \sim \begin{pmatrix} 6 & 8 \\ 0 & 8 \\ 0 & -40 \end{pmatrix} \xrightarrow{+5R_2} \sim \begin{pmatrix} 6 & 0 \\ 0 & 8 \\ 0 & 0 \end{pmatrix} \sim \dots \sim \begin{pmatrix} 2 & 0 \\ 0 & 24 \\ 0 & 0 \end{pmatrix}$$

thus $\text{coker}(\psi) \simeq G \simeq \mathbb{Z} / 2\mathbb{Z} \times \mathbb{Z} / 24\mathbb{Z} \times \mathbb{Z}$.

Theorem 8.21 (Main theorem (change name of these perhaps?)). Let G be a finitely generated abelian group, $p_i > 0$ prime numbers, and $m_i \in \mathbb{Z}^+$. Then

1. $G \simeq \mathbb{Z}^n \times \mathbb{Z}_{p_1^{m_1}} \times \cdots \times \mathbb{Z}_{p_r^{m_r}}$, for some $n, r > 0$.
2. This is unique up to permutations of finite factors.

Proof. **8.5.1 Torsion**

Definition 8.22. $T(G) := \{a \in G \mid \text{order}(a) < \infty\}$

Lemma 8.23. $T(G) < G$

Proof. $e \in T(G)$. If $ka = 0$, and $mb = 0 \implies km(a+b) = 0$ then $kk = 0$ and hence $k(-k) = 0$. ?? □

Lemma 8.24. If G is a finitely generated abelian group, then $T(G)$ is finitely generated.

Proof. There exists a homomorphism $H < \mathbb{Z}^n \xrightarrow{\pi} G > T(G)$, then from 8.21 $\mathbb{Z}^m \simeq H$ for some $m \leq n$, and so $H \xrightarrow{\tau} T(G)$ with $H = \pi^{-1}(T(G))$. Thus H is finitely generated implying $T(G)$ is finitely generated. □

Corollary 8.25. Let G be a finitely generated abelian groups, then $|T(G)| < \infty$.

Proof. If $a_1, a_2, \dots, a_n$ generate $T(G)$ with order $r_1, r_2, \dots, r_n$ respectively, then any $b \in T(G)$ can be written as $b = a_1^{i_1} a_2^{i_2} \dots a_n^{i_n}$, $0 \leq i_j < r_j$. □

We know that finitely generated abelian group G implies $G \simeq \underbrace{\mathbb{Z}^k}_{G/T(G)} \times \underbrace{\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \cdots \times \mathbb{Z}_{d_m}}_{T(G)}$, making k uniquely determined. This proves our main theorem. □

8.6 Sylow theorems

Theorem 8.26. Let G be a finite group and p be a prime number with $p \mid |G|$, and $k > 0 : p^k \mid |G|$, but $p^{k+1} \nmid |G|$. Then

1. $\forall 0 \leq b \leq k$ there exists a subgroup of G of cardinality p^b .
2. Any two subgroups of G of cardinality p^k are conjugated.
3. The number of subgroups of cardinality p^k is congruent to 1 (mod p) and also divides $\frac{|G|}{p^k}$.

Terminology: A subgroup of G of cardinality p^k is called a "Sylow p -subgroup".

Remark 8.27. $H_1, H_2 < G$ are conjugated if $|H_1| = |H_2| = p^k$ and $\exists a \in G$ such that $H_1 = aH_2a^{-1}$.

Example 8.28.

$$S_3 = \left\{ \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}}_{=e}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}}_{=s}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}}_{=t}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}}_{=st}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}}_{=ts}, \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}}_{=\omega_0=sts= tst} \right\}$$

Note that $|S_3| = 6 = 3 \cdot 2$, so $p = 2$ gives the 2-subgroups: $\{e, s\}, \{e, t\}, \{e, \omega_0\}$. So we have three subgroups, and $3 \equiv 1 \pmod{2}$, and $3 \mid \frac{6}{2}$. Moreover $\omega_0 = tst \implies \{e, \omega_0\} = t\{e, s\}t$, and $\omega_0 = sts \implies \{e, \omega_0\} = s\{e, t\}s$.

For $p = 3$, the Sylow 3-group is $\{e, st, ts\}$, and $1 \equiv 1 \pmod{3}$, and $1 \mid \frac{6}{3}$.

proof of 1. The proof is done by induction on $|G|$.

Base case: $|G| = 1$, nothing to prove.

Induction step: Consider the center of G ; $Z(G)$, then

case 1: $p \mid |Z(G)|$; $p \mid |Z(G)| \implies |Z(G)| > 1$. $Z(G)$ is abelian, so there exists $H < Z(G)$, $|H| = p$ ($H < G$).

We know from assumption that $p^{k-1} \mid |G/H|$, and $p^k \nmid |G/H|$, so

$$|G/H| = \frac{|G|}{p} < |G|.$$

By induction G has a subgroups of cardinality p^b for all $0 \leq b \leq k-1$. Let $K < G/H$, $|K| = p^b$, and let $G \xrightarrow{\pi} G/H$ be the canonical projection. Define $\tilde{K} = \{g \in G \mid \pi(g) \in K\}$, then $|\tilde{K}| = p \cdot |K| = p^{b+1}$. Checking that $\tilde{K} < G$ is left as an exercise.

Case 2: $p \nmid |Z(G)|$: Let $G \curvearrowright G$ by conjugation. Orbits are then conjugation classes, where singleton orbits represent $Z(G)$. $p \nmid |Z(G)|$ & $p \mid |G|$ implies that there exists a conjugation orbit $\mathcal{O}$ such that $p \nmid |\mathcal{O}|$, $|\mathcal{O}| > 1$. Now let $G \curvearrowright \mathcal{O}$ by conjugation. For $m \in G \curvearrowright \mathcal{O}$, $\text{St}_G(m) = \{g \in G \mid g^{-1}mg = m\} < G$. This implies that $|\text{St}_G(m)| < |G|$. As $p \nmid |\mathcal{O}| = \frac{|G|}{|\text{St}_G(m)|}$ so $p^k \mid |\text{St}_G(m)|$. Now by induction we are done. $\square$

Example 8.29. 1. Sylow 2-subgroups of S_4 . $|S_4| = 24 = 4!$, so Sylow 2-subgroups have from the first Sylow theorem 8.26 $8 = 2^3$ elements. Let a, b be the order of two elements in S_4 , and recall the least common multiple

$$\text{l.c.m.} := \frac{ab}{(a, b)}.$$

$\text{l.c.m.}(a, b) = 8 \implies a = 8$ or $b = 8$. This implies that a Sylow 2-subgroup of S_4 does not have elements of order 8.

2.

$$\text{Aut} \left(\begin{array}{c} \diagup \quad \diagdown \\ 1 \quad 2 \quad 3 \quad 4 \end{array} \right)$$

is a way to realise all Sylow 2-subgroups of S_4 . All Sylow p -subgroups of S_n can be realised in this way.

9 Solvable groups

9.1 Simple groups

Recall the definition of a simple group

Definition 9.1. A groups G is called *simple* if:

1. $G \neq \{e\}$
2. If $N \triangleleft G \implies N = \{e\}$ or $N = G$.

- Example 9.2.** 1. If p is a prime number, then $\mathbb{Z}_p$ is simple.
 2. Any other abelian groups is *not* simple!
 3. The smallest non-abelian simple group is $A_5 < S_5$, that is the even permutations in S_5 .
 $|A_5| = 60$.

9.2 Subnormal series

Definition 9.3. A *subnormal series* of a group G is a series

$$\{e\} = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \cdots \triangleleft G_k = G$$

where $G_i \triangleleft G_{i+1}$, for all $i = 1, \dots, k$.

- Example 9.4.** 1. $\{e\} \triangleleft G$
 2. $\{0\} \triangleleft 2\mathbb{Z}_6 \triangleleft \mathbb{Z}_6$
 3. $\{0\} \triangleleft 3\mathbb{Z}_6 \triangleleft \mathbb{Z}_6$
 4. $\{e\} \triangleleft A_3 = \{e, st, ts\} \triangleleft S_3 = \{e, s, t, st, ts, \omega_0 = sts, tst\}$

9.3 Refinements

Let G be group with two subnormal series

$$\begin{cases} \{e\} = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \cdots \triangleleft G_k = G & (a) \\ \{e\} = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \cdots \triangleleft G_m = G & (b), \end{cases}$$

then we say that

Definition 9.5. (b) is *refinement* of (a) if $m \geq k$ and there exists $0 \leq i_1 < i_2 < \cdots < i_k \leq m$ such that $\tilde{G}_{i_s} = G_s$.

Example 9.6. $N \triangleleft G \implies \{e\} \triangleleft N \triangleleft G$ which is a refinement of $\{e\} \triangleleft G$.

Definition 9.7. A subnormal series is called a *Jordan-Hölder series* (or *composition series*) if it does not have any non-trivial refinements.

The trivial refinements of $G_i \triangleleft G_{i+1}$ are $G_i \triangleleft G_i \triangleleft G_{i+1}$ and $G_i \triangleleft G_{i+1} \triangleleft G_{i+1}$.

Remark 9.8. A subnormal series is a Jordan-Hölder-series if and only if all subquotients G_i / G_{i-1} are simple.

Theorem 9.9 (Jordan-Hölder theorem). Let G be a finite group and

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_k = G$$

$$\{e\} = \tilde{G}_0 \triangleleft \tilde{G}_1 \triangleleft \cdots \triangleleft \tilde{G}_m = G$$

are two Jordan-Hölder-series for G . Then

1. $k = m$

2. there exists $\sigma \in S_k$ such that for all $i = 1, \dots, k$, $G_i / G_{i-1} \simeq \tilde{G}_{\sigma(i)} / \tilde{G}_{\sigma(i)-1}$.

Proof. In my notes there is a citation about the proof of the theorem: "will take non-trivial amount of time". $\square$

9.4 Solvable groups

Definition 9.10. A group G is *solvable* if it has a subnormal series with abelian subquotients

- Example 9.11.**
1. Any abelian group is solvable.
 2. S_3 is solvable.
 3. A_5 is *not* solvable!
 4. Any finite non-abelian simple group is *not* solvable.

- Remark 9.12.**
- G is solvable $\iff$ it has a subnormal series with abelian subquotients.
 - G is solvable $\iff$ its Jordan-Hölder-series has only $\mathbb{Z}_p$'s as subquotients.

9.5 Commutator subgroups/derived subgroup

Definition 9.13. Let G be a group with $a, b \in G$, then

$$[a, b] := aba^{-1}b^{-1}$$

is the *commutator* of a and b .

If $ab = ba$ then $ba^{-1} = a^{-1}b \implies [a, b] = aba^{-1}b^{-1} = e$.

The commutator can be thought of as some measure of how non-commutative a and b are.

Definition 9.14. The *commutator subgroup* / *derived subgroup* of a group G is

$$[G, G] := \text{the subgroup of } G \text{ generated by all } [a, b] \text{ for } \forall a, b \in G.$$

Remark 9.15. $[G, G] < G$ by definition

Example 9.16. G is abelian $\iff [G, G] = \{e\}$

Lemma 9.17.

- (a) $[G, G] \triangleleft G$.
- (b) $G / [G, G]$ is abelian.

Proof. (a) Let $\varphi : G \rightarrow G$ be any automorphism. Then $[\varphi(a), \varphi(b)] = \varphi(a)\varphi(b)\varphi(a)^{-1}\varphi(b)^{-1} = \varphi(aba^{-1}b^{-1}) = \varphi([a, b]) \implies \varphi([G, G]) \subset [G, G] \implies [G, G] \triangleleft G$.

(b) $[a[G, G], b[G, G]] = a[G, G]b[G, G]a[G, G]^{-1}[G, G]b^{-1}[G, G] = aba^{-1}b^{-1}[G, G] = [a, b][G, G] = [G, G]$. $\square$

Definition 9.18. $G^{(0)} := G$, $G^{(i)} := [G^{(i-1)}, G^{(i-1)}]$, $i \geq 1$.

So $\dots \triangleleft G^{(2)} \triangleleft G^{(1)} \triangleleft G^{(0)} = G$.

Warning: $[A_5, A_5] = A_5!$

Example 9.19. $\{e\} \triangleleft [A_3, A_3] \triangleleft A_3 \triangleleft [S_3, S_3] \triangleleft S_3$, that is the same as $S_3^{(4)} \triangleleft S_3^{(3)} \triangleleft S_3^{(2)} \triangleleft S_3^{(1)} \triangleleft S_3^{(0)} = S_3$.

Theorem 9.20. A finite group G is solvable $\iff \exists n \in \mathbb{N}$ such that $G^{(n)} = \{e\}$.

Proof. If $G^{(n)} = \{e\}$ then $\{e\} = G^{(n)} \triangleleft \dots \triangleleft G^{(1)} \triangleleft G^{(0)} = G$ is a subnormal series with abelian subquotients, hence solvable.

If G is solvable, then there exists a subnormal series with abelian subquotients

$$\{e\} = H_0 \triangleleft H_1 \triangleleft \dots \triangleleft H_m = G$$

where H_m / H_{m-1} is abelian. Then $\forall a, b \in H_m [aH_{m-1}, bH_{m-1}] = [a, b]H_{m-1} = H_{m-1} \implies [a, b] \in H_{m-1} \implies [H_m, H_m] \subset H_{m-1}$, so $G^{(1)} = [G^{(0)}, G^{(0)}] \subset H_{m-1}$. Similarly $[H_{m-1}, H_{m-1}] \subset H_{m-2}$ that is $G^{(2)} = [G^{(1)}, G^{(1)}] \subset H_{m-2}$, thus $G^{(i)} \subset H_{m-i} \implies G^{(m)} = \{e\}$. $\square$

Corollary 9.21 (1). Every subgroups of a finite solvable group is solvable.

Proof. $H < G$, $H^{(i)} \subset G^{(i)} \implies G^{(m)} = \{e\} \implies H^{(m)} = \{e\}$. $\square$

Corollary 9.22 (2). Every quotient group of a finite solvable group is solvable.

Proof. If G is solvable then by theorem 9.20 there exists an $n \in \mathbb{N}$ such that $G^{(n)} = \{e\} \iff m$ -multicommutators are all $\{e\}$ swe (??) is done ni any quotient. $\square$

Corollary 9.23 (3). Let G be a finite group and $H \triangleleft G$. Then G is solvable $\iff H$ and G/H are solvable.

Proof. " $\implies$ " by corollary 1 and 2.

" $\Leftarrow$ " G/H is solvable $\iff G/H^{(m)} = \{e\} \iff G^{(m)} \subset H$. H solvable $\implies H^{(k)} = \{e\} \implies G^{(m+k)} = \{e\}$ hence G is solvable. $\square$

Example 9.24. Every Sylow p -group is solvable! If p is a prime and G is a group with $|G| = p^k$, then G is a p -group. If $Z(G) \neq \{e\}$ then $\{e\} \triangleleft Z(G) \triangleleft G$. Note that $|G/Z(G)| < |G|$.

10 Rings

Definition 10.1. A *monoid* is a tuple $(M, *, e)$ which consists of

- a set M
- a binary operator $*$
- and an identity element $e \in G$

which should satisfy the following:

- $(a * b) * c = a * (b * c), \forall a, b \in G$ (associativity)
- $e * a = a * e = a, \forall a \in G$ (identity)

Example 10.2.

1. $(\mathbb{N}, +, 0)$
2. $(\mathbb{Z}^+, \cdot, 1)$
3. $(\mathbb{Z}, \cdot, 1)$

Remark 10.3. In the context of the course one can assume that

1. a ring is a unital ring and
2. if $*$ is commutative then the ring R is also commutative.

Definition 10.4. A *ring* is a tuple $(R, +, \cdot, 0, 1)$ such that

- (a) $(R, +, 0)$ is an abelian group.
- (b) $(R, \cdot, 1)$ is a monoid.
- (c) $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(b + c) \cdot a = b \cdot a + c \cdot a$ for all $a, b, c \in R$.

Notation 10.5. " R " = " $(R, +, \cdot, 0, 1)$ "

Example 10.6.

1. $(\mathbb{Z}, +, \cdot, 0, 1)$
2. Let R be a commutative ring, then $R[x] = \{a_0 + a_1x + \dots + a_nx^n \mid n \in \mathbb{N}, a_i \in R\}$ with $+$ defined coefficient wise and multiplication $x^i x^j = x^{i+j}$ paired with distribution defines the commutative *polynomial ring* with coefficients in R .
3. $(R[x])[y] = R[x, y]$.

10.1 Ring homomorphism

Setup: Let R_i denote the ring $(R_i, +_i, \cdot_i, 0_i, 1_i)$ for $i = 1, 2$.

Definition 10.7. $\varphi : R_1 \rightarrow R_2$ is a *ring homomorphism* if

- (a) $\varphi(a +_1 b) = \varphi(a) +_2 \varphi(b)$
- (b) $\varphi(a \cdot_1 b) = \varphi(a) \cdot_2 \varphi(b)$
- (c) $\varphi(0_1) = 0_2$ (redundant)
- (d) $\varphi(1_1) = 1_2$.

Example 10.8.

1. $\text{Id}_R : R \rightarrow R$.
2.
$$\begin{array}{ccc} i: R & \longrightarrow & R[x] \\ \varphi & & \varphi \\ a & \longmapsto & a \end{array}$$
3.
$$\begin{array}{ccc} \varphi_a: R[x] & \longrightarrow & R \\ \varphi & & \varphi \\ f(x) & \longmapsto & f(a) \end{array} \quad \text{for a fixed } a \in R.$$

10.2 Ideals

Definition 10.9. A *left (right / two-sided) ideal* $I \triangleleft R$ is an additive subgroup such that:

- left: $RI \subset I$
- right: $IR \subset I$
- two-sided: $RIR \subset I$.

Remark 10.10.

1. When we say "ideal" we mean "two-sided ideal" and
2. if R is commutative then "left" = "right" = "two-sided".

Example 10.11. $\text{Mat}_{n \times n}(R)$ is not commutative for $n \neq 1$.

10.3 Ring quotients

Let $I \triangleleft R$ be an ideal and consider the cosets of abelian groups R/I . We claim that:

Definition 10.12. $(R/I, \overline{+}, \overline{\cdot}, I, 1 + I)$ with

$$(a + I) \overline{+} (b + I) := (a + b) + I$$

$$(a + I) \overline{\cdot} (b + I) := (ab) + I$$

is a *quotient ring* of R modulo I .

Remark 10.13. We assume $0 \neq 1$.

Theorem 10.14 (First isomorphism theorem). If $\varphi : R \rightarrow S$ is a ring homomorphism, then

- (a) $\ker(\varphi) \triangleleft R$
- (b) $\text{im}(\varphi)$ is a subring of S .
- (c) $R / \ker \varphi \simeq \text{im}(\varphi)$.

10.4 Zero divisors

Let R be a ring that is not necessary commutative.

Definition 10.15. $a \in R$ is a left (right) zero divisor if $\exists b \in R$ such that $b \neq 0$ but $ab = 0$ (repectively $ba = 0$).

Example 10.16. $0 \cdot 1 = 1 \cdot 0 = 0$ so 0 is a left/right zero divisor.

Definition 10.17.

1. A *domain* is a ring without non-zero zero divisors.
2. An *integral domain* is a domain that is commutative.

Example 10.18.

1. $\mathbb{Z}$ is an integral domain.
2. $\forall n \in \mathbb{Z} \ n\mathbb{Z} \triangleleft \mathbb{Z} \implies \mathbb{Z}_n := \mathbb{Z} / n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$.
3. $\bar{k} \in \mathbb{Z}_n, \bar{k} \neq \bar{0}$ is a zero divisor $\iff d = (k, n) > 1$, because if $(k, n) = 1$ then $n \mid \underbrace{\bar{k} \cdot \frac{\bar{n}}{\bar{d}}}_{\neq 0} = \bar{0}$,
because $n \mid k \cdot a \implies n \mid a \implies \bar{a} = \bar{0}$.
4. If R is an integral domain then $R[X]$ is a domain.

10.5 Prime ideals

Let R be a commutative ring.

Definition 10.19. An ideal $I \triangleleft R$ is a *prime ideal* if $\forall a, b \in R \ ab \in I \iff a \in I$ or $b \in I$.

Remark 10.20. R a domain $\iff \{0\}$ is a prime ideal.

Example 10.21.

1. $n\mathbb{Z} \triangleleft \mathbb{Z}$ is a prime ideal $\iff n$ is prime.

Definition 10.22. The *principal ideal* generated by $a \in R$ is

$$(a) := Ra.$$

2. $(f(x)) \triangleleft \mathbb{C}[x]$ is a prime ideal $\iff$

$$f(x) = \begin{cases} x \\ ax + b, a \neq 0 \end{cases}.$$

Moreover, if $\deg f > 1$ then $f(x) = \prod_{i=1}^n g_i(x)$ where $\deg g_i = 1$ for all $i = 1, \dots, n$.

Proposition 10.23. $I \triangleleft R$ is a prime ideal $\iff R/I$ is a domain.

Proof. For $a, b \notin I \implies a + I, b + I \neq 0$ in R/I . Then $(a + I)(b + I) = (ab) + I \neq 0$ in $R/I \iff R/I$ is a domain. $\square$

10.6 Inverses

Let R be a ring and $a \in R$.

Definition 10.24. $b \in R$ is a left (right) *inverse* to a if $ba = 1$ (respectively $ab = 1$).

Remark 10.25. There exists rings R where some left inverse exists but not right. For example $\mathbb{C}[\langle a, b \mid ab = e \rangle]$.

Definition 10.26. $a \in R$ is *invertible* if it has both a left and a right inverse.

Remark 10.27. If a is invertible, then any left inverse of a coincides with any right inverse $\implies$ denoted a^{-1} .

Proof. $ba = e, ac = e$, then $b = be = bac = ec = c$. $\square$

Remark 10.28. If R is a domain and $ab = e = ac \iff ab - ac = 0 \iff a(b - c) = 0, a \neq 0 \implies b = c$.

Proposition 10.29. Let R be a finite commutative ring, $a \in R$, then a is either invertible or a zero-divisor.

Proof. Let $\varphi : R \rightarrow R, x \mapsto ax$. If φ is not a bijection, then φ is not an injection, and hence $\exists x \neq y$ such that $ax = ay \implies a(x - y) = 0 \implies a \mid 0$. If φ is a bijection then $\exists x$ such that $ax = 1 \implies x = a^{-1}$. $\square$

10.7 Principal ideal domains

Definition 10.30. A commutative domain R in which all ideals have the form (a) for some $a \in R$ are called *principal ideal domain* (PID).

Example 10.31. $\mathbb{Z}, \mathbb{C}[x]$, and any Euclidian domain is a PID.

Theorem 10.32. Let R be a principal ideal domain. Then

- (a) $a \in R$ is irreducible $\iff$ it is a prime ideal.
- (b) Any $x \in R$ such that $x \neq 0$ and not invertible can be written as $x = a_1 \cdot a_2 \cdot \dots \cdot a_k$, where a_i is irreducible. Moreover, uniquely in the following sense that if $x = a_1 \cdot a_2 \cdot \dots \cdot a_k = b_1 \cdot b_2 \cdot \dots \cdot b_m$ where all a_i and b_i are irreducible, then $k = m$. Furthermore, there exists $\sigma \in S_k$ and an

invertible $\xi_i, i = 1, \dots, k$ such that $a_i = \xi_i b_{\sigma(i)}$ for all $i = 1, \dots, k$.

10.8 Fields

Definition 10.33. A commutative ring R is a *field* if any non-zero element in R is invertible.

We will always assume in this course that $0 \neq 1$.

Corollary 10.34. If $\varphi : \mathbb{K} \rightarrow \mathbb{F}$ is a *field* homomorphism, then φ is injective.

By *field* homomorphism we mean a ring homomorphism where the rings are fields.

Proof. We know from the first isomorphism theorem 4.1 (a) that $\ker(\varphi) \triangleleft \mathbb{K}$. If $\ker(\varphi) = \{0\}$ then φ is injective. Else $\ker(\varphi) = \mathbb{K}$, but then $\varphi(1) = 0 \neq 1$ which is a contradiction. $\square$

Example 10.35.

1. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields.
2. p is prime, then $\mathbb{Z}_p = \mathbb{Z} / \mathbb{Z}_p$ is a field.
3. Let K be a field, then $\mathbb{K}(x) := \{ \frac{f(x)}{g(x)} \mid f, g \in \mathbb{K}[x], g \neq 0 \} / \sim$, where $\frac{f}{g} \sim \frac{f'}{g'} \iff fg' = f'g$ is a *field of rational functions*.

Definition 10.36. Let R be a ring and $I \triangleleft R$. We say that I is *maximal* if $I \neq R$ and $\forall J \not\triangleleft R, I \subset J \implies I = J$.

Proposition 10.37. $I \triangleleft R$ is maximal $\iff R/I$ is a field.

Remark 10.38. R is a commutative ring $\iff \{0\}$ and R are the only ideals.

10.8.1 Prime fields

Definition 10.39. A field $\mathbb{F}$ is called *prime* if the only subfield of $\mathbb{F}$ is $\mathbb{F}$ itself.

Example 10.40.

1. Let p be prime, then $\mathbb{Z}_p$ is a prime field: If $\mathbb{F} \subset \mathbb{Z}_p$ is a subfield $\implies 0, 1 \in \mathbb{F} \implies 1 + 1 \in \mathbb{F} \implies 1 + 1 + 1 \in \mathbb{F} \implies \dots \implies \mathbb{Z}_p \subset \mathbb{F}$. Hence $\mathbb{F} = \mathbb{Z}_p$.
2. $\mathbb{Q}$ is prime: $\mathbb{F} \subset \mathbb{Q} \implies 0, 1 \in \mathbb{F} \implies \mathbb{Z} \subset \mathbb{F} \implies \{ \frac{1}{n} \mid n \in \mathbb{Z} \} \subset \mathbb{F} \implies \mathbb{Q} \subset \mathbb{F}$.

Theorem 10.41.

1. Any field $\mathbb{F}$ contains a unique minimal (with respect to inclusions) subfield $\mathbb{K}$.
2. $\mathbb{K} \simeq \mathbb{Z}_p$ or $\mathbb{Q}$ (for some prime p).

Proof. 1.

$$\mathbb{K} = \bigcap_{\{0,1\} \subset \mathbb{k} \subset \mathbb{F}} \mathbb{k} \neq \emptyset$$

$\implies \mathbb{K} \subset \mathbb{F}$ is a unique and minimal subfield by construction.

2. Let G be the additive subgroup of F generated by 1, that is $G = \{0, \pm m \cdot 1 \mid m \in \mathbb{Z}\}$. Then define the ring homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{F}$ where $m \mapsto m \cdot 1 = 1 + \cdots + 1 \cdot \text{sign}(m)$. Ideals in $\mathbb{Z}$ are $k\mathbb{Z}$ for

$k \in \mathbb{Z} \implies \ker(\varphi) = k\mathbb{Z}$ for some $k \in \mathbb{Z}$.

Case 1: $k = 0 \implies \varphi$ is injective.

Case 2: $k \neq 0 \implies$ by first isomorphism theorem 4.1 that $\text{im}(\varphi) \simeq \mathbb{Z} / k\mathbb{Z} \simeq \mathbb{Z}_k \subset \mathbb{F}$. If k is not prime, then $k = ab$, $a, b < k$ meaning that a and b are zero-divisors in $\mathbb{F}$, which is a contradiction.

Back to case 1: $\text{im}(\varphi) \simeq \mathbb{Z} \subset \mathbb{F}$. $\forall n \in \mathbb{Z}, n \neq 0$, there exists $\frac{1}{n} \in F$. Define $\mathbb{K} = \{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\} \subset F$. So this subfield is isometric to $\mathbb{Q}$. □

10.8.2 Characteristic of a field

Definition 10.42. Let $\mathbb{F}$ be a field. We define the *characteristic* of the field $\mathbb{F}$ as

$$\text{char}(\mathbb{F}) := \begin{cases} p & \text{if } \mathbb{Z}_p \subset \mathbb{F} \text{ for some prime } p, \\ 0 & \text{if } \mathbb{Q} \subset \mathbb{F}. \end{cases}$$

Example 10.43.

1. $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ have characteristic 0.
2. $\mathbb{Z}_p$ has characteristic p .

11 Field extensions

Definition 11.1. A *field extension* is a pair $\mathbb{K} \subset \mathbb{F}$ where $\mathbb{F}$ is a field and $\mathbb{K}$ is a subfield.

Example 11.2. On field extensions.

1. Any $\mathbb{k} \subset \mathbb{K}$.
2. $\mathbb{Q} \subset \mathbb{R}, \mathbb{Q} \subset \mathbb{C}$ and $\mathbb{R} \subset \mathbb{C}$.
3. $\mathbb{K} \subset \mathbb{K}(x)$.

Remark 11.3. If $\mathbb{K} \subset \mathbb{F}$ is a field extension, then $\text{char}(\mathbb{K}) = \text{char}(\mathbb{F})$.

11.1 Algebraic & transcendental elements

Let $\mathbb{K} \subset \mathbb{F}$ be a field extension.

Definition 11.4. An element $a \in \mathbb{F}$ is called *algebraic over* $\mathbb{K}$ provided that $\exists f \in \mathbb{K}[x]$ such that $f(a) = 0$.

Example 11.5.

1. For $\mathbb{K} \subset \mathbb{K}$, any $a \in \mathbb{K}$ is algebraic over $\mathbb{K}$: $x - a \in \mathbb{K}[x]$.
2. For $\mathbb{R} \subset \mathbb{C}$, $i \in \mathbb{C}$ is algebraic over $\mathbb{R}$: $x^2 + 1 \in \mathbb{R}[x]$.
3. For $\mathbb{Q} \subset \mathbb{R}$, π, e are not algebraic over $\mathbb{Q}$.

Definition 11.6. An element $a \in \mathbb{F}$ that is not algebraic over $\mathbb{K}$ is called *transcendental over $\mathbb{K}$* .

Example 11.7. π, e are transcendental over $\mathbb{Q}$.

Definition 11.8. An extension $\mathbb{K} \subset \mathbb{F}$ is *algebraic* if every element $a \in \mathbb{F}$ is algebraic over $\mathbb{K}$.

Example 11.9.

1. $\mathbb{K} \subset \mathbb{K}$ is algebraic.
2. $\mathbb{R} \subset \mathbb{C}$ is algebraic: For $a + bi \in \mathbb{C}$, $a, b \in \mathbb{R}$. Take $f(x) = (x - a)^2 + b^2 = 0$, then $f(a + bi) = 0$.

Proposition 11.10. If $\mathbb{K} \subset \mathbb{F}$ is a field extension and $a \in \mathbb{F}$ is transcendental over $\mathbb{K}$, then the mapping

$$\mathbb{K} \ni \frac{f(x)}{g(x)} \mapsto \frac{f(a)}{g(a)} \in \mathbb{F}$$

is a field homomorphism. In particular, if

$$\mathbb{F} = \bigcup_{\substack{\mathbb{K} \subset \mathbb{K} \subset \mathbb{F} \\ a}} \mathbb{K}$$

is the minimal subfield of $\mathbb{F}$ containing both $\mathbb{K}$ and a , then $\mathbb{F} \simeq \mathbb{K}(x)$.

Proof. The mapping (call it φ) is well-defined, since $g(a) \neq 0$ because a is transcendental, and $\mathbb{K} \subset \mathbb{F}$ is a field extension. $\square$

Observation: $\mathbb{F}$ is a vector space over $\mathbb{K}$.

Definition 11.11. We say that the field extension $\mathbb{K} \subset \mathbb{F}$ is *finite* if $\dim_{\mathbb{K}} \mathbb{F} < \infty$.

Proposition 11.12. Any finite extension is algebraic.

Proof. Assume $\dim_{\mathbb{K}} \mathbb{F} < \infty$, $a \in \mathbb{F}$, then $a, a^2, \dots, a^n$, $n > \dim_{\mathbb{K}} \mathbb{F} \implies$ they are linearly dependent $\implies \exists \alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{K}$ such that $\alpha_1 a + \alpha_2 a^2 + \dots + \alpha_n a^n = 0$. Let $f(x) = \alpha_1 x + \alpha_2 x^2 + \dots + \alpha_n x^n$, then $\varphi(a) = 0 \implies a$ is algebraic. $\square$

11.2 Intermediate subfields

Let $\mathbb{K} \subset \mathbb{F}$ be a field extension.

Definition 11.13. The subfield $\mathbb{L} \subset \mathbb{F}$ is called an *intermediate subfield* of $\mathbb{K} \subset \mathbb{F}$ if $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$.

$\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ is sometimes called a *chain*.

Theorem 11.14. $\mathbb{K} \subset \mathbb{F}$ is algebraic $\iff \forall a \in \mathbb{F}$ there exists an intermediate subfield $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ such that $a \in \mathbb{L}$ and $\dim_{\mathbb{K}} \mathbb{L} < \infty$.

Proof. " $\Leftarrow$ " follows from proposition 11.12.

" $\Rightarrow$ ": Let $\mathbb{K} \subset \mathbb{F}$ be algebraic and $a \in \mathbb{F}$. Define

$$\varphi_a : \mathbb{K}[x] \rightarrow \mathbb{F}$$

$$f(x) \mapsto a$$

Then since a is algebraic over $\mathbb{K}$, $\{0\} \neq \ker(\varphi_a) \triangleleft_{\text{ideal}} \mathbb{K}[X]$. Then there exists a non-zero polynomial called the *minimal polynomial* of a $m_a(x)$ that generates $\ker(\varphi_a)$ i.e. $\ker(\varphi_a) = (m_a(x))$. By the first isomorphism theorem 10.14,

$$\mathbb{K}[x] / (m_a(x)) \simeq \text{im}(\varphi_a) \subset \mathbb{F}.$$

Observe that $m_a(x)$ is irreducible! If not, then $m_a(x) = b(x)d(x)$ where $\deg b, d > 0 \implies \underset{\neq 0}{b(a)}\underset{\neq 0}{d(a)} = m_a(a) = 0$ is a contradiction since F has no zero-divisors $\implies (m_a(x))$ is minimal $\implies a \in \mathbb{K}[x] / (m_a(x)) \subset F$ is a field. Moreover

$$\dim_{\mathbb{K}} (\mathbb{K}[x] / (m_a(x))) = \deg m_a < \infty.$$

□

Corollary 11.15. If $\mathbb{K} \subset \mathbb{F}$ is a field extension and $a \in \mathbb{F}$, then a is algebraic over $\mathbb{K} \iff \exists$ an intermediate subfield $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ such that $a \in \mathbb{L}$ and $\dim_{\mathbb{K}} \mathbb{L} < \infty$.

Example 11.16.

1. $\mathbb{K} \subset \mathbb{K} \forall a \in \mathbb{K} m_a(x) = x - a$.
2. $\mathbb{R} \subset \mathbb{C}, i \in \mathbb{C}, m_i(x) = x^2 + 1$. Moreover $m_{a+bi}(x) = (x-a)^2 + b^2, b \neq 0$.

Corollary 11.17. Let $\mathbb{K} \subset \mathbb{F} \subset \mathbb{L}$ be a chain of field extensions, then $\mathbb{K} \subset \mathbb{L}$ is algebraic $\iff \mathbb{K} \subset \mathbb{F}$ and $\mathbb{F} \subset \mathbb{L}$ are algebraic.

Proof. " $\Leftarrow$ ": $a \in \mathbb{L} \implies \exists f \in \mathbb{F}[x]$ such that $f(a) = 0, f(x) = \alpha_1 + \alpha_2 x + \dots + \alpha_{k+1} x^k$. Each α_i is algebraic over $\mathbb{K} \implies$ for all α_i there exists an intermediate subfield

$$\mathbb{K} \subset \overset{\alpha_i}{\underset{\dim < \infty}{\mathbb{M}}} \subset \mathbb{F}$$

a is algebraic over $\mathbb{M}$,

$$a \in \mathbb{K} \subset \underset{\dim < \infty}{\mathbb{M}} \subset \mathbb{K} \subset \overset{a}{\underset{\dim < \infty}{\tilde{\mathbb{M}}}} \subset \mathbb{L}.$$

" $\Rightarrow$ " is left as an exercise.

□

Theorem 11.18. Let $\mathbb{K} \subset \mathbb{F}$ be a field extension, $a \in \mathbb{F}$ and assume that $\mathbb{F} = \mathbb{K}\langle a \rangle$ i.e.

$$\mathbb{F} = \bigcup_{\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}} \mathbb{L}.$$

Then

1. $\mathbb{F} \simeq \mathbb{K}(x) \simeq \mathbb{K}(a)$ if a is transcendental.
2. $\dim_{\mathbb{K}} \mathbb{F} = \deg m_a(x)$ if a is algebraic.

Example 11.19. $\mathbb{Q} \subset \mathbb{Q}\langle \sqrt{2} \rangle \simeq \mathbb{Q}[x] / (x^2 - 2)$. Here $x^2 - 2$ is the minimal polynomial $m_{\sqrt{2}}(x)$. Then $\dim_{\mathbb{Q}} \mathbb{Q}\langle \sqrt{2} \rangle = \deg x^2 - 2 = 2$. The basis is $1, \sqrt{2} \implies \mathbb{Q}\langle \sqrt{2} \rangle = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$. Is $\frac{1}{a+b\sqrt{2}} \in \mathbb{Q}\langle \sqrt{2} \rangle$? Yes!

$$\frac{1}{a + b\sqrt{2}} \cdot \frac{a - b\sqrt{2}}{a - b\sqrt{2}} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2} \sqrt{2}.$$

11.3 Adjoining roots of polynomials

Let $\mathbb{K}$ be a field, $f(x) \in \mathbb{K}[x]$, $\deg f > 0$. In order to construct a field extension $\mathbb{K} \subset \mathbb{F}$ such that $f(x)$ has a root in $\mathbb{F}$, we write $f(x) = f_1(x) \cdot f_k(x)$, where each $f_i(x)$ is irreducible. This exists since $\mathbb{K}$ is a principal ideal domain. Notice $\deg f \geq 1$, now consider the field $\mathbb{F} = \frac{\mathbb{K}[x]}{(f_1(x))}$ and $\bar{x} = x + (f_1(x))$ which satisfies $f_1(\bar{x}) = 0 \implies f(\bar{x}) = 0$. Over $\mathbb{F}$ we can write $f(y) = (y - \bar{x})g(y)$ where $\deg g = \deg f - 1$.

Definition 11.20. We say that $f \in \mathbb{K}[x]$, $\deg f > 0$ *splits* if $\exists a_1, a_2, \dots, a_{\deg f} \in \mathbb{F}$, $\alpha \in \mathbb{F}$ with

$$f(x) = \alpha \prod_{i=1}^{\deg f} (x - a_i).$$

Corollary 11.21. For all fields $\mathbb{K}$, and for all $f \in \mathbb{K}[x]$, $\deg f > 0$, there exists a finite field extension $\mathbb{K} \subset \mathbb{F}$ such that f splits over $\mathbb{F}$.

Proof. Apply induction on the previous observation above and use what we did in the exercise 11.19. $\mathbb{K} \subset \mathbb{F} \subset \mathbb{L}$, $\dim_{\mathbb{K}} \mathbb{F} = m$, $\dim_{\mathbb{F}} \mathbb{L} = n \implies \dim_{\mathbb{K}} \mathbb{L} = n \cdot m$. Then let $\langle a_1, \dots, a_m \rangle$ be a basis for $\mathbb{F} / \mathbb{K}$ and $\langle b_1, \dots, b_n \rangle$ be a basis for $\mathbb{L} / \mathbb{F}$ then $\langle a_1, b_1 \rangle$ is a basis for $\mathbb{L} / \mathbb{K}$. $\square$

11.4 Splitting fields

Let $\mathbb{K}$ be a field and $f(x) \in \mathbb{K}[x]$, $\deg f > 0$.

Definition 11.22. A field extension $\mathbb{K} \subset \mathbb{F}$ is called the *splitting field* for $f(x)$ if

- (a) it splits over $\mathbb{F}$
- (b) $\dim_{\mathbb{K}} \mathbb{F}$ is minimal possible.

Theorem 11.23. The splitting field is unique up to isomorphism.

Proof.

Theorem 11.24 (on extensions of homomorphisms). Let $\mathbb{K} \subset \mathbb{F}$ be a field extension and $\varphi : \mathbb{K} \rightarrow \mathbb{L}$ a field homomorphism. Then there exists a field extension $\mathbb{L} \subset \mathbb{M}$ and a field homomorphism $\psi : \mathbb{F} \rightarrow \mathbb{M}$ such that

$$\begin{array}{ccc} \mathbb{L} & \xhookrightarrow{\beta} & \mathbb{M} \\ \varphi \uparrow & & \uparrow \psi \\ \mathbb{K} & \xhookrightarrow{\alpha} & \mathbb{F} \end{array} \quad \psi \circ \varphi = \beta \circ \alpha.$$

Proof. Assume $\mathbb{F} = \mathbb{K}\langle a \rangle$. If we can prove this case, then the theorem follows by induction (possible transfinite).

Case 1: a is transcendental $\implies \mathbb{F} = \mathbb{K}\langle a \rangle$. Define $\mathbb{M} := \mathbb{L}(x)$, $\psi(\frac{f(a)}{g(a)}) = \frac{\varphi(f(x))}{\varphi(g(x))}$.

Case 2: a is algebraic, then $m_a(x) \in \mathbb{K}[x]$ is the minimal polynomial. $\varphi(m_a(x)) \in \mathbb{L}[x]$. Let $\mathbb{M}$ be a field extension of $\mathbb{L}$ where $\varphi(m_a(x))$ has a root β . $\mathbb{F} = \mathbb{K}[x] / (m_a(x))$, $\mathbb{K}[x] \ni f(x) \mapsto \varphi(f(\beta))$, $\varphi(m_a(\beta)) = 0$. This gives the homomorphism $\mathbb{F} \rightarrow \mathbb{L}$. $\square$

Now for uniqueness: Let $\mathbb{K} \subset \mathbb{F}$ and $\mathbb{K} \subset \mathbb{L}$ be the splitting fields

$$\begin{array}{ccc} \mathbb{L} & \hookrightarrow & \exists \mathbb{M} \\ \varphi \uparrow & & \uparrow \exists \psi \\ \mathbb{K} & \hookrightarrow & \mathbb{F} \end{array} \quad \text{by the extension theorem ??}$$

Let $a_1, a_2, \dots, a_k$ be all roots of $f(x)$ in $\mathbb{L} \implies \mathbb{L} = \mathbb{K}\langle a_1, a_2, \dots, a_k \rangle$. $\mathbb{M} \ni \psi(a_i)$ must be a root of $f(x) \implies \psi(a_i) \in \mathbb{F}$ i.e. ψ restricts to $\hat{\psi} : \mathbb{L} \rightarrow \mathbb{F}$. By symmetry there exists $\eta : \mathbb{F} \rightarrow \mathbb{L}$, where both $\hat{\psi}$ and η are the identity on $\mathbb{K}$. Moreover $\dim_{\mathbb{K}} \mathbb{L}, \dim_{\mathbb{K}} \mathbb{F} < \infty$, $\hat{\psi}$ and η are injective functions $\implies \dim_{\mathbb{K}} \mathbb{L} = \dim_{\mathbb{K}} \mathbb{F}$, and $\hat{\psi}, \eta$ are bijections $\implies$ isometric. $\square$

Example 11.25.

1. $\deg f = 1 \implies \mathbb{K} \subset \mathbb{K}$ is the splitting field.
2. $\mathbb{R} \subset \mathbb{C}$ is the splitting field of $x^2 + 1$. Note that the field extension $\mathbb{K} \subset \mathbb{F}$, $\mathbb{K} \subset \mathbb{L}$, $\mathbb{L} \simeq \mathbb{F}$ don't necessarily have a unique isomorphism. Here $\exists \mathbb{C} \xrightarrow{\sim} \mathbb{C}$ by the identity $\text{id}_{\mathbb{C}}$, but also by the complex conjugation $\bar{\cdot}$.
3. $\mathbb{Q} \ni f(x) = x^3 - 1 = (x-1)(x^2 + x + 1)$ which has roots $\frac{-1 \pm i\sqrt{3}}{2}$. $\mathbb{Q}(\frac{-1 \pm i\sqrt{3}}{2}) \simeq \mathbb{Q}[x] / (x^2 + x + 1)$. Hence $\dim_{\mathbb{Q}} \mathbb{Q}(\frac{-1 \pm i\sqrt{3}}{2}) = 2$.

11.5 Linear independence of homomorphism

Theorem 11.26. Let G be a group, $\mathbb{F}$ a field and $\varphi_1, \dots, \varphi_n : G \rightarrow \mathbb{F}^* = (\mathbb{F} \setminus \{0\}, \cdot, 1)$ are pairwise different group homomorphisms. Then $\varphi_1, \dots, \varphi_n$ are linearly independent as functions i.e. if $a_1, \dots, a_n \in \mathbb{F}$, $\sum_{i=1}^n \varphi_i(g) a_i = 0 \forall g \in G \implies a_1 = \dots = a_n = 0$.

Proof. Proof is done by induction on n .

Base case: $n = 1$, $a_1 \varphi_1(e) = a_1 \cdot 1 = a_1 = 0 \implies a_1 = 0$.

Induction step: Assume $\sum_{i=1}^n \varphi_i(g) a_i = 0 \forall g \in G$. Assume $a_1 \neq 0$. $\varphi_n \neq \varphi_1 \implies \exists h \in G$ such that $\varphi_n(h) \neq \varphi_1(h)$. Then

$$\sum_{i=1}^n a_i \varphi_i(h) \varphi_i(g) = \sum_{i=1}^n a_i \varphi_i(hg) = 0$$

$$\begin{aligned}
\sum_{i=1}^n a_i \varphi_n(h) \varphi_i(g) &= \varphi_n(h) \sum_{i=1}^n a_i \varphi_i(g) = 0 \\
\implies \sum_{i=1}^n a_i [\varphi_n(h) - \varphi_i(h)] \varphi_i(g) &= 0 \\
\implies \sum_{i=1}^{n-1} a_i [\varphi_n(h) - \varphi_i(h)] \varphi_i(g) &= 0.
\end{aligned}$$

But then $a_1[\varphi_n(h) - \varphi_1(h)] \neq 0 \implies \varphi_1, \dots, \varphi_{n-1}$ are linearly dependent, a contradiction! $\square$

Corollary 11.27. Let $\varphi_1, \dots, \varphi_n$ be pairwise different homomorphism from a field $\mathbb{F}$ to a field $\mathbb{L}$. Define $\mathbb{K} := \{a \in \mathbb{F} \mid \varphi_i(a) = \varphi_j(a) \forall i, j\}$ (subfield of $\mathbb{F}$). Then $\dim_{\mathbb{K}} \mathbb{F} \geq n$.

Notation 11.28. $[\mathbb{F} : \mathbb{K}] = \dim_{\mathbb{K}} \mathbb{F}$.

Example 11.29. $\mathbb{R} \subset \mathbb{C}$, $\text{id}_{\mathbb{C}} : \mathbb{C} \rightarrow \mathbb{C}$, $\bar{\cdot} : \mathbb{C} \rightarrow \mathbb{C}$ (complex conjugation). $[\mathbb{C} : \mathbb{R}] = 2 \implies \text{id}_{\mathbb{C}}$ and $\bar{\cdot}$ are the only automorphisms of $\mathbb{C}$ that fixes $\mathbb{R}$.

Proof. Assume $[\mathbb{F} : \mathbb{K}] = m < n$. Consider the finite dimensional vector space $\mathbb{L}^m$. Let $b_1, \dots, b_m$ be a basis of $\mathbb{F} / \mathbb{K}$.

$$\mathbb{L}^m \ni v_i = \begin{pmatrix} \varphi_i(b_1) \\ \vdots \\ \varphi_i(b_m) \end{pmatrix} \quad i = 1, \dots, n.$$

$v_1, \dots, v_n$ are linearly dependent, that is $\exists a_1, \dots, a_n \in \mathbb{L}$ not all zero, such that $\sum_{i=1}^n a_i v_i = 0$. Take $\mathbb{F} \ni d = \sum_{j=1}^m d_j b_j$, $d_j \in \mathbb{K}$. Then

$$\begin{aligned}
\sum_{i=1}^n a_i \varphi_i(d) &= \sum_{i=1}^n a_i \sum_{j=1}^m \varphi_i(d_j) \varphi_i(b_j) = \\
&= \sum_{j=1}^m \sum_{i=1}^n a_i \varphi_i(d_j) \varphi_i(b_j) = \sum_{j=1}^m \varphi_i(d_j) \sum_{i=1}^n a_i \varphi_i(b_j) = 0.
\end{aligned}$$

A contradiction! Linear dependent as φ_i 's are different homomorphisms $\mathbb{F}^* \rightarrow \mathbb{L}^*$. $\square$

11.6 Derivative

Let $\mathbb{K}$ be a field and $f(x) \in \mathbb{K}[x]$. The derivative is defined so that it behaves like one is used to.

Definition 11.30. The *derivative* $(_) ' = \frac{d}{dx} = D : \mathbb{K}[x] \rightarrow \mathbb{K}[x]$ is linear and $(x^n)' := nx^{n-1}$.

Properties:

1. Linear
2. $(fg)' = f'g + fg'$ (Libnitz rule)
3. $\text{char } \mathbb{K} = 0 \implies (f)' = 0 \iff f(x) = c \in \mathbb{K}$.

If $\text{char } \mathbb{K} = p > 0$ prime, then $(x^p)' = 0$ (Unusual).

11.7 separable polynomials

Let $\mathbb{K}$ be a field and $f(x) \in \mathbb{K}[x]$.

Definition 11.31. If f is irreducible, we say f is *separable* if $f' \neq 0$. A general polynomial is separable if it's all irreducible factors are.

Example 11.32.

1. $\text{char } \mathbb{K} = 0$ anything is separable.
2. $\text{char } \mathbb{K} = 2$, $\mathbb{K}(t) \supset \mathbb{K}(t^2)$. $x^2 - t^2 \in \mathbb{K}(t^2)[x]$ $(x^2 - t^2) = 0 = 2x$, so root t has multiplicity 2. Moreover $(x^2 - t^2) = (x + t)^2$ is irreducible over $\mathbb{K}(t^2) \not\ni t$.

Remark 11.33. If $\mathbb{K}$ is a field over characteristic p , then $\alpha \mapsto \alpha^p$ (Frobenius map) has the properties

- $(\alpha\beta)^p = \alpha^p\beta^p$
- $(\alpha + \beta)^p = \alpha^p + \beta^p$, as

$$(\alpha + \beta)^p = \sum_{i=0}^p \alpha^i \beta^{p-i} \frac{p!}{i!(p-i)!} = \beta^p + \cancel{\alpha^1 \beta^{p-1} \frac{p!}{1!(p-1)!}} + \cdots + \cancel{\alpha^{p-1} \beta^1 \frac{p!}{(p-1)!1!}} + \alpha^p.$$

Let $\mathbb{K} \subset \mathbb{F}$ be a field extension, and $a \in \mathbb{F}$ an algebraic element.

Definition 11.34. a is *separable* if $m_a(x)$ is separable.

Definition 11.35. $\mathbb{K} \subset \mathbb{F}$ is *separable* if all elements $a \in \mathbb{F}$ are separable and algebraic.

Example 11.36. Any algebraic extension in characteristic 0 is separable.

Remark 11.37. A field $\mathbb{K}$ is *perfect* if all irreducible polynomials over $\mathbb{K}$ are separable.

Theorem 11.38 (On separable extensions). Let $\mathbb{K} \subset \mathbb{F}$ be a field extension with $[\mathbb{F} : \mathbb{K}] = n < \infty$. Then the following are equivalent:

1. This extension is separable.
2. $\mathbb{F} = \mathbb{K}\langle a_1, \dots, a_m \rangle$, a_i is separable.
3. $\forall$ homomorphisms $\mathbb{K} \xrightarrow{\varphi} \mathbb{K}$ there exists an extension $\mathbb{K} \hookrightarrow \mathbb{L}$ such that φ admits n different extensions to a homomorphism $\mathbb{F} \rightarrow \mathbb{L}$.
4. There is an extension $K \subset \mathbb{M}$ and n different homomorphisms $\mathbb{F} \rightarrow \mathbb{M}$ that are the identity when restricted to $\mathbb{K}$.

Proof. 1) $\implies$ 2) and 3) $\implies$ 4) are left as exercise. We'll start by proving 4) $\implies$ 1).

Assume $a \in \mathbb{F}$ is not separable, that is $\deg m_a(x) = d$ (not separable), hence

$$\overbrace{\mathbb{K} \subset \mathbb{K}\langle a \rangle \subset \mathbb{F}}^{\deg=n}$$

$$\underbrace{\hspace{1.5cm}}_{\deg=d}$$

so $[\mathbb{F} : \mathbb{K}\langle a \rangle] = \frac{n}{d}$. Define the homomorphism $\psi : \mathbb{K}\langle a \rangle \rightarrow \mathbb{M}$, where ψ is uniquely determined by $\psi(a)$ such that $\psi|_{\mathbb{K}} = \text{id}_{\mathbb{K}}$. $\psi(a)$ is a root of $m_a(x) \implies \psi(a)$ can be in r ways ($r < d$). r being the number of different roots of $m_a(x)$. Now by corollary ??, any $\psi : \mathbb{K}\langle a \rangle \rightarrow \mathbb{M}$ can be extended to a homomorphism $\mathbb{F} \rightarrow \mathbb{M}$ in (at most) $[\mathbb{F} : \mathbb{K}\langle a \rangle] = \frac{n}{d}$ ways $\implies \mathbb{K} \rightsquigarrow \mathbb{F}$ we have at most $r \frac{n}{d} < n$ which is a contradiction.

For 2) $\implies$ 3) we do $\mathbb{F} = \mathbb{K}\langle a \rangle$ where a is separable (general case is left as an exercise). $m_a(x)$ is separable and has degree d . $\varphi : \mathbb{K} \rightarrow \mathbb{K}$, $\varphi(m_a(x)) \in \mathbb{K}[x]$. Let $\mathbb{L}$ be the splitting field.

Lemma 11.39. An irreducible polynomial $f(x) \in \mathbb{K}[x]$ is separable $\iff f$ does not have multiple roots.

Proof. $\mathbb{K} \subset$ "splitting field of $f(x)$ ", where $f(x) = c \cdot \prod_{i=1}^d (x - \alpha_i)$. We can write $f(x) = (x - \alpha)^2 g(x)$. Then $0 \neq f'(x) = (x - \alpha)g(x) + (x - \alpha)^2 g'(x) \implies f'(\alpha) = 0$, hence $(f, f') = 1 \implies f'$ does not have multiple roots.

The other direction is left as an exercise. □

$\implies \alpha_1, \dots, \alpha_d$ are different roots of $\varphi(m_a(x))$. $\forall s = 1, \dots, d$,

$$\varphi_s : \frac{\mathbb{K}[x]}{(m_a(x))} \rightarrow \mathbb{L}$$

$$\varphi_s|_{\mathbb{K}} = \varphi \quad \varphi_s(x) = \alpha_s.$$

□

11.8 Normal extensions

Definition 11.40. A field extension $\mathbb{K} \subset \mathbb{F}$ is *normal* if for all irreducible $f(x) \in \mathbb{K}[x]$ if $\exists a \in \mathbb{F}$ such that $f(a) = 0$, then $f(x)$ splits over $\mathbb{F}$.

Definition 11.41 (Alternative). A field extension $\mathbb{K} \subset \mathbb{F}$ is *normal* if for all algebraic $a \in \mathbb{F}$, $\mathbb{F}$ contains the splitting field of $m_a(x)$.

Example 11.42.

1. $K \subset \mathbb{K}$ is normal.
2. If $\mathbb{F}$ is algebraically closed, then $\mathbb{K} \subset \mathbb{F}$ is normal. E.g. $\mathbb{R} \subset \mathbb{C}$ is normal.
3. We claim that for $x^3 - 2 \in \mathbb{Q}[x]$, $x^3 - 2$ is irreducible, for which we need the help of the following criterium:

Eisenstein's criterium 11.43. Let $f(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ such that $a_n \neq 0$. If there exists a prime p such that $p \nmid a_n$, $p \mid a_0, a_1, \dots, a_{n-1}$ and $p^2 \nmid a_0$. Then $f(x)$ is irreducible over $\mathbb{Q}$.

So for $(-2) + x^3$, $p = 2$ apply E.C. We then have $\mathbb{Q} \subset \frac{\mathbb{Q}[x]}{(x^3-2)} =: \mathbb{F}$. $[\mathbb{F} : \mathbb{Q}] = 3$. We now claim that this extension is *not* normal! In fact, $x^3 - 2$ does not split over $\mathbb{F}$. We have that $\mathbb{Q} \subset \mathbb{R} \ni \sqrt[3]{2}$, $(\sqrt[3]{2})^3 - 2 = 0$. So there exists $\varphi : \mathbb{F} \rightarrow \mathbb{R}$ with $\varphi|_{\mathbb{Q}} = \text{id}_{\mathbb{Q}} \implies$ we can view $\mathbb{F}$ as a subfield of $\mathbb{R}$ via φ . For $\mathbb{R} \subset \mathbb{C}$, $x^3 - 2$ splits over $\mathbb{C}$. The roots are $\sqrt[3]{2}$, $\sqrt[3]{2} \cdot \epsilon$, $\sqrt[3]{2} \cdot \epsilon^2$, where $\epsilon^3 = 1$, $\epsilon = e^{\frac{2\pi i}{3}} = \frac{1+i\sqrt{3}}{2} \implies x^3 - 2$ does not split over $\mathbb{Q}(\sqrt[3]{2}) \subset \mathbb{R}$.

Theorem 11.44 (on normal extensions). Let $\mathbb{K} \subset \mathbb{F}$ be a finite field extension. The following are equivalent:

1. The extension is normal
2. $\mathbb{F}$ is the splitting field of some $f(x) \in \mathbb{K}[x]$.
3. For all extensions $\mathbb{F} \subset \mathbb{L}$, $\forall \varphi : \mathbb{F} \rightarrow \mathbb{L}$ such that $\varphi|_{\mathbb{K}} = \text{id}_{\mathbb{K}} \implies \varphi(\mathbb{F}) \subset \mathbb{F}$.

Proof. 2) $\implies$ 3): Let $\mathbb{F} = \mathbb{K}\langle a_1, \dots, a_m \rangle$ where $a_1, \dots, a_m$ are all roots of $f(x)$. Let $\varphi : \mathbb{F} \rightarrow \mathbb{L}$ be such that $\varphi|_{\mathbb{K}} = \text{id}_{\mathbb{K}}$ then $\varphi(f) = f \implies \varphi$ send roots $\{a_i\}$ to roots $\{a_i\} \implies \varphi(a_i) \in \mathbb{F} \implies \varphi : \mathbb{F} \rightarrow \mathbb{F}$.

3) $\implies$ 1): Let $g(x) \in \mathbb{K}[x]$ be irreducible and $a \in \mathbb{F}$ such that $g(a) = 0$. Let $\mathbb{L}$ be a splitting field of g over $\mathbb{F}$. Let $b \in \mathbb{L}$ be a root of g . There exists $\mathbb{K}\langle a \rangle = \frac{\mathbb{K}[x]}{(g(x))} \xrightarrow{\varphi} \mathbb{L}$, $\varphi(a) = b$. By the extension of homomorphisms theorem ?? φ extends to $\psi : \mathbb{F} \rightarrow \mathbb{L}\langle b \rangle = \mathbb{L} \implies \varphi(\mathbb{F}) \subset \mathbb{F}$ by 3) $\implies b \in \mathbb{F}$.

1) $\implies$ 2): Let $a_1, \dots, a_m$ be a basis of $\mathbb{F}$ over $\mathbb{K}$. Then for all $s = 1, \dots, m$ $m_{a_s}(x)$ splits over $\mathbb{F} \implies \mathbb{F}$ is the splitting field of $\prod_{s=1}^m m_{a_s}(x)$. $\square$

11.9 Automorphism intersections of field extensions

Definition 11.45. The automorphism groups $\text{Aut}(\mathbb{F}/\mathbb{K})$ of $\mathbb{K} \subset \mathbb{F}$ is

$$\text{Aut}(\mathbb{F}/\mathbb{K}) := \{\varphi : \mathbb{F} \rightarrow \mathbb{F} \mid \varphi \text{ is an automorphism and } \varphi|_{\mathbb{K}} = \text{id}_{\mathbb{K}}\}.$$

Remark 11.46. $\text{Aut}(\mathbb{F}/\mathbb{K}) < \text{Aut}(\mathbb{F})$.

Example 11.47.

1. For $\mathbb{R} \subset \mathbb{C}$ $\text{Aut}(\mathbb{C}/\mathbb{R}) = \{\text{id}_{\mathbb{C}}, \bar{\cdot}\}$, $[\mathbb{C} : \mathbb{R}] = 2$. We know that if $G < \text{Aut}(\mathbb{F})$, $|G| < \infty$, then $\mathbb{F}^G := \{x \in \mathbb{F} \mid \varphi(x) = x \forall \varphi \in G\}$, $[\mathbb{F} : \mathbb{F}^G] \geq |G| \implies |\text{Aut}(\mathbb{C}/\mathbb{R})| \leq 2$.
2. For the field extension $\mathbb{Q} \subset \mathbb{R}$ we claim that $\text{Aut}(\mathbb{R}/\mathbb{Q}) = \{\text{id}_{\mathbb{R}}\}$.

Proof.

- (a) For any $\varphi \in \text{Aut}(\mathbb{R})$ that satisfies $\varphi|_{\mathbb{Q}} = \text{id}_{\mathbb{Q}}$, $\varphi(1) = 1 \implies \varphi(n) = n \forall n \in \mathbb{Z}$. So $\varphi(\frac{n}{m}) = \frac{\varphi(n)}{\varphi(m)} = \frac{n}{m} \forall n, m \in \mathbb{Z}, m \neq 0 \implies \text{Aut}(\mathbb{R}/\mathbb{Q}) = \text{Aut}(\mathbb{R})$.
- (b) Let $\varphi \in \text{Aut}(\mathbb{R})$, $a \in \mathbb{R}_{>0} \implies \varphi(a) \in \mathbb{R}_{>0}$. $a \in \mathbb{R}$ is $> 0 \iff x^2 - a$ has a solution in $\mathbb{R}$. Any automorphism of $\mathbb{R}$ preserves this property.
- (c) $a, b \in \mathbb{R}$, $a < b \implies \varphi(a) < \varphi(b)$, $\varphi \in \text{Aut}(\mathbb{R})$: $a < b \iff b - a > 0 \implies \varphi(b - a) > 0 \implies \varphi(b) > \varphi(a)$.
- (d) $a \in \mathbb{R}$

$$a^- := \{b \in \mathbb{Q} \mid b < a\}$$

$$a^+ := \{b \in \mathbb{Q} \mid a < b\}.$$

a is uniquely determined by a^- and a^+ . $\{z \in \mathbb{R} \mid z > b \forall b \in a^-, z < b \forall b \in a^+\} = \{a\}$. Now (a) and (b) together gives $\varphi(a^-) = a^-$ and $\varphi(a^+) = a^+ \implies \varphi(a) = a$.

$\square$

12 Galois extensions

Definition 12.1. An algebraic field extension $\mathbb{K} \subset \mathbb{F}$ is called a *Galois extension* (or *Galois*) provided that it is normal and separable.

Definition 12.2. If $\mathbb{K} \subset \mathbb{F}$ is Galois, then $\text{Aut}(\mathbb{F}/\mathbb{K})$ is called the *Galois group* and is denoted $\text{Gal}(\mathbb{F}/\mathbb{K})$.

Example 12.3. $\mathbb{R} \subset \mathbb{C}$ is Galois. $\text{char} = 0 \implies$ separable, and $\mathbb{C}$ is algebraically closed $\implies$ normal. The Galois group $\text{Gal}(\mathbb{C}/\mathbb{R}) = \{\text{id}_{\mathbb{C}}, \bar{\cdot}\}$.

Theorem 12.4 (Artin, characterism of finite Galois extensions). Let $\mathbb{K} \subset \mathbb{F}$ be a finite field extension. The following are equivalent.

1. This extension is Galois.
2. $\mathbb{F}$ is the splitting field of the polynomial $\varphi(x) \in \mathbb{K}[x]$ with separable and irreducible factors.
3. $[\mathbb{F}, \mathbb{K}] = |\text{Aut}(\mathbb{F}/\mathbb{K})|$.
4. There exists a finite $G < \text{Aut}(\mathbb{F})$ such that $\mathbb{K} = \mathbb{F}^G$.
5. $\mathbb{K} = \mathbb{F}^{\text{Aut}(\mathbb{F}/\mathbb{K})}$.

Proof. Both $1 \iff 2$ and $4 \iff 5$ are clear. (According to my notes)

$3 \implies 1$: Assume that $[\mathbb{F} : \mathbb{K}] = |\text{Aut}(\mathbb{F}/\mathbb{K})| = n$. We have n different automorphisms and $\dim_{\mathbb{K}} \mathbb{F} = n \implies \mathbb{K} \subset \mathbb{F}$ is separable. We know that for all $\mathbb{F} \subset \mathbb{L}$ there exists at most $n = [\mathbb{F} : \mathbb{K}]$ homomorphisms $\mathbb{F} \rightarrow \mathbb{L}$ that are the identity when restricted to $\mathbb{K}$. But for all $\varphi \in \text{Aut}(\mathbb{F}/\mathbb{K})$, $\mathbb{F} \xrightarrow{\varphi} \mathbb{F} \hookrightarrow \mathbb{L}$ is a homomorphism [like that? (can't read my notes)], and there are n many such homomorphisms $\implies$ they are all. They all map $\mathbb{F} \rightarrow \mathbb{F} \implies \mathbb{K} \subset \mathbb{F}$ is normal.

$1 \implies 5$: To prove $\forall a \in \mathbb{F}/\mathbb{K} \exists \varphi \in \text{Aut}(\mathbb{F}/\mathbb{K})$ such that $\varphi(a) \neq a$ consider the irreducible polynomial $m_a(x) \in \mathbb{K}[x]$ with $\deg \geq 2$. separable $\implies m_a(x)$ has a root $b \neq a \implies \mathbb{K}\langle a \rangle \xrightarrow{\psi} \mathbb{F} \quad a \mapsto b$, and ψ is the identity restricted to $\mathbb{K}$.

$$\begin{array}{ccc} \mathbb{K}\langle a \rangle & \xhookrightarrow{\quad} & \mathbb{F} \\ \psi \downarrow & \swarrow & \downarrow \\ \mathbb{F} & \xrightarrow{\quad} & \mathbb{L} \end{array}$$

which gives $\varphi \in \text{Aut}(\mathbb{F}/\mathbb{K})$ such that $\varphi(a) = b$.

$5 \implies 3$: Assume $\mathbb{K} = \mathbb{F}^{\text{Aut}(\mathbb{F}/\mathbb{K})}$. We have that $[\mathbb{F} : \mathbb{K}] \geq |\text{Aut}(\mathbb{F}/\mathbb{K})|$, so we want to prove $[\mathbb{F} : \mathbb{K}] \leq |\text{Aut}(\mathbb{F}/\mathbb{K})|$. To start, we prove that for all finite subgroups G of $\text{Aut}(\mathbb{F})$ ($\mathbb{F}$ a field), $[\mathbb{F}, \mathbb{F}^G] = |G|$. To prove $[\mathbb{F}, \mathbb{F}^G] \leq |G|$, i.e. $\forall m > n \forall a_1, \dots, a_m \in \mathbb{F}$ linear independent over $\mathbb{K}$, and $G = \{g_1, \dots, g_n\}$, for $i = 1, \dots, m$ consider $v_i \in \mathbb{F}^n$, $(v_i)_j := g_j(a_i)$, creating an $n \times m$ matrix where the columns $v_1, \dots, v_m$ are linear dependent over $\mathbb{F}^n$, that is, there exists $c_1, \dots, c_m \in \mathbb{F}$ (not all zero) such that $c_1 v_1 + \dots + c_m v_m = 0$. Choose c_1 (not zero) such that $g_1(c_1)^{-1} + \dots + g_n^{-1}(c_1) \neq 0$, $\forall j = 1, \dots, n : c_1 g_j(a_1) + \dots + c_m g_j(a_n) = 0$. Now apply g_j^{-1} :

$$\begin{aligned} & \forall j = 1, \dots, n : g_j^{-1}(c_1) a_1 + \dots + g_j^{-1}(c_m) a_m \\ & \left[\sum_{j=1}^n g_j^{-1}(c_1) a_1 \right] + \dots + \left[\sum_{j=1}^n g_j^{-1}(c_m) a_m \right] = 0 \end{aligned}$$

$$h\left(\sum_{g \in G} g(x)\right) = \sum_{g \in G} hg(x) = \sum_{g \in G} g(x)$$

$\implies a_1, \dots, a_m$ are linearly dependent over $\mathbb{K}$. □

Example 12.5. Consider $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}, i)$ ($i^2 = -1$). Then $m_{\sqrt{2}}(x) = x^2 - 2$ is the minimal polynomial, as Eisenstein's criterion tells us that it is irreducible. For i the minimal polynomial is $m_i(x) = x^2 + 1$. We note that this is an algebraic extension.

$$\mathbb{Q} \xrightarrow{\deg 2} \mathbb{Q}(\sqrt{2}) \xrightarrow{\deg 2} \mathbb{Q}(\sqrt{2}, i)$$

so $[\mathbb{Q}(\sqrt{2}, i), \mathbb{Q}] = 2 \cdot 2 = 4$. As $\text{char} = 0$ $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}, i)$ is separable, and as $\mathbb{Q}(\sqrt{2}, i)$ is the splitting field of $(x^2 - 2)(x^2 + 1)$ hence $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}, i)$ is normal $\implies \mathbb{Q} \subset \mathbb{Q}(\sqrt{2}, i)$ is Galois.

$$|\text{Gal}(\mathbb{Q}(\sqrt{2}, i)/\mathbb{Q})| = \text{'degree of the extension'} = 4.$$

The basis of $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$ is $1, \sqrt{2}$ as $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$. Likewise, the basis of $\mathbb{Q}(\sqrt{2}, i)$ over $\mathbb{Q}(\sqrt{2})$ is $1, i$ as $\mathbb{Q}(\sqrt{2}, i) = \{a + bi \mid a, b \in \mathbb{Q}(\sqrt{2})\}$, so the basis for $\mathbb{Q}(\sqrt{2}, i)$ is then $1, \sqrt{2}, i, \sqrt{2}i$. Moreover, any $\varphi \in \text{Gal}(\mathbb{Q}(\sqrt{2}, i)/\mathbb{Q})$ is uniquely determined by what it does to the basis: φ sends $1 \mapsto 1, \sqrt{2} \mapsto \pm\sqrt{2}, i \mapsto \pm i, \sqrt{2}i \mapsto \pm\sqrt{2}i$. The reason why φ cannot send for example $\sqrt{2} \mapsto i$ is left as a fun exercise on homomorphisms. We note that $\varphi(\sqrt{2}i) = \varphi(\sqrt{2})\varphi(i)$, so what φ does to $\sqrt{2}i$ is determined by what it does to $\sqrt{2}$ and i . We then get the following table:

	1	$\sqrt{2}$	i	$\sqrt{2}i$
φ_1	+	+	+	+
φ_2	+	-	+	-
φ_3	+	+	-	-
φ_4	+	-	-	+

$$\implies \text{Gal}(\mathbb{Q}(\sqrt{2}, i)/\mathbb{Q}) = \{\varphi_1, \varphi_2, \varphi_3, \varphi_4\} \simeq \mathbb{Z}_2 \times \mathbb{Z}_2.$$

From Artin's theorem, if $\mathbb{K} \subset \mathbb{F}$ is a finite field extension and $H < \text{Aut}(\mathbb{F}/\mathbb{K})$, then $\mathbb{F}^H$ is a subfield of $\mathbb{F}$. So $\mathbb{K} \subset \mathbb{F}^H \subset \mathbb{F}$ it's an intermediate subfield. Conversely for all intermediate subfields $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$, $\text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{L}} := \{\varphi \in \text{Aut}(\mathbb{F}/\mathbb{K}) \mid \varphi|_{\mathbb{L}} = \text{id}_{\mathbb{L}}\} < \text{Aut}(\mathbb{F}/\mathbb{K})$.

Example 12.6.

1. $\mathbb{F}^{\{e\}} = \mathbb{F}, e \in \text{Aut}(\mathbb{F}/\mathbb{K})$.
2. If $\mathbb{K} \subset \mathbb{F}$ is Galois, then $\mathbb{F}^{\text{Aut}(\mathbb{F}/\mathbb{K})} = \mathbb{K}$.
3. $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}, \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{K}} = \text{Aut}(\mathbb{F}/\mathbb{K})$.
4. $\mathbb{K} \subset \mathbb{F} \subset \mathbb{F}, \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{F}} = \{e\}$.

12.1 Fundamental theorem of Galois theory

Theorem 12.7 (Galois, 1830). Let $\mathbb{K} \subset \mathbb{F}$ be a finite Galois extension, then:

- (a) $H \mapsto \mathbb{F}^H, \mathbb{L} \mapsto \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$ are mutually inverse bijections between subgroups of $\text{Aut}(\mathbb{F}/\mathbb{K})$

and intermediate subfields. Both maps reverse inclusions.

- (b) For all $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ intermediate subfields, the extension $\mathbb{L} \subset \mathbb{F}$ is Galois with $\text{Gal}(\mathbb{F}/\mathbb{L}) = \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$.
- (c) For all $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ intermediate subfields, the extension $\mathbb{L} \subset \mathbb{F}$ is Galois $\iff \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}} \triangleleft \text{Gal}(\mathbb{F}/\mathbb{K})$. If this condition is satisfied, then

$$\text{Gal}(\mathbb{L}/\mathbb{K}) \simeq \text{Gal}(\mathbb{F}/\mathbb{K}) / \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$$

Proof. Let $\mathbb{K} \subset \mathbb{F}$ be a finite Galois extension. (b): If $H < \mathbb{K} < \text{Aut}(\mathbb{F}/\mathbb{K})$, then $\mathbb{F}^{\mathbb{K}} \subset \mathbb{F}^H \implies H \mapsto \mathbb{F}^H$ is order reversing. If $\mathbb{K} \subset \mathbb{L}_1 \subset \mathbb{L}_2 \subset \mathbb{F}$, $\text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{L}_2} < \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{L}_1} \implies \mathbb{L} \mapsto \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$ is also order reversing. Now $\mathbb{K} \subset \mathbb{F}$ being Galois implies by Artin's theorem 12.4 $\mathbb{F}$ is the splitting field over $\mathbb{K}$ of some $f(x) \in \mathbb{K}[x]$ with separable irreducible factors. $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ some $f(x) \in \mathbb{L}[x]$ and $\mathbb{F}$ is the splitting field of $f(x)$ over $\mathbb{L}$, then by Artin's theorem 12.4 $\mathbb{L} \subset \mathbb{F}$ is Galois, and $\text{Gal}(\mathbb{F}/\mathbb{L}) = \text{Aut}(\mathbb{F}/\mathbb{L}) = \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$.

(a): $\mathbb{L} \mapsto \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{L}} \mapsto \mathbb{F}^{\text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}} = \mathbb{L}$, $H < \text{Aut}(\mathbb{F}/\mathbb{K})$, $\mathbb{F}^H \subset \mathbb{F}$ is Galois by Artin's theorem 12.4 $\implies \text{Aut}(\mathbb{F}/\mathbb{F}^H) = \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{F}^H} = H$, hence $H \mapsto \mathbb{F}^H \mapsto \text{Aut}(\mathbb{F}/\mathbb{K})_{\mathbb{F}^H} = H$.

$$\begin{array}{ccc} & \xrightarrow{H \mapsto \mathbb{F}^H} & \\ \{H \mid H < \text{Gal}(\mathbb{F}/\mathbb{K})\} & & \{\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}\} \\ & \xleftarrow{\text{Gal}(\mathbb{F}/\mathbb{K}_{\mathbb{L}}) \leftarrow \mathbb{L}} & \end{array}$$

so (a) and (b) are proved.

For (c), Let $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ be an intermediate subfield, $\mathbb{K} \subset \mathbb{L}$ is Galois when it is normal and separable, and $\mathbb{K} \subset \mathbb{F}$ is separable so $\mathbb{K} \subset \mathbb{L}$ is separable. Moreover $\mathbb{K} \subset \mathbb{L}$ is normal $\iff \forall K \subset \mathbb{L} \subset \mathbb{H}$, $\forall \varphi \in \text{Aut}(\mathbb{H}/\mathbb{K})$, $\varphi(\mathbb{L}) \subset \mathbb{L}$

$$\begin{array}{ccc} & \mathbb{F} & \\ \cup & & \cup \\ \mathbb{K} \subset \mathbb{L} & & \tilde{\mathbb{H}} \\ \cup & & \cup \\ & \mathbb{H} & \end{array}$$

Any automorphism φ of $\tilde{\mathbb{H}}$ which fixes $\mathbb{K}$ and sends $\mathbb{F}$ to $\mathbb{F} \implies \varphi|_{\mathbb{F}} \in \text{Aut}(\mathbb{F}/\mathbb{K}) = \text{Gal}(\mathbb{F}/\mathbb{K})$. $\mathbb{K} \subset \mathbb{L}$ is normal $\iff \forall \varphi \in \text{Gal}(\mathbb{F}/\mathbb{K})$ $\varphi(\mathbb{L}) \subset \mathbb{L} \iff \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}} \triangleleft \text{Gal}(\mathbb{F}/\mathbb{K})$. What is then left to check is that $\forall \varphi \in \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$, $\forall \sigma \in \text{Gal}(\mathbb{F}/\mathbb{K})$, $\sigma^{-1}\varphi\sigma \in \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$. If we let $a \in \mathbb{L}$ then what is left to check is $\sigma^{-1}\varphi\sigma(a) = a \iff \varphi(\sigma(a)) = \sigma(a)$. If any $\sigma \in \text{Gal}(\mathbb{F}/\mathbb{K})$ sends $\mathbb{L}$ to $\mathbb{L}$ then $\sigma(a) \in \mathbb{L} \implies \varphi(\sigma(a)) = \sigma(a)$ because $\varphi \in \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$ which implies $\text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}} \triangleleft \text{Gal}(\mathbb{F}/\mathbb{K})$.

Conversely, if $\text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}} \triangleleft \text{Gal}(\mathbb{F}/\mathbb{K})$, then $\varphi(\sigma(a)) = \sigma(a) \forall a \in \mathbb{L} \implies \sigma(a) \in \mathbb{L}$ because it is fixed by all $f \in \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$. $\mathbb{L} \subset \mathbb{F}$ is Galois, so any $a \notin \mathbb{L}$ is not fixed by some φ .

If $K \subset \mathbb{L}$ is Galois, then $\forall \sigma \in \text{Gal}(\mathbb{F}/\mathbb{K})$ sends $\mathbb{L} \rightarrow \mathbb{L}$. We then define the restriction map:

$$\begin{array}{ccc} \text{Res}_{\mathbb{K}}^{\mathbb{F}}: \text{Gal}(\mathbb{F}/\mathbb{K}) & \longrightarrow & \text{Gal}(\mathbb{L}/\mathbb{K}) \\ \cup & & \cup \\ \sigma & \longmapsto & \sigma|_{\mathbb{K}} \end{array}$$

Then $\ker \text{Res}_{\mathbb{K}}^{\mathbb{F}} = \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$. It is surjective by the extension property. By the first isomorphism theorem 10.14: $\text{Gal}(\mathbb{F}/\mathbb{K}) \simeq \text{Gal}(\mathbb{F}/\mathbb{K}) / \text{Gal}(\mathbb{F}/\mathbb{K})_{\mathbb{L}}$. $\square$

Example 12.8. $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2} + \sqrt{3}) =: \mathbb{F}$. We want to show that this extension is Galois, determine the Galois-group, and all intermediate subfields.

We note that $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ is algebraic, so $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ is algebraic. Then

$$\mathbb{Q} \underset{\deg 2}{\subset} \mathbb{Q}(\sqrt{2}) \underset{\deg 2}{\subset} \mathbb{Q}(\sqrt{2}, \sqrt{3})$$

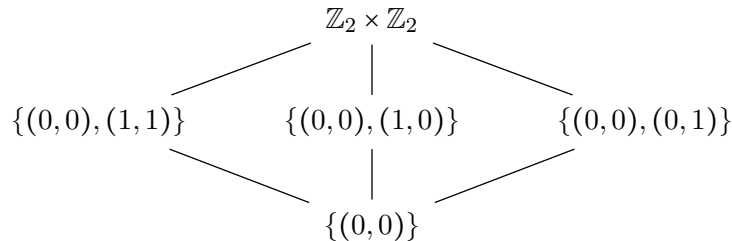
so $[\mathbb{Q}(\sqrt{2}, \sqrt{3}), \mathbb{Q}] = 2 \cdot 2 = 4 \implies |\text{Aut}(\mathbb{F}/\mathbb{Q})| = 4$. Basis of $\mathbb{F}$ over $\mathbb{Q}$ is: $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$. The $\varphi \in \text{Aut}(\mathbb{F}/\mathbb{Q})$ is uniquely determined by what it does to the basis, so similarly to example 12.5 we get a table, but this time we will present it slightly different. We note that $1 \mapsto 1$ always, and that $\varphi(\sqrt{6}) = \varphi(\sqrt{2} \cdot \sqrt{3})$ is determined by the sign of $\pm\sqrt{2}$ and $\pm\sqrt{3}$, call these signs ϵ_1 and ϵ_2 respectively. The table can then be written as

	ϵ_1	ϵ_2
φ_1	+	+
φ_2	+	-
φ_3	-	+
φ_4	-	-

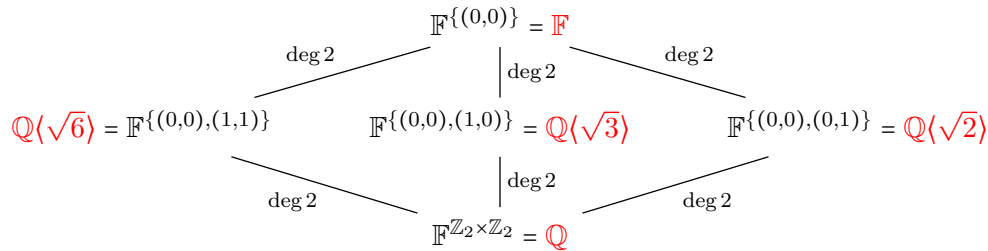
so $\text{Aut}(\mathbb{F}/\mathbb{Q}) \simeq \mathbb{Z}_2 \times \mathbb{Z}_2$. The characteristic is 0, so the extension is separable. $\mathbb{F}$ is the splitting field of $(x^2 - 2)(x^2 - 3)$ over $\mathbb{Q} \implies$ the extension is normal, thus the extension is Galois. Now for subgroups for $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0,0), (0,1), (1,0), (1,1)\}$ are:

$$\mathbb{Z}_2 \times \mathbb{Z}_2, \quad \{(0,0)\}, \quad \{(0,0), (0,1)\}, \quad \{(0,0), (1,0)\}, \quad \{(0,0), (1,1)\}$$

This is:



Subfields:



where $(0,0)$ is φ_1 , $(0,1)$ is φ_2 , $(1,0)$ is φ_3 , and $(1,1)$ is φ_4 .

Claim: Any field extension of degree 2 and characteristic 0 is Galois.

Proof. The extension is separable as characteristic is 0. $\mathbb{K} \subset \mathbb{F}$ extension of degree 2 and characteristic 0. Take any $a \in \mathbb{F}$, $a \notin \mathbb{K}$. Then $\mathbb{F} = \mathbb{K}(a)$ so $m_a(x) \in \mathbb{K}[x]$ is irreducible. $\deg m_a(x) = 2$, so $m_a(x)$ splits into $m_a(x) = (x - a)(x - b)$ $b \in \mathbb{F}$. So $\mathbb{F}$ is the splitting field of $m_a(x)$, which implies that the extension is normal, hence Galois. Moreover $\text{Gal}(\mathbb{F}/\mathbb{K}) \simeq \mathbb{Z}_2$, where φ maps $1 \mapsto 1$ and $a \mapsto b$. $\square$

$$\{\text{id}_{\mathbb{F}}, \varphi\}$$

12.2 Finite fields

Let $\mathbb{F}$ be a field with $|\mathbb{F}| < \infty$ then $\text{char } \mathbb{F} = p$, some prime $\implies \mathbb{F} \supset \mathbb{Z}_p$. We define the *Frobenius map* to be

$$\begin{array}{ccc} \Phi: \mathbb{F} & \longrightarrow & \mathbb{F} \\ \cup & & \cup \\ x & \longmapsto & x^p \end{array}$$

which is an automorphism since $|\mathbb{F}| < \infty$. Then

Theorem 12.9.

- (a) For all primenumbers p , and all $n \geq 1$, there exists a unique (up to an isomorphism) field $\mathbb{F}_{p^n}$ such that $|\mathbb{F}_{p^n}| = p^n$.
- (b) $k \mid n \implies \mathbb{F}_{p^n}$ has a unique subfield isometric to $\mathbb{F}_{p^k}$.
- (c) $k \mid n \implies \mathbb{F}_{p^k} \subset \mathbb{F}_{p^n}$ is a Galois extension whose $\text{Gal}(\mathbb{F}_{p^n} / \mathbb{F}_{p^k})$ is cyclic and is generated by Φ^n .
- (d) $\mathbb{F}_{p^n}^*$ is cycle $\implies \mathbb{F}_{p^n} = \mathbb{F}_{p^k} \langle \alpha \rangle$, the α -generator of $\mathbb{F}_{p^n}^*$.

Remark 12.10.

1. $\mathbb{F}_p = \mathbb{Z}_p$.
2. $\mathbb{F}_{p^k} \subset \mathbb{F}_{p^n} \implies k \mid n$ because $p^n = (p^k)^m = p^{p^m}$.

Proof. Fix p and n and consider $g_n(x) = x^{p^n} - x \in \mathbb{Z}_p[x]$. Let $\mathbb{F}$ be the splitting field of $g_n(x)$ over $\mathbb{Z}_p$. Then $|\mathbb{F}| < \infty$. Consider $\mathbb{F} \supset \mathbb{L} = \{a \in \mathbb{F} \mid g_n(a) = 0\} \ni 0, 1$. Then $g'_n(x) = -1 \implies (g_n, g'_n) = 1 \implies g_n$ does not have multiple roots $\implies |\mathbb{L}| = p^n$. Note that $\mathbb{L} = \{a \in \mathbb{F} \mid \Phi^n(a) = a\}$, thus $\mathbb{L}$ are the fixed points of Φ^n meaning that this is a subfield of $\mathbb{F}$. Then as $g_n(x)$ splits over $\mathbb{L}$, $\mathbb{F}$ must equal $\mathbb{L}$ by the normality of a splitting field. This implies the existence.

For uniqueness: we let $\mathbb{F}$ be a field with p^n elements. Then $\mathbb{F}^*$ is an abelian group of order $p^n - 1$. Lagrange's theorem 2.32 then implies for $\forall a \in \mathbb{F}$, $a^{p^n-1} = 1$, $a^{p^n} = a$, so all elements of $\mathbb{F}$ are roots of $g_n(x)$ which means that the splitting field of $g_n(x)$ is $\mathbb{F}$, implying that $\mathbb{F}$ is unique up to an isomorphism. This implies (a).

For (b), let $k \mid n$, and consider $\mathbb{F}_{p^n}$. We then claim that any root of $g_k(x)$ is a root of $g_n(x)$.

Proof.

$$a^{p^m} = a \implies a^{p^n} = a^{p^{k \cdot m}} = a^{\underbrace{p^k p^k \dots p^k}_m} = (a^{p^k})^{\underbrace{p^k p^k \dots p^k}_{m-1}} = a^{\underbrace{p^k p^k \dots p^k}_m} = a.$$

So the set of all roots of $g_k(x)$ over $\mathbb{F}_{p^k}$ is a subset of all roots of $g_n(x)$ over $\mathbb{F}_{p^n}$. This $\mathbb{F}_{p^k} \subset \mathbb{F}_{p^n}$, which is unique up to some isomorphism (as a subset). $\square$

(c) we assume $k \mid n$ and $\mathbb{F}_{p^k} \subset \mathbb{F}_{p^n}$ is a finite field extension. $\mathbb{F}_{p^n}$ is the splitting field of the separable polynomial $g_n(x)$ which implies that the extension is separate and also normal as it is a splitting field. Thus the extension is Galois. Moreover $|\text{Gal}(\mathbb{F}_{p^n} / \mathbb{F}_{p^k})| = [\mathbb{F}_{p^n} : \mathbb{F}_{p^k}] = \frac{n}{k}$. $\Phi^k \in \text{Gal}(\mathbb{F}_{p^n} / \mathbb{F}_{p^k})$ equals $\text{id}_{\mathbb{F}_{p^k}}$, and $\text{id}_{\mathbb{F}_{p^n}} = \Psi^n$ which thus equals $(\Phi^k)^{\frac{n}{k}}$. Then the order of Φ^k in $\text{Gal}(\mathbb{F}_{p^n} / \mathbb{F}_{p^k})$ is $\frac{n}{k} \implies \text{Gal}(\mathbb{F}_{p^n} / \mathbb{F}_{p^k}) = \langle \Phi^k \rangle \implies$ (c).

(d) we consider $\mathbb{F}_{p^n}^*$ and note that $|\mathbb{F}_{p^n}^*| = p^n - 1$, $\mathbb{F}_{p^n}^*$ is a finite abelian group and $p^n - 1 = q_1^{k_1} \dots q_l^{k_l}$, where $q_i \neq q_j$ are primes. Now let G_i be the sylow q_i -subgroup of $\mathbb{F}_{p^n}^*$, so then $\mathbb{F}_{p^n}^* \simeq G_1 \times \dots \times G_l$, where

$G_i \simeq \mathbb{Z}_{q_i^{r_1}} \times \cdots \times \mathbb{Z}_{q_i^{r_{f_i}}}$. What is then left to prove is that G_i is cyclic as then by the Chinese remainder theorem 8.6 implies $\mathbb{F}_{p^n}^*$ is cyclic. We look for the maximal order of an element in $\mathbb{Z}_{q_i^{r_1}} \times \cdots \times \mathbb{Z}_{q_i^{r_{f_i}}}$ which is $q_i^{\max(r_1, \dots, r_{f_i})}$. Then $\forall a \in \mathbb{F}_{p^n}^*, a^{\prod_{i=1}^l \max(r_1^{(i)}, \dots, r_{f_i}^{(i)})} = 1 \implies a^{1 + \prod_{i=1}^l \max(r_1^{(i)}, \dots, r_{f_i}^{(i)})} = a \implies 1 + \prod_{i=1}^l \max(r_1^{(i)}, \dots, r_{f_i}^{(i)}) \geq p^n \implies \prod_{i=1}^l \max(r_1^{(i)}, \dots, r_{f_i}^{(i)}) \geq p^n - 1 \implies r_{f_i}^{(i)} = 1$ and $r_1^{(i)} = k_i \implies$ (d). $\square$

Corollary 12.11. For all primes p , and all $n, k \geq 1$, there exists an irreducible polynomial of degree k over $\mathbb{F}_{p^n}$.

Proof. The field extension $\mathbb{F}_{p^n} \subset \mathbb{F}_{p^{nk}}$ is of degree k . Then choose $\alpha \in \mathbb{F}_{p^{nk}}$ such that $\langle \alpha \rangle = \mathbb{F}_{p^{nk}}^*$ which implies $\mathbb{F}_{p^{nk}} = \mathbb{F}_{p^n} \langle \alpha \rangle$. Then $m_\alpha(x) \in \mathbb{F}_{p^n}[x]$ is irreducible and of degree k . $\square$

Example 12.12.

- Let us consider the finite field $\mathbb{F}_4$. $4 = 2^2$ so $\mathbb{Z}_2 \subset \mathbb{F}_4$ is a field extension of degree 2. $\mathbb{Z}_2 = \{0, 1\}$ and the addition and multiplication tables of $\mathbb{Z}_2$ are.

+	0	0
0	0	1
1	1	0

·	0	0
0	0	0
1	0	1

then $\mathbb{F}_4 \simeq \mathbb{Z}_2[x] / (f(x))$, where $f(x)$ is some irreducible polynomial of degree 2. The polynomials of degree 2 over $\mathbb{Z}_2$ are

$$x^2 \quad x^2 + 1 \quad x^2 + x \quad x^2 + x + 1$$

We can immediately cross out $x^2 = x \cdot x$, $x^2 + x = x(x + 1)$, and also $x^2 + 1 = (x + 1)^2$. We can check if $x^2 + x + 1$ is irreducible by testing with $x = 0, 1$: $0^2 + 0 + 1 = 1 \neq 0$ and $1^2 + 1 + 1 = 1 \neq 0$. No root, hence irreducible, thus $\mathbb{F}_4 \simeq \mathbb{Z}_2[x] / (x^2 + x + 1) = \{\bar{0}, \bar{1}, \bar{x}, \overline{x+1}\}$. The addition and multiplication table for $\mathbb{F}_4$ are then

+	$\bar{0}$	$\bar{1}$	$\bar{x}$	$\overline{x+1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{x}$	$\overline{x+1}$
$\bar{1}$	$\bar{1}$	$\bar{0}$	$\overline{x+1}$	$\bar{x}$
$\bar{x}$	$\bar{x}$	$\overline{x+1}$	$\bar{0}$	$\bar{1}$
$\overline{x+1}$	$\overline{x+1}$	$\bar{x}$	$\bar{1}$	$\bar{0}$

·	$\bar{0}$	$\bar{1}$	$\bar{x}$	$\overline{x+1}$
$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{1}$	$\bar{0}$	$\bar{1}$	$\bar{x}$	$\overline{x+1}$
$\bar{x}$	$\bar{0}$	$\bar{x}$	$\overline{x+1}$	$\bar{1}$
$\overline{x+1}$	$\bar{0}$	$\overline{x+1}$	$\bar{1}$	$\bar{x}$

so $\mathbb{F}_4^* = \langle \bar{x} \rangle = \langle \overline{x+1} \rangle = \{\bar{1}, \bar{x}, \overline{x+1}\} \simeq \mathbb{Z}_3$. The equalities come from the fact that $\bar{x} \cdot \bar{x} = \overline{x^2} = \overline{x+1}$ and $\bar{x} \cdot \overline{x+1} = \bar{1}$. So the ideal $(y^2 + y + 1) = (y - \bar{x})(y - \overline{x+1})$ (writing y instead of x in order to avoid confusion). Moreover $\Phi = (\bar{x}) = \overline{x^2} = \overline{x+1}$.

- $\mathbb{Z}_2 \subset \mathbb{F}_8$. Similar to the first example, we look for an irreducible polynomial of degree 3 ($8 = 2^3$) over $\mathbb{Z}_2$ which are

$$\begin{array}{ll}
 x^3 & x^3 + x^2 \\
 x^3 + 1 & x^3 + x^2 + x \\
 x^3 + x & x^3 + x^2 + 1 \\
 x^3 + x + 1 & x^3 + x^2 + x + 1
 \end{array}$$

the easiest method in this situation for finding the irreducible polynomials would be to plug in 0 or 1 and see if the result is 0 or not. By doing that we get that $x^3 + x + 1$ and $x^3 + x^2 + 1$ are the only irreducible ones of degree 3 over $\mathbb{Z}_2$. Then $\mathbb{F}_8 \simeq \mathbb{Z}_2[x] / (x^3 + x + 1)$ (one can choose any of the two). Leaving the addition table as an exercise, the multiplication table then becomes

$\cdot$	$\bar{0}$	$\bar{1}$	$\bar{x}$	$\overline{x+1}$	$\overline{x^2}$	$\overline{x^2+1}$	$\overline{x^2+x}$	$\overline{x^2+x+1}$
$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{1}$	$\bar{0}$	$\bar{1}$	$\bar{x}$	$\overline{x+1}$	$\overline{x^2}$	$\overline{x^2+1}$	$\overline{x^2+x}$	$\overline{x^2+x+1}$
$\bar{x}$	$\bar{0}$	$\bar{x}$	$\overline{x^2}$	$\overline{x^2+x}$	$\overline{x+1}$	$\bar{1}$	$\overline{x^2+x+1}$	$\overline{x^2+1}$
$\overline{x+1}$	$\bar{0}$	$\overline{x+1}$	$\overline{x^2+x}$	$\overline{x^2+1}$	$\overline{x^2+x+1}$	$\overline{x^2}$	$\bar{1}$	$\bar{x}$
$\overline{x^2}$	$\bar{0}$	$\overline{x^2}$	$\overline{x+1}$	$\overline{x^2+x+1}$	$\overline{x^2+x}$	$\bar{x}$	$\overline{x^2+x}$	$\bar{1}$
$\overline{x^2+1}$	$\bar{0}$	$\overline{x^2+1}$	$\bar{1}$	$\overline{x^2}$	$\bar{x}$	$\overline{x^2+x+1}$	$\overline{x+1}$	$\overline{x^2+x}$
$\overline{x^2+x}$	$\bar{0}$	$\overline{x^2+x}$	$\overline{x^2+x+1}$	$\bar{1}$	$\overline{x^2+1}$	$\overline{x+1}$	$\bar{x}$	$\overline{x^2}$
$\overline{x^2+x+1}$	$\bar{0}$	$\overline{x^2+x+1}$	$\overline{x^2+1}$	$\bar{x}$	$\bar{1}$	$\overline{x^2+x}$	$\overline{x^2}$	$\overline{x+1}$

hence $\mathbb{F}_8^* \simeq \mathbb{Z}_7$, where all elements except $\bar{1}$ generate $\mathbb{F}_8^*$ as $\bar{x}ab = 1 \implies ab = \overline{x^2+1}$, and so $(y^3 + y + 1) = (y - \bar{x})(y - a)(y - b)$ splits as the extension is Galois (separable and normal). Moreover $(y^3 + y + 1) = (y - \overline{x+1})(y - d)(y - f)$ etc. (the rest are left as an exercise).

13 Algebraic closure

Recall that a field $\mathbb{K}$ is algebraically closed if for all extensions $\mathbb{K} \subset \mathbb{F}$, we have $\mathbb{F} = \mathbb{K}$, or the alternative definition: any $f(x) \in \mathbb{K}[x]$, $\deg f > 0$ has a root in $\mathbb{K}$.

Definition 13.1. A field extension $\mathbb{K} \subset \mathbb{F}$ is called an *algebraic closure* (of $\mathbb{K}$) if

- (a) this extension is algebraic and
- (b) $\mathbb{F}$ is algebraically closed.

Example 13.2.

1. $\mathbb{R} \subset \mathbb{C}$ is an algebraic closure.
2. $\mathbb{C} \subset \mathbb{C}$ is an algebraic closure.
3. $\mathbb{Q} \subset \mathbb{C}$ is not an algebraic closure. It is not algebraic.

Theorem 13.3. Any field admits an algebraic closure. Moreover it is unique up to an isomorphism that is the identity on the original field.

If $\mathbb{K}$ is a field algebraic closure set of $\mathbb{K}$ is the set of all algebraic elements over $\mathbb{K}$. In order to prove the theorem we need the Axiom of Choice which states

Axiom of Choice 13.4. If x_i , $i \in I$ are non-empty sets, then the Cartesian product $\prod_{i \in I} x_i$ is non-empty.

Without giving a proof, the Axiom of Choice turns out to be equivalent to

Lemma 13.5 (Zorn's lemma). Let $(M, \leq)$ be a partially ordered set. Assume that $\forall N < M$ are totally ordered, that is $\forall a, b \in N$ either $a \leq b$ or $b \leq a$, and that there $\exists q \in M$ such that $a \leq q \forall a \in N$. Then M contains a maximal element. I.e. there exist $m \in M$ such that $\forall a \in M, m \leq a \implies a = m$.

and also equivalent to

Theorem 13.6 (Well-ordering theorem). Every set can be totally ordered such that each non-empty subset has a minimal element.

proof of the first theorem using Zorn's lemma. The existence of algebraic closure: Let $\mathbb{K}$ be a field and $|\mathbb{K}| = \kappa$ some cardinal. Define $\mathcal{M}$ to be the set of all monic irreducible polynomials over $\mathbb{K}$, and $\mathcal{Q}$ to be the set of all roots of all these polynomials in $\mathcal{M}$.

Remark 13.7. $|\mathbb{K}| < \infty \implies |\mathcal{Q}| = \omega$ and $|\mathbb{K}| < \infty \implies |\mathcal{Q}| = \kappa$

$|\mathcal{Q}| \leq \max\{\omega, \kappa\}$. Let $\mathcal{S}$ be a set of cardinality ν such that $\nu > \omega, \kappa$. Let $\mathcal{F}$ be the set of all algebraic extensions of $\mathbb{K}$ for which the underlying set is $\mathcal{S}$, then $\mathcal{F}$ is a coset with respect to inclusions. For any totally ordered set $\mathbb{F}_i$, with $\mathbb{K} \subset \mathbb{F}_i$ being an algebraic extension where $\mathbb{F}_i \subset \mathbb{F}_j$ $i \leq j$,

$$\mathbb{K} \subset \mathbb{F}_1 \subset \mathbb{F}_2 \subset \dots \subset \bigcup_i \mathbb{F}_i \subseteq \mathcal{S}$$

which is an upperbound $\{\mathbb{F}_i\}$. By Zorn's lemma 13.5, there exists a maximal element $\mathbb{K} \subset \mathbb{F}$ in $\mathcal{F}$, we then claim that this is an algebraic closure.

Proof.

$|\mathbb{F}| \leq |\mathcal{Q}| \leq \max\{\omega, \kappa\} \leq \nu = |\mathcal{S}|$. If a is algebraic over $\mathbb{K}$ and $a \notin \mathbb{F}$, then we can write the field extension $\mathbb{K} \subset \mathbb{F} \subset \mathbb{F}\langle a \rangle$, where $|\mathbb{F}\langle a \rangle| < \nu$. We can realise the extension inside $\mathcal{S}$. I.e. we can assume that $\mathbb{K} \subset \mathbb{F}\langle a \rangle \in \mathcal{F}$, which is a contradiction to the maximality of $\mathbb{F}$. This implies that $\mathbb{F}$ contains all elements over $\mathbb{K}$ and any $z \in \mathbb{F}$ is algebraic over $\mathbb{K} \implies \mathbb{K} \subset \mathbb{F}$ is an algebraic closure. $\square$

For uniqueness: we assume $\mathbb{K} \subset \mathbb{F}$ and $\mathbb{K} \subset \mathbb{L}$ are algebraic closures. By the extension of homomorphisms theorem 11.24, there exists $\mathbb{L} \hookrightarrow \mathbb{L}'$ and $\mathbb{F} \hookrightarrow \mathbb{L}$ such that

$$\begin{array}{ccc} \mathbb{L} & \xrightarrow{\delta} & \mathbb{L}' \\ \beta \uparrow & & \uparrow \gamma \\ \mathbb{K} & \xrightarrow{\alpha} & \mathbb{F} \end{array}$$

commutes. Moreover $\gamma(\mathbb{F})$ is algebraic over $\gamma \circ \alpha(\mathbb{K})$, $\delta(\mathbb{L})$ is algebraic over $\delta \circ \beta(\mathbb{K}) = \gamma \circ \alpha(\mathbb{K})$, implying $\delta(\mathbb{L}) \subset \gamma(\mathbb{F})$ and $\gamma(\mathbb{F}) \subset \delta(\mathbb{L}) \implies \gamma(\mathbb{F}) = \delta(\mathbb{L}) \implies \mathbb{F} \simeq \mathbb{L}$. $\square$

Notation 13.8. $\overline{\mathbb{K}}$ is the algebraic closure of $\mathbb{K}$.

Remark 13.9. If $|\mathbb{K}| = \infty$ then $|\overline{\mathbb{K}}| = \infty$, and if $|\mathbb{K}| < \infty$ then $|\overline{\mathbb{K}}| = \omega$.

Remark 13.10. $|\overline{\mathbb{Q}}| = |\mathbb{Q}| = \omega$, $\overline{\mathbb{Q}} \subset \mathbb{C}$.

We let $\bar{\cdot} : \text{Fields} \rightarrow \text{Fields}$ be the *closure operator* mapping $\mathbb{K} \mapsto \overline{\mathbb{K}}$.

Proposition 13.11.

1. $\overline{\overline{\mathbb{K}}} = \overline{\mathbb{K}}$.
2. $\mathbb{K} \subset \overline{\mathbb{K}}$.
3. $\mathbb{K} \subset \mathbb{F} \implies \overline{\mathbb{K}} \subset \overline{\mathbb{F}}$.

Together they imply that $\bar{\cdot}$ is a (well defined) *closure operator*.

Let $\mathbb{K} \subset \mathbb{F}$ be a field extension.

Definition 13.12. A subset $Q \subset \mathbb{F}$ is called *algebraically independent over $\mathbb{K}$* if all finite, non-empty sets

$$\{q_1, q_2, \dots, q_k\} \subset Q \quad q_i \neq q_j \quad i \neq j$$

and all non-zero polynomials $f(x_1, x_2, \dots, x_k) \in \mathbb{K}[x_1, \dots, x_k]$, we have $f(q_1, q_2, \dots, q_k) \neq 0$.

Example 13.13.

1. $q \in \mathbb{F}$ is algebraically independent if and only if it's transcendental.
2. $\mathbb{K} \subset \mathbb{K}(x) \subset (\mathbb{K}(x))(y) = \mathbb{K}(x, y)$, $\{x, y\}$ is algebraic independent.

13.1 Transcendence degree

Let $\mathbb{K} \subset \mathbb{F}$ be a field extension.

Definition 13.14. The *transcendence degree* of this extension is the maximal cardinality of an algebraic independent set.

Notation 13.15. $\text{Trdeg}_{\mathbb{K}}(\mathbb{F})$

Example 13.16.

1. $\text{Trdeg}_{\mathbb{K}}(\mathbb{F}) = 0 \iff \mathbb{K} \subset \mathbb{F}$ is algebraic.
2. $\text{Trdeg}_{\mathbb{K}}(\mathbb{K}(x)) = 1$.

Lemma 13.17 (Steinitz Exchange lemma). Let $\mathbb{K} \subset \mathbb{F}$ be a field extension, and $Q_1, Q_2 \subset \mathbb{F}$ be two algebraic independent sets that are maximal (with respect to inclusions). Then $|Q_1| = |Q_2|$.

If $\mathbb{K} \subset \mathbb{F}$ is a field extension and $Q \subset \mathbb{F}$ is an algebraic independent set that is maximal with respect to inclusions, then

$$\mathbb{K} \subset \mathbb{K}\langle Q \rangle \underset{\text{transcendental}}{=} \mathbb{L} \subset \mathbb{F} \underset{\text{algebraic}}{.}$$

Moreover, if $\mathbb{Q}$ or $\mathbb{Z}_p \subset \mathbb{K} \subset \bar{\mathbb{K}}$, then all algebraically closed fields $\mathbb{F}$ are the algebraic closure of some purely transcendental extension

$$\mathbb{Q} \text{ or } \mathbb{Z}_p \subset \mathbb{Q} \text{ or } \mathbb{Z}_p(M)$$

where M is an algebraically independent set.

Corollary 13.18. For any uncountable cardinality κ , and any characteristic, there exists and unique algebraically closed field of that cardinality and that characteristic.

Example 13.19.

1. The field extension $\mathbb{Q} \subset \mathbb{C}$:

$$\mathbb{Q} \subset \mathbb{Q}(M) = \mathbb{L} \subset \mathbb{C} \underset{\text{algebraic}}{\quad} \quad |\mathbb{L}| = |\mathbb{C}| = c \implies |M| = c.$$

2. For $\mathbb{C} \subset \mathbb{C}(x) \subset \overline{\mathbb{C}(x)}$ each admits an algebraic closure, where $|\overline{\mathbb{C}(x)}| = |\mathbb{C}(X)| = |\mathbb{C}| = c$. This implies that $\mathbb{C} \simeq \overline{\mathbb{C}(x)}$, but $\mathbb{C} \not\subset \overline{\mathbb{C}(x)} \simeq \mathbb{C}$. Takeaway is that $\mathbb{C}$ has many endomorphisms.

Assume $\mathbb{F}$ and $\mathbb{L}$ are algebraically closed, have cardinality w and both have characteristic 0. Does this imply $\mathbb{F} \simeq \mathbb{L}$?

Example 13.20. $\overline{\mathbb{Q}}$ satisfies these three conditions, same with $\overline{\mathbb{Q}(x)}$ and $\overline{\overline{\mathbb{Q}(x)}}$. Can these be isomorphic? $\mathbb{Q} \subset \overline{\mathbb{Q}}$ is algebraic, but $\mathbb{Q} \subset \overline{\mathbb{Q}(x)}$ is not an algebraic extension, as $\overline{\mathbb{Q}} \not\subset \overline{\mathbb{Q}(x)}$.

14 Simple radical extensions

Let $\mathbb{K} \subset \mathbb{F}$ be a field extension.

Definition 14.1. $\mathbb{K} \subset \mathbb{F}$ is called a *simple radical extension* (SRE) if there exists some $b \in \mathbb{F}$, $a \in \mathbb{K}$ and $m \in \mathbb{Z}^+$ such that

1. $\mathbb{F} = \mathbb{K}\langle b \rangle$.
2. $b^m = a$.

We do not require m to be minimal.

Example 14.2.

1. $\mathbb{K} \subset \mathbb{K}$ is an SRE.
2. $\mathbb{R} \subset \mathbb{C}$ is also an SRE. Take $b = i$, $a = -1$ and $m = 2$.

Remark 14.3. Any SRE is finite and algebraic.

14.1 Radical extension

Definition 14.4. A field extension $\mathbb{K} \subset \mathbb{F}$ is a *radical extension* (RE) if there exists a flag of intermediate subfield

$$\mathbb{K} \subset \mathbb{L}_0 \subset \mathbb{L}_1 \subset \mathbb{L}_2 \subset \cdots \subset \mathbb{L}_r = \mathbb{F}$$

such that each $\mathbb{L}_i \subset \mathbb{L}_{i+1}$ is a SRE.

Remark 14.5.

1. A simple radical extension is a radical extension.
2. A radical extension is finite and algebraic.
3. If $\mathbb{K} \subset \mathbb{L}$ and $\mathbb{L} \subset \mathbb{F}$ are radical extensions, then $\mathbb{K} \subset \mathbb{F}$ is a radical extension.

Proposition 14.6. $\mathbb{K} \subset \mathbb{F}$ is a radical extension and $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ be an intermediate subfield, then $\mathbb{L} \subset \mathbb{F}$ is a radical extension.

Proof. $\mathbb{K} \subset \mathbb{F}$ is a radical extension then there exists a flag: $\mathbb{K} \subset \mathbb{M}_0 \subset \cdots \subset \mathbb{M}_r = \mathbb{F}$ such that $\mathbb{M}_i \subset \mathbb{M}_{i+1}$ is a simple radical extension, that is $\mathbb{M}_{i+1} = \mathbb{M}_i\langle b_i \rangle$, where $b_i^{m_i} = a_i$. To show $\mathbb{L} \subset \mathbb{F}$ is a RE take

$$\mathbb{L} = \langle \mathbb{L}, \mathbb{K} \rangle \subset \langle \mathbb{L}, \mathbb{M}_1 \rangle \subset \cdots \subset \langle \mathbb{L}, \mathbb{F} \rangle = \mathbb{F}$$

where $\langle \mathbb{L}, \mathbb{K} \rangle$ is the minimal subfield generated by $\mathbb{L}$ and $\mathbb{K}$. The trivial steps are simple radical extensions. For the non-trivial steps, take the same b_1 , a_i and m_i . Then

$$\langle L, \mathbb{M}_i \rangle \subset \langle \mathbb{L}, M_{i+1} \rangle$$

is a simple radical extension. □

Proposition 14.7. Let $\mathbb{K} \subset \mathbb{F}$ be a field extension, and $\mathbb{K} \subset \mathbb{L} \subset \mathbb{F}$ & $\mathbb{K} \subset \mathbb{M} \subset \mathbb{F}$ intermediate subfields. Moreover let $\mathbb{T} = \langle \mathbb{L}, \mathbb{M} \rangle$. If both $\mathbb{K} \subset \mathbb{L}$ and $\mathbb{K} \subset \mathbb{M}$ are radical extensions, then $\mathbb{K} \subset \mathbb{T}$ is a radical extension as well.

Proof. Since we can "glue" together radical extensions, it is enough to show that $\mathbb{L} \subset \mathbb{T}$ is a radical extension. If $\mathbb{K} \subset \mathbb{M}$ is a radical extension, then there exists a flag

$$\mathbb{K} = \mathbb{Y}_0 \subset \mathbb{Y}_1 \subset \cdots \subset \mathbb{Y}_r = \mathbb{M}$$

of simple radical extensions ($\mathbb{Y}_i \subset \mathbb{Y}_{i+1}$). That is $\mathbb{Y}_{i+1} = \mathbb{Y}_i \langle b_i \rangle$, where $b_i^{m_i} = a_i$, by definition.

$$\mathbb{L} = \langle \mathbb{L}, \mathbb{K} \rangle \subset \langle \mathbb{L}, \mathbb{Y}_1 \rangle \subset \cdots \subset \langle \mathbb{L}, \mathbb{M} \rangle = \mathbb{T}$$

where

$$\langle L, \mathbb{Y}_i \rangle \subset \langle \mathbb{L}, \mathbb{Y}_{i+1} \rangle$$

is a simple radical extension by taking b_i , a_i and m_i , thus each field extension is a simple radical extension. □

14.2 Solvable by radicals

Let $\mathbb{K}$ be a field and $f \in \mathbb{K}[x]$ be some polynomial. Moreover, let $\mathbb{F}$ be the splitting field of $f(x)$. that is $\mathbb{F} = \mathbb{K}(\text{"all roots of } f(x)\text{"})$

Definition 14.8. We say that $f(x) = 0$ is *solvable by radicals* if $\mathbb{F}$ is an intermediate subfield of some radical extension $\mathbb{K} \subset \mathbb{L}$.

Remark 14.9. There is no expectation that there is an algorithm to find these roots.

Theorem 14.10 (Galois Criterion). Let $\mathbb{K}$ be a field, $f(x) \in \mathbb{K}[x]$, and $\mathbb{F}$ the splitting field of $f(x)$. Then $f(x) = 0$ is solvable by radicals if and only if $\text{Gal}(\mathbb{F}/\mathbb{K})$ is solvable.

Theorem 14.11 (Lagrange's theorem). Assume that $\mathbb{K} \subset \mathbb{F}$ is a Galois extension such that

1. $\text{Gal}(\mathbb{F}/\mathbb{K})$ is cyclic of order n .
2. $\mathbb{K}$ contains a primitive n -th root of 1, say ζ , where $\zeta^n = 1$ but $\zeta^i \neq 1$ for $i < n$.

Then $\mathbb{K} \subset \mathbb{F}$ is a simple radical extension.

Proof. Let $\sigma \in \text{Gal}(\mathbb{F}/\mathbb{K})$ be a generator, that is $\sigma^n = \text{id}$ and $\sigma^i \neq \text{id}$ for all $i < n$. Then we know that $\sigma^1, \sigma^2, \dots, \sigma^{n-1}$ are linearly independent, so there exists some $a \in \mathbb{F}$ such that $0 \neq \theta = a + \zeta\sigma(a) + \zeta^2\sigma^2(a) + \cdots + \zeta^{n-1}\sigma^{n-1}(a)$. Then

$$\sigma(\theta) = \sigma\left(\sum_{i=1}^{n-1} \zeta^i \sigma^i(a)\right) = \sigma(a) + \zeta\sigma^2(a) + \zeta^2\sigma^3(a) + \cdots + \zeta^{n-1}\sigma^n(a) = \zeta^{-1}\theta$$

$\implies \sigma^i(\theta) = \zeta^{-i}\theta \neq \theta$ for all $1 \leq i < n$. Thus the only element of $\text{Gal}(\mathbb{F}/\mathbb{K})$ that fixes θ is id. This implies by the Galois theorem 12.7 that $\mathbb{F} = \mathbb{K}(\theta)$.

$$\sigma(\theta^n) = \sigma(\theta)^n = \zeta^n \theta = \theta$$

$\implies \theta^n$ is fixable by all $\text{Gal}(\mathbb{F}/\mathbb{K}) \implies \theta^n \in \mathbb{K} \implies \mathbb{K} \subset \mathbb{F}$ is a radical extentsion

□

Corollary 14.12. Let $\mathbb{K} \subset \mathbb{F}$ be a finite Galois extension with $[\mathbb{F} : \mathbb{K}] = n$ and assume $\zeta \in \mathbb{K}$ where $\zeta^n = 1$ but $\zeta^i \neq 1$ for $1 \leq i < n$. Then $\mathbb{K} \subset \mathbb{F}$ is a radical extension if and only if $\text{Gal}(\mathbb{F}/\mathbb{K})$ is solvable.

Proof. For " $\Leftarrow$ ", we assume that $\text{Gal}(\mathbb{F}/\mathbb{K}) = G$ is solvable, that is, there exists a subnormal series:

$$\{e\} = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \dots \triangleleft G_r = G$$

with abelian subquotients, and also with cycling quotients. The Galois correspondents are

$$\mathbb{K} = \mathbb{F}_0 \subset \underset{\mathbb{F}^{G_r}}{\mathbb{F}_1} \subset \dots \subset \underset{\mathbb{F}^{G_1}}{\mathbb{F}_r} = \mathbb{F}$$

where $\mathbb{F}_i \subset \mathbb{F}_{i+1}$ corresponds to $G_j \triangleleft G_{j+1}$.

$$[\mathbb{F}_{i+1} : \mathbb{F}_i] = \underbrace{\frac{|G_{i+1}|}{|G_i|}}_{=l} \mid |G| = n$$

$\mathbb{K}$ contains a primitive n -th root of 1 and so $\mathbb{K}$ also contains a primitive l -th root of 1 as $l \mid n$. By Lagranges theorem 14.11 $\mathbb{F}_i \subset \mathbb{F}_{i+1}$ is a simple radical extension, and so $\mathbb{K} \subset \mathbb{F}$ is a radical extension.

" $\implies$ " is at the moment left as an exercise. □

Remark 14.13. $x^n - 1$ is always solvable by radicals. $\mathbb{K} \subset \mathbb{F}$ is a simple radical extension, where $\mathbb{F}$ is $\mathbb{K}$ together with the root of the primitive n -th root of 1 of the n -th cyclotomic polynomial to $\mathbb{K}$.

$\varphi \in \text{Gal}(\mathbb{F}/\mathbb{K})$ is uniquely determined by $\varphi(\zeta)$ where $\zeta^n = 1$ but $\zeta^i \neq 1$ for $1 \leq i < n$. $\varphi(\zeta) = \zeta^d$, where $(d, n) = 1$. So $\text{Gal}(\mathbb{F}/\mathbb{K})$ is abelian. Then $\varphi \mapsto d \in \mathbb{Z}_n^*$ is a homomorphism. This implies that $\text{Gal}(\mathbb{F}/\mathbb{K}) < \mathbb{Z}_n^*$, where $|\mathbb{Z}_n^*| = \text{"Euler function of } n, \text{ that is the number of elements relatively prime to } n$.

We are now ready to prove Galois criterium 14.10.

Proof. Assume G is solvable and $|G| = n$. Define $\mathbb{L} := \mathbb{K}(\zeta)$ where $\zeta^n = 1$ but $\zeta^i \neq 1$ for $1 \leq i < n$. Then $\mathbb{K} \subset \mathbb{L}$ is a radical extension. Then there exists some $\mathbb{P}$

$$\begin{array}{ccc} & \mathbb{L} & \\ \swarrow & & \searrow \\ \mathbb{K} & & \mathbb{P} \\ \swarrow & & \searrow \\ & \mathbb{F} & \end{array}$$

and let $\mathbb{F}'(\mathbb{F}, \mathbb{L}) \subset \mathbb{P}$, then $\mathbb{L} \subset \mathbb{F}'$ is Galois with $\text{Gal}(\mathbb{F}'/\mathbb{L}) = \text{Gal}(\mathbb{F} \cap \mathbb{L}/\mathbb{K})$ being solvable. Then $\text{Gal}(\mathbb{F} \cap \mathbb{L}/\mathbb{K}) \subset \text{Gal}(\mathbb{F}/\mathbb{K})$ is also solvable with

$$|\text{Gal}(\mathbb{F} \cap \mathbb{L}/\mathbb{K})| \mid |G| = n.$$

Moreover $\mathbb{K} \subset \mathbb{L}$ is a radical extension and $\mathbb{L} \subset \mathbb{F}'$ is a Galois extension, where $\mathbb{L}$ contains the primitive root of 1 whose degree divides the order of $G \implies \mathbb{K} \subset \mathbb{L} \subset \mathbb{F}'$ is also a radical extension by corollary 14.12, implying $\mathbb{K} \subset \mathbb{F}'$ is a radical extension. Then $\mathbb{F} \subset \mathbb{F}' \implies \varphi(x) = 0$ is solvable by radicals.

Now assume $g(x) = 0$ is solvable by radicals, then $\mathbb{K} \subset \mathbb{F} \subset \mathbb{L}$ is a radical extension. We can then write the flag of simple radical extensions

$$\mathbb{K} := \mathbb{L}_0 \subset \mathbb{L}_1 \subset \dots \subset \mathbb{L}_r \subset \mathbb{L}.$$

Assume $[\mathbb{L}, \mathbb{K}] = n$ and $\mathbb{K}$ contains the primitive n -th root of 1. We then do induction on r :

$r = 1 \implies \mathbb{K} \subset \mathbb{L}$ is a simple radical extension. $x^m = a$. This implies $\text{Gal}(\mathbb{L}/\mathbb{K})$ is abelian, thus solvable.

Induction step: Let $\mathbb{L}'_1 = \langle \mathbb{L}_1, \mathbb{F} \rangle$ then $\mathbb{L}_1 \text{ is Galois with } \text{Gal}(\mathbb{L}'_1/\mathbb{L}_1) \simeq \text{Gal}(\mathbb{F}/\mathbb{L}_1 \cap \mathbb{F}) =: \mathbb{H}$. $\mathbb{K} \subset \mathbb{L}_1$ is Galois with abelian Galois group implying $\mathbb{H} \triangleleft G = \text{Gal}(\mathbb{F}/\mathbb{K})$, with G/H being abelian. By induction H is solvable by radicals. As $\mathbb{K}$ is arbitrary, consider $\mathbb{K} \subset \mathbb{K}(\zeta)$ and $\mathbb{F} \subset \mathbb{F}(\zeta)$, where ζ is the primitive n -th root. Then $\mathbb{K}(\zeta) \subset \mathbb{F}(\zeta)$ is Galois and gives the group $\text{Gal}(\mathbb{F}/\mathbb{K} \cap \mathbb{F})$. Then $\mathbb{K} \subset \mathbb{K}(\zeta)$ is Galois and abelian group thus also solvable. $\square$

14.3 Polynomials with $\text{Gal} \simeq S_n$

Let $\mathbb{K} \subset \mathbb{K}\langle t_1, \dots, t_n \rangle = \mathbb{L}$ be a field extension and $f(x) = x^n + t_1 x^{n-1} + t_2 x^{n-2} + \dots + t_n \in \mathbb{L}[x]$. We want $f(x) = 0$ over $\mathbb{L}$. Let $\mathbb{F}$ be the splitting field, and assume we have solutions $\alpha_1, \dots, \alpha_n$. We now claim that $\alpha_1, \dots, \alpha_n$ are algebraically independent implying $\mathbb{F} \simeq \mathbb{L}\langle \alpha_1, \dots, \alpha_n \rangle$

Vietta's formula 14.14.

$$\begin{aligned} -t_1 &= \alpha_1 + \alpha_2 + \dots + \alpha_n \\ t_2 &= \sum_{i < j} \alpha_i \alpha_j \\ -t_3 &= \sum_{i < j < k} \alpha_i \alpha_j \alpha_k \\ &\vdots \end{aligned}$$

Any permutation of $\alpha_1, \dots, \alpha_n$ gives automorphisms of $\mathbb{F}$ which fixes all t_i , thus $S_n \subseteq \text{Gal}(\mathbb{F}/\mathbb{L})$.
(but actually $\simeq$)

15 Ruler and compass construction

With ruler and compass we consider the plane $\mathbb{R}^2$ and some finite set of points $P \in \mathbb{R}^2$. We allow ourselves to

- draw a line through two points in P , and
- draw a circle with a center and a radius, that is a point in P and a distance between two points in P respectively.

A new point is created as lines/circles intersect. $P \subset P \cup \{\text{new points}\}$ and so on.

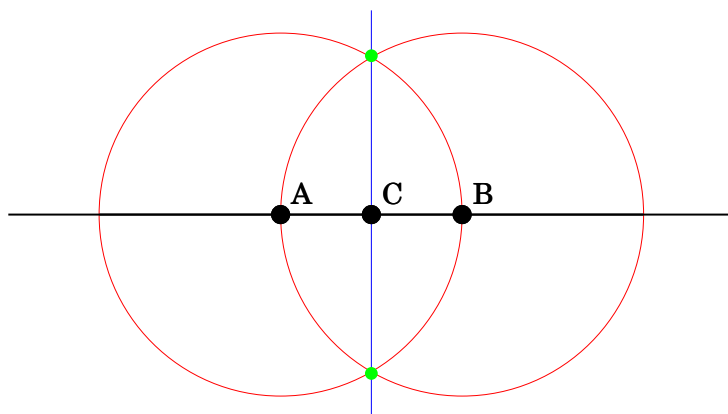
Definition 15.1. We say that a point is *constructable* if we can construct it in a finite number of steps starting from P .

If we let $\overline{P} := \{\text{all constructable points}\}$, then a natural question is: what is $\overline{P}$?
Hopefully it's clear that $|\overline{P}| \leq \omega$.

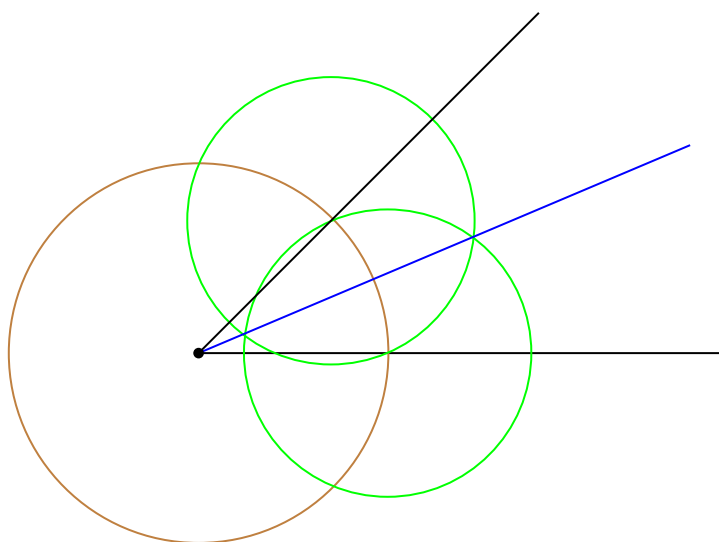
Example 15.2.

1. Starting from two points A and B , we want to create a middle point C . This can be done by first drawing a line between A and B and then creating two circles with center A intersecting B and a circle with center B intersecting A creating two new points where the circle

intersect. Draw a line between the two new points that will orthogonally intersect the line between A and B creating the middle point C .



2. If we start with a point A and an angle from A , we can create half the angle by drawing a circle with center in and an arbitrary radius creating two intersections one each line of the angle. Now we construct two circles between these new points and a line intersecting the new points the two circles made, just like the blue line in example 1.



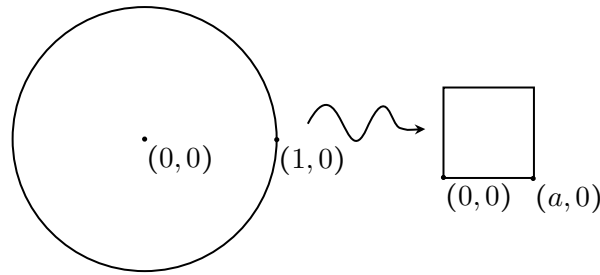
$\mathbb{R}^2$ comes equipped with coordinates, that is we can describe every point $p \in P = \{(x_i, y_i) \mid i \in I\}$ where I is finite.

Now let $\mathbb{K} :=$ "the minimal field generated by $\mathbb{Q}$ and all x_i, y_i ". Then if $u \in \overline{P}$, say $u = (a, b)$, then let $\mathbb{F} := \mathbb{K}\langle a, b \rangle \implies \mathbb{K} \subset \mathbb{F}$, and this is something in Galois theory. Question is now what we can say about $\mathbb{K} \subset \mathbb{F}$? Assume that u is constructable in 1 step. The line equation is of degree 1 and the circle equation is of degree 2, thus a and b are obtained by elements in $\mathbb{K}$ by solving polynomial equations of degree 1 and 2, that is $[\mathbb{F} : \mathbb{K}] = 1$ or 2.

Corollary 15.3. For all $u \in \overline{P}$ $[\mathbb{F} : \mathbb{K}] = 2^m$ for some m .

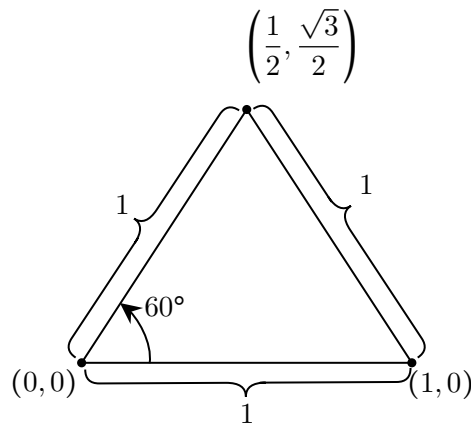
Example 15.4.

- **Application 1:** Squaring the circle. Problem: construct a square whose area equals the area of a circle of radius 1.

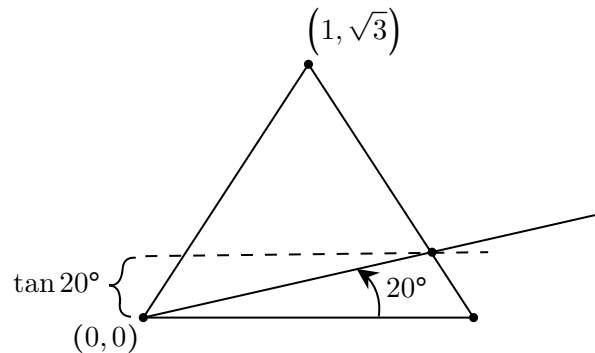


where $a^2 = \pi$ and $a = \sqrt{\pi}$. Thus we have $\mathbb{Q} \subset \mathbb{Q}(\sqrt{\pi})$, but we know that π is transcendental and this $\sqrt{\pi}$ is also transcendental, so this construction is impossible!

➤ **Application 2:** Trisecting the angle:



So we then claim that a 20° angle is not constructable. Let $\mathbb{K} = \mathbb{Q}(\sqrt{3})$. If we can construct 20° then we can construct the point $u = (0, \tan 20^\circ)$ as



and as $\tan(3\theta) = \frac{3\tan(\theta) - \tan^3(\theta)}{1 - 3\tan^2(\theta)}$. $\theta = 20^\circ \implies 3\theta = 60^\circ \implies \tan 3\theta = \sqrt{3}$. Letting $a = \tan(20^\circ)$ gives the polynomial $a^3 - 3\sqrt{3}a^2 - 3a + \sqrt{3} = 0$. We claim that this polynomial is irreducible over $\mathbb{Q}(\sqrt{3})$: that is a polynomial of degree 3 is irreducible $\iff$ it has no roots in $\mathbb{Q}(\sqrt{3})$.

$$\mathbb{Q}(\sqrt{3}) = \{u + v\sqrt{3} \mid u, v \in \mathbb{Q}\}$$

Then $a^3 - 3\sqrt{3}a^2 - 3a + \sqrt{3} = 0$ can be written as

$$(u + v\sqrt{3})^3 - 3\sqrt{3}(u + v\sqrt{3})^2 - 3(u + v\sqrt{3}) + \sqrt{3} = 0$$

$$(u^3 + 9uv^2 - 18uv - 3u) + (3u^3v + 3v^3 + 3u^2 - 9v^2 - 3v + 1)\sqrt{3} = 0$$

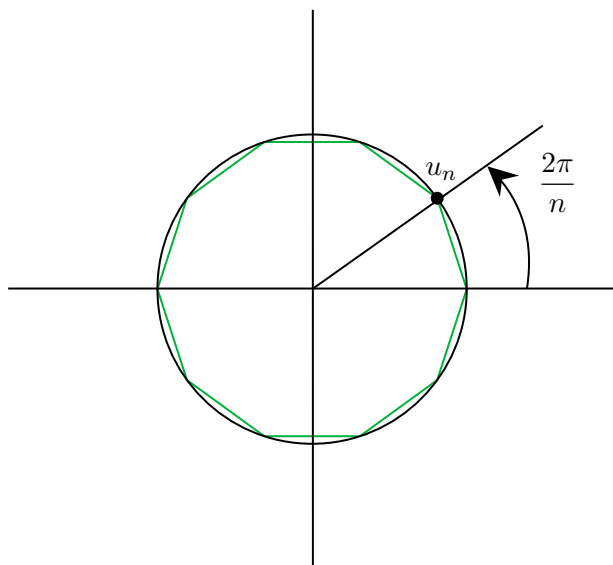
where both terms must be 0 as they are linear independent (because of the $\sqrt{3}$ at the end).

Case 1: $u = 0 \implies 3v^3 - 9v^2 - 3v + 1 = 0$. Using the rational root test?? we get the rational candidates ± 1 and $\pm \frac{1}{3}$, which non are roots, hence no rational roots.

Case 2: $u \neq 0 \implies u^2 = -9v^2 + 18v + 3$. We then get that $-3v^3 + 9v^2 - 6v - 1 = 0$ gives the rational candidates ± 1 and $\pm \frac{1}{3}$, which again are not roots. Thus we can conclude that $f(a)$ is irreducible implying that $[\mathbb{F} : \mathbb{K}] = 3 \neq 2^m$

15.1 Construction of regular polygons

Problem 15.5. For which $n \in \{1, 2, \dots\}$ can one construct from the points $(0, 0)$ and $(1, 0)$ a regular n -gon inscribed in the unit circle?



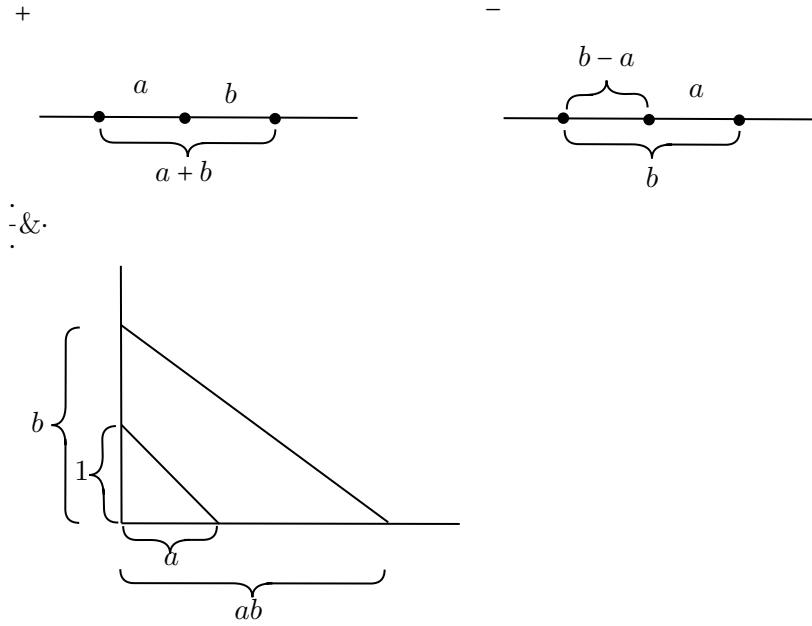
where $u_n = (\cos(\frac{2\pi}{n}), \sin(\frac{2\pi}{n}))$.

Theorem 15.6 (Wantzel, 1837 (conjectured by Gauss)). u_n is constructable *if and only if* any odd prime factor of n is square free and of the form $2^{2^m} + 1$.

Such primes are exactly the Fermat primes. Some known are 3, 5, 17, 257, 65537.

Observe 15.7.

1. $\mathbb{Q}$ is constructable. We show using three figures how all the operations are well defined.



2. u_n being constructable implies $\forall m \mid n$ u_m is also constructable. $n = m \cdot k$, repeat $\frac{2\pi}{n}$ k times to get $\frac{2\pi}{m}$.
3. u_n and u_m being constructable implies $u_{\frac{n \cdot m}{(n, m)}}$ is constructable. By Eulers theorem ?? $(n, m) = an + bm$ for some $a, b \in \mathbb{Z}$. Thus $\frac{(n, m)}{nm} = \frac{a}{m} + \frac{b}{n}$, so just repeat $\frac{2\pi}{m}$ a times, and then $\frac{2\pi}{n}$ b times.

Corollary 15.8. If $n = p_1^{k_1} \cdots p_r^{k_r}$ where $p_i \neq p_j, i \neq j$, then u_n is constructable $\iff u_{p_i^{k_i}}$ is constructable for all $1 \leq i \leq r$.

Proof. " $\implies$ " by 2. and " $\impliedby$ " by 3. □

4. u_{2^m} is constructable for all m we can divide any angle in 2.
5. If p is prime and ζ is the primitive p -th root of 1 over $\mathbb{Q}$, so $m_\zeta(x) = x^{p-1} + x^{p-2} + \cdots + 1$, and ζ is a root of $x^p - 1 = (x - 1)(x^{p-1} + x^{p-2} + \cdots + 1)$. If we change variable so that $x \rightarrow x + 1$, then $x^2 + x + 1 \rightarrow (x + 1)^2 + (x + 1) + 1 = x^2 + 3x + 3$ which is irreducible by Eisenstein's criterium ??. Without giving a proof, we claim that this generalises to $m_\zeta(x)$. Then $[Q(\zeta) : \mathbb{Q}] = p - 1$.
6. If p is an odd prime and ζ is the primitive p^2 root of 1, then $m_\zeta(x) = x^{p(p-1)} + x^{p(p-2)} + \cdots + x^p + 1$.

Corollary 15.9. u_{p^2} is not constructable.

This also implies u_{p^r} is not constructable for any $r > 1$.

Proof. $[Q(\zeta) : \mathbb{Q}] = p(p - 1) \neq 2^m$ as p is an odd prime. □

Corollary 15.10. If p is an odd prime. Then u_p being constructable implies $p = 2^{2^m} + 1$ for some m .

Proof. □

Corollary 15.11. If u_n is constructable, then u_n is as in Wantzel's theorem 15.6

Remark 15.12. If $p = 2^{2^m} + 1$ is a prime, then u_p is constructable.

Proof. Let ζ be the primitive p -th root of 1. Then $[\mathbb{Q}(\zeta) : \mathbb{Q}] = p - 1$ (power of 2). The field extension $\mathbb{Q} \subset \mathbb{Q}(\zeta)$ is separable as the characteristic is 0 and the splitting field of $x^p - 1$ hence normal, so this is a Galois extension. Let $G = \text{Gal}(\mathbb{Q}(\zeta) / \mathbb{Q})$ the Galois group. Then $|G| = \text{"some power of 2"}$ meaning that G is solvable. That is, there exists some subnormal chain

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \dots \triangleleft G_k = G$$

where G_i / G_{i-1} is abelian. Refine such that $|G_i / G_{i-1}| = 2$, then

$$\mathbb{Q} = \mathbb{K} = \mathbb{F}^G \subset \underbrace{\mathbb{F}^{G_{k-1}} \subset \mathbb{F}^{G_{k-1}}}_{\text{deg}=2} \subset \dots \subset \mathbb{F}^{\{e\}} = \mathbb{Q}(\zeta).$$

Each extension we have we solve equations of degree 2, hence constructable. □

16 Appendix